HARSH PATEL, Final Project, ITMD 555, May 5th, 2026

1. Application code *(Java Version: JRE system library JavaSE-21)*

1.1. MainActivity.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import android.content.Intent;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;

public class MainActivity extends AppCompatActivity {

    FirebaseAuth tempUserAuth;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Code for : Check for broken pipe errors and exit gracefully.
        Thread.setDefaultUncaughtExceptionHandler((thread, throwable) -> {
            if (throwable.getMessage() != null &&
throwable.getMessage().contains("EPIPE")) {
                android.util.Log.e("NexusDebug", "Caught Broken Pipe error, ignoring.");
            } else {
                android.util.Log.e("NexusDebug", "Uncaught Exception", throwable);
                System.exit(1);
            }
        });

        EdgeToEdge.enable(this);
        // Layout file: activity_main.xml
        setContentView(R.layout.activity_main);

        // Note: Edge-to-edge padding handling
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        // Firebase Auth check
        tempUserAuth = FirebaseAuth.getInstance();
        // get all the users' details.
        FirebaseUser user = tempUserAuth.getCurrentUser();

        // if user is not null, then login activities.
        if (user != null) {
            // Go to LoginActivity.class
```

```
            android.util.Log.d("NexusDebug", "User found: " + user.getUid());
            // Force login every time as requested
            Intent intent = new Intent(MainActivity.this, LoginActivity.class);
            // start the LoginActivity.class
            startActivity(intent);
        } else {         // if user is null, then sign up activities.
            // Go to SignupActivity.class
            android.util.Log.d("NexusDebug", "No user found, redirecting to Signup");
            // start the SignupActivity.class
            startActivity(new Intent(MainActivity.this, SignupActivity.class));
        }

        // If then, close this activity.
        finish();
    }
}
```

1.2. Comments.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import com.google.firebase.firestore.Exclude;

// Comment.java
public class Comment {
    private String userName;
    private String userId;
    private String text;
    private long timestamp;
    private String userAvatarUrl;

    private int upvotes = 0;
    private int downvotes = 0;
    private String parentCommentId; // null if top-level
    private boolean isExpanded = true;
    private String commentId;
    private boolean isUpvoted = false;
    private boolean isDownvoted = false;

    // Required for Firebase
    public Comment() {
        // Required for Firebase
    }

    // Constructor for top-level comments
    public Comment(String userName, String userId, String text, long timestamp) {
        this.userName = userName;
        this.userId = userId;
        this.text = text;
        this.timestamp = timestamp;
    }

    // Getter-setter
    public String getUserName() { return userName; }
    public String getUserId() { return userId; }
    public String getText() { return text; }
    public long getTimestamp() { return timestamp; }
    public String getUserAvatarUrl() { return userAvatarUrl; }

    public String getCommentId() { return commentId; }
    public void setCommentId(String commentId) { this.commentId = commentId; }
```

```
    public int getUpvotes() { return upvotes; }
    public void setUpvotes(int upvotes) { this.upvotes = upvotes; }

    public int getDownvotes() { return downvotes; }
    public void setDownvotes(int downvotes) { this.downvotes = downvotes; }

    public String getParentCommentId() { return parentCommentId; }
    public void setParentCommentId(String parentCommentId) { this.parentCommentId =
parentCommentId; }

    public boolean isExpanded() { return isExpanded; }
    public void setExpanded(boolean expanded) { isExpanded = expanded; }

    @Exclude
    public boolean isUpvoted() { return isUpvoted; }
    @Exclude
    public void setUpvoted(boolean upvoted) { isUpvoted = upvoted; }

    @Exclude
    public boolean isDownvoted() { return isDownvoted; }
    @Exclude
    public void setDownvoted(boolean downvoted) { isDownvoted = downvoted; }
}
```

1.3. CommentAdapter.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import android.text.format.DateUtils;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.core.content.ContextCompat;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

// CommentAdapter.java
public class CommentAdapter extends
RecyclerView.Adapter<CommentAdapter.CommentViewHolder> {

    private List<Comment> tempAllComments;
    private OnCommentActionListener tempActionListener;

    // Interface for handling comment actions
    public interface OnCommentActionListener {
        // Called when a comment is replied to
        void onReply(Comment parentComment);
        // Called when a comment is upvoted or downvoted
        void onVote(Comment comment, boolean isUpvote);
    }

    // Constructor
    public CommentAdapter(List<Comment> commentList, OnCommentActionListener listener) {
        // Initialize the adapter with the provided data
        this.tempAllComments = commentList;
```

```
        // Initialize the listener
        this.tempActionListener = listener;
    }

    // ViewHolder
    @NonNull
    @Override
    public CommentViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType)
{
        // Inflate the layout for each item in the RecyclerView
        View tempView =
LayoutInflater.from(parent.getContext()).inflate(R.layout.item_comment, parent, false);
        // Create a new ViewHolder with the inflated view
        return new CommentViewHolder(tempView);
    }

    // Bind data to the ViewHolder
    @Override
    public void onBindViewHolder(@NonNull CommentViewHolder tempLikeVoteHolder, int
position) {
        // Get the comment at the current position
        Comment tempComment = tempAllComments.get(position);

        // Set padding based on parent-child relationship
        int paddingStart = tempComment.getParentCommentId() != null ? 32 : 0;
        // Set padding
        tempLikeVoteHolder.itemView.setPadding(paddingStart,
tempLikeVoteHolder.itemView.getPaddingTop(),
                                tempLikeVoteHolder.itemView.getPaddingRight(),
tempLikeVoteHolder.itemView.getPaddingBottom());

        // Set the text for each view in the ViewHolder
        tempLikeVoteHolder.tvCommentUser.setText(tempComment.getUserName() != null ?
tempComment.getUserName() : "Anonymous");
        // Set the text for each view in the ViewHolder
        tempLikeVoteHolder.tvCommentText.setText(tempComment.getText());

        // Set the time for each view in the ViewHolder
        if (tempComment.getTimestamp() > 0) {
            // Set the time for each view in the ViewHolder
            CharSequence timeAgo = DateUtils.getRelativeTimeSpanString(
                    // Set the time for each view in the ViewHolder
                    tempComment.getTimestamp(),
                    // Set the time for each view in the ViewHolder
                    System.currentTimeMillis(),
                    // Set the time for each view in the ViewHolder
                    DateUtils.MINUTE_IN_MILLIS);
            // Set the time for each view in the ViewHolder
            tempLikeVoteHolder.tvCommentTime.setText(timeAgo);
        } else {
            tempLikeVoteHolder.tvCommentTime.setText("");
        }

        // Set the vote counts for each view in the ViewHolder

tempLikeVoteHolder.tvCommentUpvotes.setText(String.valueOf(tempComment.getUpvotes()));

tempLikeVoteHolder.tvCommentDownvotes.setText(String.valueOf(tempComment.getDownvotes()))
;

        // Update vote icons UI
        if (tempComment.isUpvoted()) {
            // Update ivCommentUpvote icons UI
```

```
tempLikeVoteHolder.ivCommentUpvote.setColorFilter(ContextCompat.getColor(tempLikeVoteHold
er.itemView.getContext(), R.color.upvote_green));
            // Update tvCommentUpvotes text color

tempLikeVoteHolder.tvCommentUpvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHolde
r.itemView.getContext(), R.color.upvote_green));
            // Update ivCommentDownvote icons UI
            tempLikeVoteHolder.ivCommentDownvote.clearColorFilter();
            // Update tvCommentDownvotes text color

tempLikeVoteHolder.tvCommentDownvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHol
der.itemView.getContext(), R.color.subtitleText));
        } else if (tempComment.isDownvoted()) {

tempLikeVoteHolder.ivCommentDownvote.setColorFilter(ContextCompat.getColor(tempLikeVoteHo
lder.itemView.getContext(), R.color.downvote_red));

tempLikeVoteHolder.tvCommentDownvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHol
der.itemView.getContext(), R.color.downvote_red));
            tempLikeVoteHolder.ivCommentUpvote.clearColorFilter();

tempLikeVoteHolder.tvCommentUpvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHolde
r.itemView.getContext(), R.color.subtitleText));
        } else {
            tempLikeVoteHolder.ivCommentUpvote.clearColorFilter();
            tempLikeVoteHolder.ivCommentDownvote.clearColorFilter();

tempLikeVoteHolder.tvCommentUpvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHolde
r.itemView.getContext(), R.color.subtitleText));

tempLikeVoteHolder.tvCommentDownvotes.setTextColor(ContextCompat.getColor(tempLikeVoteHol
der.itemView.getContext(), R.color.subtitleText));
        }

        tempLikeVoteHolder.ivCommentUpvote.setOnClickListener(v -> {
            // Handle upvote
            tempActionListener.onVote(tempComment, true);
            // Update the UI
            notifyItemChanged(tempLikeVoteHolder.getAdapterPosition());
        });
        tempLikeVoteHolder.ivCommentDownvote.setOnClickListener(v -> {
            // Handle downvote
            tempActionListener.onVote(tempComment, false);
            // Update the UI
            notifyItemChanged(tempLikeVoteHolder.getAdapterPosition());
        });
        tempLikeVoteHolder.tvCommentReply.setOnClickListener(v ->
tempActionListener.onReply(tempComment));

        // Toggle visibility of children (simplified)
        tempLikeVoteHolder.itemView.setOnClickListener(v -> {
            // Toggle the expanded state
            tempComment.setExpanded(!tempComment.isExpanded());
            // Update the UI
            notifyDataSetChanged();
        });

        // Hide if parent is collapsed (basic logic)
        if (shouldHide(tempComment)) {
            // Hide the view
            tempLikeVoteHolder.itemView.setVisibility(View.GONE);
            // Set the layout params to 0
```

```
            tempLikeVoteHolder.itemView.setLayoutParams(new RecyclerView.LayoutParams(0,
0));
        } else {
            // Show the view
            tempLikeVoteHolder.itemView.setVisibility(View.VISIBLE);
            // Set the layout params to match parent
            tempLikeVoteHolder.itemView.setLayoutParams(new RecyclerView.LayoutParams(
                    ViewGroup.LayoutParams.MATCH_PARENT,
ViewGroup.LayoutParams.WRAP_CONTENT));
        }
    }

    private boolean shouldHide(Comment comment) {
        // Check if the comment is not expanded
        String tempParentId = comment.getParentCommentId();
        // If not expanded, return true
        while (tempParentId != null) {
            // Find the parent comment
            for (Comment tempCommentIdx : tempAllComments) {
                // If found, check if it's expanded
                if (tempCommentIdx.getCommentId().equals(tempParentId)) {
                    // If not expanded, return true
                    if (!tempCommentIdx.isExpanded()) return true;
                    // Update tempParentId
                    tempParentId = tempCommentIdx.getParentCommentId();
                    // Break the loop
                    break;
                }
            }
            // If not found, break the loop
            if (tempParentId == null) break;
        }
        // Return false
        return false;
    }

    // Get the number of items in the list
    @Override
    public int getItemCount() {
        return tempAllComments.size();
    }

    // ViewHolder
    public static class CommentViewHolder extends RecyclerView.ViewHolder {
        TextView tvCommentUser, tvCommentText, tvCommentTime, tvCommentUpvotes,
tvCommentDownvotes, tvCommentReply;
        ImageView ivCommentUpvote, ivCommentDownvote;

        // Constructor
        public CommentViewHolder(@NonNull View itemView) {
            super(itemView);
            tvCommentUser = itemView.findViewById(R.id.tvCommentUser);
            tvCommentText = itemView.findViewById(R.id.tvCommentText);
            tvCommentTime = itemView.findViewById(R.id.tvCommentTime);
            tvCommentUpvotes = itemView.findViewById(R.id.tvCommentUpvotes);
            tvCommentDownvotes = itemView.findViewById(R.id.tvCommentDownvotes);
            tvCommentReply = itemView.findViewById(R.id.tvCommentReply);
            ivCommentUpvote = itemView.findViewById(R.id.ivCommentUpvote);
            ivCommentDownvote = itemView.findViewById(R.id.ivCommentDownvote);
        }
    }
}
```

1.4. Community.java

```java
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import com.google.firebase.firestore.PropertyName;

// Community.java
public class Community {

    // name
    @PropertyName("name")
    public String name;

    // icon
    @PropertyName("icon")
    public String icon;

    // summary
    @PropertyName("summary")
    public String summary;

    // timestamp
    @PropertyName("timestamp")
    public long timestamp;

    // communityId
    public String communityId;
    // selected
    public boolean selected;

    // Constructor
    public Community() {}

    public String getCommunityId() {
        // Return the communityId
        return communityId;
    }
    public void setCommunityId(String communityId) {
        // Set the communityId
        this.communityId = communityId;
    }

    // Constructor
    public Community(String name, String icon) {
        // name
        this.name = name;
        // icon
        this.icon = icon;
        // selected
        this.selected = false;
    }

    // Getters and Setters
    @PropertyName("name")
    public String getName() {
        return name;
    }

    // Setters
    @PropertyName("name")
    public void setName(String name) {
```

```
        this.name = name;
    }

    // Getters and Setters
    @PropertyName("icon")
    public String getIcon() {
        return icon;
    }

    // Setters
    @PropertyName("icon")
    public void setIcon(String icon) {
        this.icon = icon;
    }

    // Getters and Setters
    @PropertyName("summary")
    public String getSummary() { return summary; }

    // Setters
    @PropertyName("summary")
    public void setSummary(String summary) {
        this.summary = summary;
    }

    // Getters and Setters
    @PropertyName("timestamp")
    public long getTimestamp() {
        return timestamp;
    }

    // Setters
    @PropertyName("timestamp")
    public void setTimestamp(long timestamp) {
        this.timestamp = timestamp;
    }

    // Getters and Setters
    public boolean isSelected() {
        return selected;
    }

    // Setters
    public void setSelected(boolean selected) {
        this.selected = selected;
    }
}
```

1.5. CommunityAdapter.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
```

```java
import java.util.ArrayList;
import java.util.HashSet;
import java.util.List;
import java.util.Set;

// CommunityAdapter
public class CommunityAdapter extends RecyclerView.Adapter<CommunityAdapter.ViewHolder> {

    // Context, List<Community>, OnCommunitySelectedListener, Set
    Context tempContext;
    List<Community> tempCommunityList;
    List<Community> tempCommunityListFull;
    OnCommunitySelectedListener tempListener;
    Set<String> tempSelectedCommunities = new HashSet<>();

    // Interface for handling community selection
    public interface OnCommunitySelectedListener {
        // Called when the selection count changes
        void onSelectionChanged(int count);
    }

    // Constructor
    public CommunityAdapter(Context context, List<Community> communityList,
OnCommunitySelectedListener listener) {
        // tempContext
        this.tempContext = context;
        // tempCommunityList
        this.tempCommunityList = communityList;
        // tempCommunityListFull
        this.tempCommunityListFull = new ArrayList<>(communityList);
        // tempListener
        this.tempListener = listener;
    }

    // ViewHolder
    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        // Inflate the layout for each item in the RecyclerView
        View tempView = LayoutInflater.from(tempContext)
                // Inflate the layout for each item in the RecyclerView
                .inflate(R.layout.community_item, parent, false);
        // Create a new ViewHolder with the inflated view
        return new ViewHolder(tempView);
    }

    // Bind data to the ViewHolder
    @Override
    public void onBindViewHolder(@NonNull ViewHolder holder, int position) {
        // Get the community at the current position
        Community tempCommunity = tempCommunityList.get(position);
        // Set the text for each view in the ViewHolder
        holder.name.setText(tempCommunity.getName());

        // Set the icon for each view in the ViewHolder
        int iconRes = 0;
        // Set the icon for each view in the ViewHolder
        if (tempCommunity.getIcon() != null && !tempCommunity.getIcon().isEmpty()) {
            // Set the icon for each view in the ViewHolder
            iconRes = tempContext.getResources().getIdentifier(
                    // Set the icon for each view in the ViewHolder
                    tempCommunity.getIcon(),
```

```
                // Set the icon for each view in the ViewHolder
                "drawable",
                // Set the icon for each view in the ViewHolder
                tempContext.getPackageName()
        );
    }

    // Set the icon for each view in the ViewHolder
    if (iconRes != 0) {
        // Set the icon for each view in the ViewHolder
        holder.icon.setImageResource(iconRes);
    } else {
        // Set the icon for each view in the ViewHolder
        holder.icon.setImageResource(R.drawable.ic_launcher_foreground);
    }

    // highlight selected
    if (tempSelectedCommunities.contains(tempCommunity.getCommunityId())) {
        // highlight color.
        holder.cardView.setStrokeColor(android.graphics.Color.parseColor("#4CAF50"));
        // highlight width
        holder.cardView.setStrokeWidth(4);
        // highlight visibility
        holder.checkMark.setVisibility(View.VISIBLE);
    } else {
        // highlight color.
        holder.cardView.setStrokeColor(android.graphics.Color.parseColor("#E6E8F0"));
        // highlight width
        holder.cardView.setStrokeWidth(2);
        // highlight visibility
        holder.checkMark.setVisibility(View.GONE);
    }

    holder.itemView.setOnClickListener(v -> {
        // toggle selected
        String tempId = tempCommunity.getCommunityId();
        // tempSelectedCommunities
        if (tempSelectedCommunities.contains(tempId)) {
            // tempID
            tempSelectedCommunities.remove(tempId);
            // toggle selected
        } else {
            tempSelectedCommunities.add(tempId);
        }

        // update UI
        notifyItemChanged(position);

        // update count
        if (tempListener != null) {
            // update count
            tempListener.onSelectionChanged(tempSelectedCommunities.size());
        }
    });
}

// Get the number of items in the list
@Override
public int getItemCount() {
    return tempCommunityList.size();
}

// Get the selected communities
```

```java
    public Set<String> getTempSelectedCommunities() {
        return tempSelectedCommunities;
    }

    // Set the selected communities
    public void setTempSelectedCommunities(List<String> selectedList) {
        // Clear the list
        if (selectedList != null) {
            // Clear the list
            this.tempSelectedCommunities = new HashSet<>(selectedList);
            // Notify the adapter
            notifyDataSetChanged();
            // Update the count
            updateCount();
        }
    }

    // Update the count
    private void updateCount() {
        // Update the count
        if (tempListener != null) {
            // Update the count
            int count = tempSelectedCommunities != null ? tempSelectedCommunities.size()
: 0;
            // Update the count
            tempListener.onSelectionChanged(count);
        }
    }

    // Filter the list of communities
    public void filter(String text) {
        // Filter the list
        List<Community> tempFilteredList = new ArrayList<>();
        // If no text, show all communities
        if (text == null || text.isEmpty()) {
            // Add all communities to the list
            tempFilteredList.addAll(tempCommunityListFull);
        } else {
            // Filter the list
            String tempQuery = text.toLowerCase().trim();
            // Loop through the list
            for (Community item : tempCommunityListFull) {
                // if filterPattern
                if (item.getName() != null &&
item.getName().toLowerCase().contains(tempQuery)) {
                    // add into tempFilteredList
                    tempFilteredList.add(item);
                }
            }
        }
        // Update the list
        tempCommunityList.clear();
        // Update the list
        tempCommunityList.addAll(tempFilteredList);
        // Notify the adapter
        notifyDataSetChanged();
        // Keep count in sync after filter
        updateCount();
    }

    public void updateList(List<Community> newList) {
        // Update the list
        this.tempCommunityListFull = new ArrayList<>(newList);
```

```
        // Update the list with clear
        this.tempCommunityList.clear();
        // add all
        this.tempCommunityList.addAll(newList);
        // Notify the adapter
        notifyDataSetChanged();
        // Keep count in sync after filter
        updateCount();
    }

    // ViewHolder
    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView name;
        ImageView icon;
        com.google.android.material.card.MaterialCardView cardView;
        ImageView checkMark;

        // Constructor ViewHolder
        public ViewHolder(View itemView) {
            // Constructor ViewHolder
            super(itemView);
            // Initialize name
            name = itemView.findViewById(R.id.communityName);
            // Initialize icon
            icon = itemView.findViewById(R.id.communityIcon);
            // Initialize cardView
            cardView = itemView.findViewById(R.id.cardView);
            // Initialize checkMark
            checkMark = itemView.findViewById(R.id.checkMark);
        }
    }
}
```

1.6. CustomizedFeedActivity.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import android.content.Intent;
import android.os.Bundle;
import android.util.Log;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.GridLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import com.google.firebase.firestore.DocumentSnapshot;
import com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.SetOptions;

import com.google.android.material.dialog.MaterialAlertDialogBuilder;
import androidx.appcompat.app.AlertDialog;
import android.view.View;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Set;
```

```java
public class CustomizeFeedActivity extends AppCompatActivity {

    RecyclerView tempRecyclerView;
    List<Community> tempCommunityList;
    CommunityAdapter tempAdapter;

    FirebaseFirestore firebaseDb;
    Button tempContinueBtn;

    TextView tempSelectedCountText;

    private String tempCurrentUserName;
    private String tempCurrentUserEmail;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        // Layout file: activity_customize_feed.xml
        super.onCreate(savedInstanceState);
        // Layout file: activity_customize_feed.xml
        setContentView(R.layout.activity_customize_feed);

        tempRecyclerView = findViewById(R.id.communityRecycler);
        tempContinueBtn = findViewById(R.id.startFeedBtn);
        tempSelectedCountText = findViewById(R.id.selectedCountText);

        tempRecyclerView.setLayoutManager(new GridLayoutManager(this, 2));

        // User profile
        findViewById(R.id.btnUserProfile).setOnClickListener(v ->
showUserProfileDialog());

        // Initialize adapter
        tempCommunityList = new ArrayList<>();
        // Set adapter
        tempAdapter = new CommunityAdapter(this, tempCommunityList, count -> {
            // Update selected count text
            Log.d("NexusDebug", "Selection count changed to: " + count);
            // Update selected count text
            tempSelectedCountText.setText(getString(R.string.selected_count, count));
        });
        // Set initial count display
        tempSelectedCountText.setText(getString(R.string.selected_count, 0));
        // Set adapter
        tempRecyclerView.setAdapter(tempAdapter);
        // Search
        android.widget.EditText searchEdit = findViewById(R.id.searchCommunities);
        // Search
        searchEdit.addTextChangedListener(new android.text.TextWatcher() {
            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int
after) {}
            // Search
            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                // Filter
                tempAdapter.filter(s.toString());
            }
            // Search
            @Override
            public void afterTextChanged(android.text.Editable s) {}
        });
```

```
        // Firebase Firestore check
        firebaseDb = FirebaseFirestore.getInstance();
        // Get userName and email from intent
        tempCurrentUserName = getIntent().getStringExtra("userName");
        tempCurrentUserEmail = getIntent().getStringExtra("userEmail");
        // Load user preferences and communities
        loadUserPreferencesAndCommunities();
        // Continue button
        tempContinueBtn.setOnClickListener(v -> saveSelection());
    }

    private void loadUserPreferencesAndCommunities() {
        // Check for user
        FirebaseUser tempCurrentUser = FirebaseAuth.getInstance().getCurrentUser();
        // If no user, load all communities
        if (tempCurrentUser == null) {
            // Load all communities
            loadCommunities(null);
            // Return
            return;
        }

        firebaseDb.collection("users").document(tempCurrentUser.getUid()).get()
                // If user exists, load user preferences and communities
                .addOnSuccessListener(documentSnapshot -> {
                    // If no preferences exist, load all communities
                    tempCurrentUserName = documentSnapshot.getString("name");
                    // If no preferences exist, load all communities
                    tempCurrentUserEmail = documentSnapshot.getString("email");

                    // If no preferences exist, load all communities
                    @SuppressWarnings("unchecked")
                    List<String> selected = (List<String>)
documentSnapshot.get("selectedCommunities");
                    // Load communities
                    loadCommunities(selected);
                })
                // If user does not exist, load all communities
                .addOnFailureListener(e -> {
                    // If user does not exist, load all communities
                    loadCommunities(null);
                });
    }

    // Load communities
    private void loadCommunities(List<String> userSelectedIds) {
        Log.d("NexusDebug", "Fetching communities...");
        // Fetch communities
        firebaseDb.collection("communities")
                .get()
                .addOnSuccessListener(queryDocumentSnapshots -> {
                    // If no communities exist, show toast
                    List<Community> fetchedCommunities = new ArrayList<>();
                    // Log number of communities found
                    Log.d("NexusDebug", "Documents found: " +
queryDocumentSnapshots.size());
                    // If no communities exist, show toast
                    if (queryDocumentSnapshots.isEmpty()) {
                        // Show toast
                        Toast.makeText(this, "No communities found",
Toast.LENGTH_SHORT).show();
                        // Return
                        return;
```

```
                    }

                    // Loop through communities
                    for (DocumentSnapshot doc : queryDocumentSnapshots) {
                        // Convert document to community
                        Community community = doc.toObject(Community.class);
                        // If community exists, add to list
                        if (community != null && community.getName() != null) {
                            // Set community id
                            community.setCommunityId(doc.getId());
                            // Add community to list
                            fetchedCommunities.add(community);
                        }
                    }

                    // Update adapter
                    tempAdapter.updateList(fetchedCommunities);
                    // Set selected communities
                    if (userSelectedIds != null) {
                        // Set selected communities
                        tempAdapter.setTempSelectedCommunities(userSelectedIds);
                        // Update selected count text
                    } else {
                        // Set selected communities
                        tempAdapter.setTempSelectedCommunities(new ArrayList<>());
                    }
                })
                // If error, show toast
                .addOnFailureListener(e -> {
                    // Show toast
                    Log.e("NexusDebug", "Load failed", e);
                    // Show toast
                    Toast.makeText(this,
                            "Load failed: " + e.getMessage(),
                            Toast.LENGTH_LONG).show();
                });
    }

    private void saveSelection() {
        // Log selection
        Log.d("NexusDebug", "Attempting to save selection...");
        // Get selected communities
        Set<String> selected = tempAdapter.getTempSelectedCommunities();
        // If no communities selected, show toast
        if (selected == null || selected.isEmpty()) {
            // Show toast
            Toast.makeText(this, "Select at least 1 community",
Toast.LENGTH_SHORT).show();
            // Return
            return;
        }
        // Get current user
        FirebaseUser currentUser = FirebaseAuth.getInstance().getCurrentUser();
        // If no user, show toast
        if (currentUser == null) {
            // Show toast
            Toast.makeText(this, "User not authenticated", Toast.LENGTH_SHORT).show();
            // Return
            return;
        }

        // Get user id
        String uid = currentUser.getUid();
```

```java
        // Disable button
        tempContinueBtn.setEnabled(false);
        // Create hashmap
        HashMap<String, Object> data = new HashMap<>();
        // Put selected communities in hashmap
        data.put("selectedCommunities", new ArrayList<>(selected));

        // Save to Firebase
        firebaseDb.collection("users")
                // Update user
                .document(uid)
                // Merge with existing data
                .set(data, SetOptions.merge())
                // If success, show toast
                .addOnSuccessListener(aVoid -> {
                    // Log success
                    Log.d("NexusDebug", "Preferences saved successfully!");
                    // Show toast
                    Toast.makeText(this,
                            // USE STRING RESOURCES
                            "Preferences saved",
                            // USE STRING RESOURCES
                            Toast.LENGTH_SHORT).show();

                    // NAVIGATE ONLY ON SUCCESS
                    Intent intent = new Intent(
                            // USE STRING RESOURCES
                            CustomizeFeedActivity.this,
                            // USE STRING RESOURCES
                            DashboardActivity.class
                    );
                    // Pass userName and email
                    intent.putExtra("userName", tempCurrentUserName);
                    intent.putExtra("userEmail", tempCurrentUserEmail);
                    // Start activity
                    startActivity(intent);
                    // Finish activity
                    finish();
                })
                // If error, show toast
                .addOnFailureListener(e -> {
                    // Log error
                    Log.e("NexusDebug", "Error writing to Firebase", e);
                    // Show toast
                    Toast.makeText(this,
                            // USE STRING RESOURCES
                            "Error: " + e.getMessage(),
                            // USE STRING RESOURCES
                            Toast.LENGTH_SHORT).show();
                    // Enable button
                    tempContinueBtn.setEnabled(true);
                });
    }

    // User profile
    private void showUserProfileDialog() {
        // Get current user
        FirebaseUser tempUser = FirebaseAuth.getInstance().getCurrentUser();
        // If no user, return
        if (tempUser == null) return;

        // Create dialog
        View tempDialogView = getLayoutInflater().inflate(R.layout.dialog_user_profile,
```

```
null);
        // Create dialog
        AlertDialog tempDialog = new MaterialAlertDialogBuilder(this)
                // Set dialog view
                .setView(tempDialogView)
                // Create dialog
                .create();
        // Set background
        if (tempDialog.getWindow() != null) {
            // Set background

tempDialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);
        }

        // Get views
        TextView txtName = tempDialogView.findViewById(R.id.profileName);
        // Get views
        TextView txtEmail = tempDialogView.findViewById(R.id.profileEmail);
        // Get views
        Button btnLogout = tempDialogView.findViewById(R.id.btnLogout);

        // Get user info

FirebaseFirestore.getInstance().collection("users").document(tempUser.getUid()).get()
                // If success, set user info
                .addOnSuccessListener(doc -> {
                    // If user exists, set user info
                    if (doc.exists()) {
                        // Get user info
                        String tempName = doc.getString("name");
                        // Get user info
                        String tempEmail = doc.getString("email");
                        // If userName exists, set userName
                        if (tempName != null) {
                            // Set userName
                            tempCurrentUserName = tempName;
                            // Set userName
                            txtName.setText(tempName);
                        }
                        if (tempEmail != null) {
                            // Set userEmail
                            tempCurrentUserEmail = tempEmail;
                            // Set userEmail
                            txtEmail.setText(tempEmail);
                        }
                    }
                });

        // Set user info
        txtName.setText(tempCurrentUserName != null ? tempCurrentUserName : "Email ID");
        // Set user info
        txtEmail.setText(tempCurrentUserEmail != null ? tempCurrentUserEmail :
tempUser.getEmail());

        // Logout
        btnLogout.setOnClickListener(v -> {
            // Logout
            FirebaseAuth.getInstance().signOut();

            // NAVIGATE ONLY ON SUCCESS
            Intent intent = new Intent(this, LoginActivity.class);
            // Set flags
            intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
```

```
Intent.FLAG_ACTIVITY_CLEAR_TASK);
            // Start activity to FLAG_ACTIVITY_CLEAR_TASK
            startActivity(intent);
            // Finish activity
            finish();
        });

        // Show dialog
        tempDialog.show();
    }
}
```

1.7. DashboardActivity.java

```
package com.example.nexus;

import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import com.google.android.material.dialog.MaterialAlertDialogBuilder;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import java.util.HashMap;
import java.util.Map;

import com.google.firebase.auth.FirebaseAuth;
import android.widget.TextView;
import com.google.firebase.firestore.*;
import com.google.firebase.messaging.FirebaseMessaging;

import java.util.ArrayList;
import java.util.List;

// Dashboard Activity
public class DashboardActivity extends AppCompatActivity {

    RecyclerView tempRecyclerView;
    FeedAdapter tempAdapter;
    List<Post> tempPostList;

    FirebaseFirestore tempFirebaseDB;

    private String tempCurrentUserName;
    private String tempCurrentUserEmail;

    private int tempPendingRequests = 0;
    private boolean tempShowingBookmarks = false;
    private List<String> tempUserCommunityIds = new ArrayList<>();
    private boolean tempIsDateFilterActive = false;

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        // Set up the activity
        super.onCreate(savedInstanceState);
        // Set the layout for the activity_dashboard.xml
        setContentView(R.layout.activity_dashboard);

        // Set up the RecyclerView
        tempRecyclerView = findViewById(R.id.feedRecycler);
        // Set the layout manager for the RecyclerView
        tempRecyclerView.setLayoutManager(new LinearLayoutManager(this));

        // Initialize the adapter
        tempPostList = new ArrayList<>();
        // Set the adapter for the RecyclerView
        tempAdapter = new FeedAdapter(this, tempPostList);
        // Set the adapter for the RecyclerView
        tempRecyclerView.setAdapter(tempAdapter);

        // Set up the search bar
        EditText tempSearchPosts = findViewById(R.id.searchPosts);
        // Set up the search bar listener
        tempSearchPosts.addTextChangedListener(new android.text.TextWatcher() {
            // Set up beforeTextChanged
            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int
after) {}
            // Set up onTextChanged
            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                // Filter the adapter
                tempAdapter.filter(s.toString());
            }
            // Set up afterTextChanged
            @Override
            public void afterTextChanged(android.text.Editable s) {}
        });

        // Initialize the Firebase Firestore
        tempFirebaseDB = FirebaseFirestore.getInstance();

        // Get the user's name and email
        tempCurrentUserName = getIntent().getStringExtra("userName");
        tempCurrentUserEmail = getIntent().getStringExtra("userEmail");

        // Set the user's name and email in the adapter
        if (tempCurrentUserName != null && tempCurrentUserEmail != null) {
            // Set the user's name and email in the adapter
            tempAdapter.setCurrentUser(tempCurrentUserName, tempCurrentUserEmail);
        }

        // Load the user's feed
        loadUserFeed();
        // Schedule the widget update
        scheduleWidgetUpdates();

        // Set up the add post button
        FloatingActionButton addPostFab = findViewById(R.id.addPostFab);
        // Set up the add post button listener
        addPostFab.setOnClickListener(v -> showCreateOptions());

        // Set up the show bookmarks button
        findViewById(R.id.btnShowBookmarks).setOnClickListener(v ->
toggleBookmarksView());
```

```
        // Set up the filter button
        findViewById(R.id.btnFilterCommunities).setOnClickListener(v -> {
            // CustomizeFeedActivity.class
            Intent intent = new Intent(this, CustomizeFeedActivity.class);
            // Pass the user's name and email to the next activity
            intent.putExtra("userName", tempCurrentUserName);
            intent.putExtra("userEmail", tempCurrentUserEmail);
            // startActivity
            startActivity(intent);
        });

        // Set up the filter date button
        findViewById(R.id.btnFilterDate).setOnClickListener(v -> {
            // Check if date filter is active
            if (tempIsDateFilterActive) {
                // Clear the date filter
                clearDateFilter();
            } else {
                // Show the date picker
                showDatePicker();
            }
        });

        // Set up the user profile button
        findViewById(R.id.btnUserProfile).setOnClickListener(v ->
showUserProfileDialog());

        // Request Notification Permission for Android 13+
        requestNotificationPermission();
    }

    private void requestNotificationPermission() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (checkSelfPermission(android.Manifest.permission.POST_NOTIFICATIONS) !=
PackageManager.PERMISSION_GRANTED) {
                requestPermissions(new
String[]{android.Manifest.permission.POST_NOTIFICATIONS}, 101);
            }
        }
    }

    // Set up the date picker
    private void showDatePicker() {
        // Set up the date picker

com.google.android.material.datepicker.MaterialDatePicker<androidx.core.util.Pair<Long,
Long>> dateRangePicker =
                // Set up the date picker

com.google.android.material.datepicker.MaterialDatePicker.Builder.dateRangePicker()
                        // Set up the title
                        .setTitleText("Filter posts by date range")
                        // Set up the theme
                        .build();

        // Set up the date picker listener
        dateRangePicker.addOnPositiveButtonClickListener(selection -> {
            // Filter the adapter
            if (selection.first != null && selection.second != null) {
                // Filter the adapter
                filterPostsByDateRange(selection.first, selection.second);
            }
```

```
        });

        // Show the date picker
        dateRangePicker.show(getSupportFragmentManager(), "DATE_RANGE_PICKER");
    }

    // Clear the date filter
    private void clearDateFilter() {
        // Clear the date filter
        resetDateFilterUI();
        // Set up the adapter
        tempAdapter.updateList(tempPostList);
        // Show all posts
        Toast.makeText(this, "Showing all posts", Toast.LENGTH_SHORT).show();
    }

    // Reset the date filter UI
    private void resetDateFilterUI() {
        // Reset the date filter UI
        tempIsDateFilterActive = false;
        // Set up the filter button
        android.widget.ImageButton btn = findViewById(R.id.btnFilterDate);
        // Set up the filter button
        btn.setColorFilter(null);
    }

    // Filter the adapter
    private void filterPostsByDateRange(long startTimestampMs, long endTimestampMs) {
        // Normalize start to start of day
        java.util.Calendar rightNow = java.util.Calendar.getInstance();
        // Set up the calendar
        rightNow.setTimeInMillis(startTimestampMs);
        // Set up the Hours
        rightNow.set(java.util.Calendar.HOUR_OF_DAY, 0);
        // Set up the Minutes
        rightNow.set(java.util.Calendar.MINUTE, 0);
        // Set up the Seconds
        rightNow.set(java.util.Calendar.SECOND, 0);
        // Set up the Milliseconds
        rightNow.set(java.util.Calendar.MILLISECOND, 0);
        // Get the time in milliseconds
        long rangeStart = rightNow.getTimeInMillis();

        // Normalize end to end of day
        rightNow.setTimeInMillis(endTimestampMs);
        // Set up the Hours
        rightNow.set(java.util.Calendar.HOUR_OF_DAY, 23);
        // Set up the Minutes
        rightNow.set(java.util.Calendar.MINUTE, 59);
        // Set up the Seconds
        rightNow.set(java.util.Calendar.SECOND, 59);
        // Set up the Milliseconds
        rightNow.set(java.util.Calendar.MILLISECOND, 999);
        // Get the time in milliseconds
        long rangeEnd = rightNow.getTimeInMillis();

        // Set up the date filter UI
        tempIsDateFilterActive = true;
        // Set up the filter button
        android.widget.ImageButton btn = findViewById(R.id.btnFilterDate);
        // Set up the filter button
        btn.setColorFilter(android.graphics.Color.parseColor("#FF9800"));
```

```
        // Filter the adapter
        List<Post> filteredList = new ArrayList<>();
        // Set up the adapter
        for (Post tempPost : tempPostList) {
            // Filter the adapter
            if (tempPost.getTimestamp() >= rangeStart && tempPost.getTimestamp() <=
rangeEnd) {
                // Add the post to the filtered list
                filteredList.add(tempPost);
            }
        }

        // Set up the adapter
        tempAdapter.updateList(filteredList);

        // Check if there are any posts
        if (filteredList.isEmpty()) {
            // Set up the adapter
            Toast.makeText(this, "No posts found for this range",
Toast.LENGTH_SHORT).show();
        }
    }

    // Set up the user profile dialog
    private void showUserProfileDialog() {
        // Get the current user
        com.google.firebase.auth.FirebaseUser user =
FirebaseAuth.getInstance().getCurrentUser();
        if (user == null) return;

        // Set up the dialog
        View dialogView = getLayoutInflater().inflate(R.layout.dialog_user_profile,
null);
        // Set up the dialog
        AlertDialog dialog = new MaterialAlertDialogBuilder(this)
                .setView(dialogView)
                .create();

        // Set up the dialog
        if (dialog.getWindow() != null) {
            // Set up the dialog window

dialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);
        }

        // Set up the tempTxtName
        TextView tempTxtName = dialogView.findViewById(R.id.profileName);
        // Set up the tempTxtEmail
        TextView tempTxtEmail = dialogView.findViewById(R.id.profileEmail);
        // Set up the tempBtnLogout
        Button tempBtnLogout = dialogView.findViewById(R.id.btnLogout);
        // Set up the tempBtnRefreshWidget
        Button tempBtnRefreshWidget = dialogView.findViewById(R.id.btnRefreshWidget);
        Button btnShowFCMToken = dialogView.findViewById(R.id.btnShowFCMToken);

        // Show FCM Token logic
        btnShowFCMToken.setOnClickListener(v -> {
            FirebaseMessaging.getInstance().getToken()
                .addOnCompleteListener(task -> {
                    if (!task.isSuccessful()) {
                        Toast.makeText(this, "Fetching FCM token failed",
Toast.LENGTH_SHORT).show();
                        return;
```

```
                    }

                    String token = task.getResult();
                    new MaterialAlertDialogBuilder(this)
                        .setTitle("FCM Registration Token")
                        .setMessage(token)
                        .setPositiveButton("Copy", (d, w) -> {
                            android.content.ClipboardManager clipboard =
(android.content.ClipboardManager)
getSystemService(android.content.Context.CLIPBOARD_SERVICE);
                            android.content.ClipData clip =
android.content.ClipData.newPlainText("FCM Token", token);
                            clipboard.setPrimaryClip(clip);
                            Toast.makeText(this, "Token copied!",
Toast.LENGTH_SHORT).show();
                        })
                        .setNegativeButton("Close", null)
                        .show();
                });
        });

        // Set up the refresh widget button
        tempBtnRefreshWidget.setOnClickListener(v -> {
            // Trigger the widget update
            triggerWidgetUpdate();
            Toast.makeText(this, "Widget update requested", Toast.LENGTH_SHORT).show();
            // Dismiss the dialog
            dialog.dismiss();
        });

        // Get the user's name and email
        FirebaseFirestore.getInstance().collection("users").document(user.getUid()).get()
                .addOnSuccessListener(doc -> {
                    // Check if the document exists
                    if (doc.exists()) {
                        String name = doc.getString("name");
                        String email = doc.getString("email");
                        // Check if the name exists
                        if (name != null) {
                            // Set up the user's name
                            tempCurrentUserName = name;
                            tempTxtName.setText(name);
                        }
                        // Check if the email exists
                        if (email != null) {
                            // Set up the user's email
                            tempCurrentUserEmail = email;
                            tempTxtEmail.setText(email);
                        }
                    }
                });

        // Set up the user's name and email
        tempTxtName.setText(tempCurrentUserName != null ? tempCurrentUserName : "Email
ID");
        // Set up the user's email
        tempTxtEmail.setText(tempCurrentUserEmail != null ? tempCurrentUserEmail :
user.getEmail());

        // Set up the logout button
        tempBtnLogout.setOnClickListener(v -> {
            // Logout the user
            FirebaseAuth.getInstance().signOut();
```

```
            // Set up the intent
            Intent intent = new Intent(this, LoginActivity.class);
            // Set up the intent
            intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
            // Set up the intent
            startActivity(intent);
            // Finish the activity
            finish();
        });

        // Show the dialog
        dialog.show();
    }

    // Toggle the bookmarks view
    private void toggleBookmarksView() {
        // Toggle the bookmarks view
        tempShowingBookmarks = !tempShowingBookmarks;
        // Set up the filter button
        android.widget.ImageButton btn = findViewById(R.id.btnShowBookmarks);
        // Set up the adapter
        tempAdapter.setTempShowingBookmarksMode(tempShowingBookmarks);

        if (tempIsDateFilterActive) {
            // Clear the date filter
            resetDateFilterUI();
        }

        if (tempShowingBookmarks) {
            // Set up the filter button
            btn.setColorFilter(android.graphics.Color.parseColor("#FF9800"));
            // Load the bookmarked posts
            loadBookmarkedPosts();
        } else {
            // Set up the filter button
            btn.setColorFilter(null);
            // Load the user's feed
            loadUserFeed();
        }
    }

    // Load the user's bookmarked posts
    private void loadBookmarkedPosts() {
        // Get the current user
        com.google.firebase.auth.FirebaseUser currentUser =
FirebaseAuth.getInstance().getCurrentUser();
        // Check if the user is logged in
        if (currentUser == null) return;
        // Get the user's ID
        String tempUID = currentUser.getUid();
        // Clear the tempPostList
        tempPostList.clear();
        // Set up the adapter
        tempAdapter.notifyDataSetChanged();

        // Check if the user has any bookmarked posts
        if (tempUserCommunityIds == null || tempUserCommunityIds.isEmpty()) {
            Toast.makeText(this, "No communities found to search bookmarks",
Toast.LENGTH_SHORT).show();
            return;
        }
```

```
        // Set up the tempUserCommunityIds.size();
        tempPendingRequests = tempUserCommunityIds.size();
        // Traverse the firebaseDB
        for (String tempCommunityId : tempUserCommunityIds) {
            // firebaseDB

tempFirebaseDB.collection("communities").document(tempCommunityId).get().addOnSuccessList
ener(commDoc -> {
                String communityName = commDoc.getString("name");
                // firebaseDB

tempFirebaseDB.collection("communities").document(tempCommunityId).collection("posts").ge
t().addOnSuccessListener(querySnapshot -> {
                    // Traverse the postDoc
                    for (DocumentSnapshot postDoc : querySnapshot.getDocuments()) {
                        // firebaseDB

postDoc.getReference().collection("bookmarks").document(tempUID).get().addOnSuccessListen
er(bookmarkDoc -> {
                            // Check if the post is bookmarked
                            if (bookmarkDoc.exists()) {
                                // Set up the post
                                Post tempPost = postDoc.toObject(Post.class);
                                // Check if the post is not null
                                if (tempPost != null) {
                                    // Set up the post ID
                                    tempPost.setPostId(postDoc.getId());
                                    // Set up the community ID
                                    tempPost.setCommunityId(tempCommunityId);
                                    // Set up the community name
                                    tempPost.setAuthorName(communityName);
                                    // Set up the bookmarked
                                    tempPost.setBookmarked(true);
                                    // Add the post to the tempPostList
                                    tempPostList.add(tempPost);
                                }
                            }
                            // Check if all bookmarks are finished
                            checkAllBookmarksFinished();
                        });
                    }
                    // Check if all bookmarks are finished
                    if (querySnapshot.isEmpty()) {
                        // Check if all bookmarks are finished
                        checkAllBookmarksFinished();
                    }
                });
            });
        }
    }

    // Check if all bookmarks are finished
    private void checkAllBookmarksFinished() {
        // Sort the tempPostList
        tempPostList.sort((p1, p2) -> Long.compare(p2.getTimestamp(),
p1.getTimestamp()));
        // Set up the adapter
        tempAdapter.updateList(tempPostList);
    }

    private void showCreatePostDialog() {
        // Set up the dialog
        MaterialAlertDialogBuilder tempBuilder = new MaterialAlertDialogBuilder(this);
```

```
        // Set up the inflater
        LayoutInflater inflater = getLayoutInflater();
        // Set up the view
        View dialogView = inflater.inflate(R.layout.dialog_create_post, null);
        // Set up the builder
        tempBuilder.setView(dialogView);
        // Set up the dialog
        AlertDialog dialog = tempBuilder.create();
        // Set up the dialog
        if (dialog.getWindow() != null) {
            // Set up the dialog window

dialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);
            // Set up the dialog's window
            dialog.getWindow().setDimAmount(0.7f);
        }

        // Set up the editTitle
        EditText editTitle = dialogView.findViewById(R.id.editPostTitle);
        // Set up the editContent
        EditText editContent = dialogView.findViewById(R.id.editPostContent);
        // Set up the communityAutoComplete
        AutoCompleteTextView communityAutoComplete =
dialogView.findViewById(R.id.communityAutoComplete);
        // Set up the btnPost
        Button btnPost = dialogView.findViewById(R.id.btnPost);
        // Set up the btnCancel
        Button btnCancel = dialogView.findViewById(R.id.btnCancel);

        // Get the current user
        com.google.firebase.auth.FirebaseUser currentUser =
FirebaseAuth.getInstance().getCurrentUser();
        // Check if the user is logged in
        if (currentUser == null) {
            return;
        }
        // Get the user's ID
        String tempUID = currentUser.getUid();
        // firebaseDB

tempFirebaseDB.collection("users").document(tempUID).get().addOnSuccessListener(documentS
napshot -> {
            @SuppressWarnings("unchecked")
            List<String> communityIds = (List<String>)
documentSnapshot.get("selectedCommunities");
            // Check if the user has any communities
            if (communityIds != null && !communityIds.isEmpty()) {
                // Set up the communityNames
                List<String> tempCommunityNames = new ArrayList<>();
                // Set up the nameToIdMap
                Map<String, String> nameToIdMap = new HashMap<>();

                // Traverse the communityIds
                for (String tempIdx : communityIds) {
                    // firebaseDB

tempFirebaseDB.collection("communities").document(tempIdx).get().addOnSuccessListener(com
mDoc -> {
                        // Set up the tempName
                        String tempName = commDoc.getString("name");
                        // Check if the tempName is not null
                        if (tempName != null) {
                            // Add the name to the tempCommunityNames
```

```
                            tempCommunityNames.add(tempName);
                            // Add the tempIdx to the nameToIdMap
                            nameToIdMap.put(tempName, tempIdx);
                        }

                        // Check if all communities are finished
                        if (tempCommunityNames.size() == communityIds.size()) {
                            // Set up the adapter
                            ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
                                    android.R.layout.simple_list_item_1,
tempCommunityNames);
                            // Set up the adapter
                            communityAutoComplete.setAdapter(adapter);
                        }
                    });
                }

                // Set up the btnPost
                btnPost.setOnClickListener(v -> {
                    // Get the title
                    String title = editTitle.getText().toString().trim();
                    // Get the content
                    String content = editContent.getText().toString().trim();
                    // Get the selected community.
                    String selectedCommName = communityAutoComplete.getText().toString();

                    // Check if the title and content are empty
                    if (title.isEmpty() || content.isEmpty() ||
selectedCommName.isEmpty() || selectedCommName.equals("Select Community")) {
                        Toast.makeText(this, "Please fill all fields",
Toast.LENGTH_SHORT).show();
                        return;
                    }

                    String commId = nameToIdMap.get(selectedCommName);
                    // Set up the btnPost
                    savePostToFirestore(title, content, commId, dialog);
                });
            }
        });

        // Set up the btnCancel
        btnCancel.setOnClickListener(v -> dialog.dismiss());
        // Show the dialog
        dialog.show();
    }

    // Show the create options dialog
    private void showCreateOptions() {
        // Set up the dialog
        View dialogView = getLayoutInflater().inflate(R.layout.dialog_select_option,
null);
        // Set up the dialog
        AlertDialog dialog = new AlertDialog.Builder(this).setView(dialogView).create();

        // Set up the dialog
        if (dialog.getWindow() != null) {
            // Set up the dialog window

dialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);

            // Set up the dialog's window
            android.view.WindowManager.LayoutParams params =
```

```
dialog.getWindow().getAttributes();
            // Set up the dialog's window
            params.gravity = android.view.Gravity.BOTTOM | android.view.Gravity.END;

            // Set up the dialog's window
            params.x = 60;
            params.y = 260;

            // Set up the dialog's window
            dialog.getWindow().setAttributes(params);
            // Set up the dialog's window
            dialog.getWindow().setDimAmount(0.2f);
        }

        // Set up the findViewById
        dialogView.findViewById(R.id.optionCreatePost).setOnClickListener(v -> {
            // Dismiss the dialog
            dialog.dismiss();
            // Show the create post dialog
            showCreatePostDialog();
        });

        // Set up the findViewById
        dialogView.findViewById(R.id.optionCreateCommunity).setOnClickListener(v -> {
            // Dismiss the dialog
            dialog.dismiss();
            // Show the create community dialog
            showCreateCommunityDialog();
        });

        dialog.show();
    }

    // Show the create community dialog
    private void showCreateCommunityDialog() {
        // Set up the dialog
        MaterialAlertDialogBuilder tempBuilder = new MaterialAlertDialogBuilder(this);
        // Set up the inflater
        LayoutInflater inflater = getLayoutInflater();
        // Set up the view
        View dialogView = inflater.inflate(R.layout.dialog_create_community, null);
        // Set up the builder
        tempBuilder.setView(dialogView);

        AlertDialog dialog = tempBuilder.create();
        // Set up the dialog
        if (dialog.getWindow() != null) {
            // Set up the dialog window

dialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);
            // Set up the dialog's window
            dialog.getWindow().setDimAmount(0.7f);
        }

        // Set up the editName
        EditText editName = dialogView.findViewById(R.id.editCommunityName);
        // Set up the editSummary
        EditText editSummary = dialogView.findViewById(R.id.editCommunitySummary);
        // Set up the btnCreate
        Button btnCreate = dialogView.findViewById(R.id.btnCreateComm);
        // Set up the btnCancel
        Button btnCancel = dialogView.findViewById(R.id.btnCancelComm);
```

```
        // Set up the btnCreate
        btnCreate.setOnClickListener(v -> {
            // Get the name
            String name = editName.getText().toString().trim();
            // Get the summary
            String summary = editSummary.getText().toString().trim();

            // Check if the name and summary are empty
            if (name.isEmpty() || summary.isEmpty()) {
                Toast.makeText(this, "Please fill all fields",
Toast.LENGTH_SHORT).show();
                return;
            }

            // Check if the name already exists
            tempFirebaseDB.collection("communities")
                    .whereEqualTo("name", name)
                    .get()
                    .addOnSuccessListener(queryDocumentSnapshots -> {
                        if (!queryDocumentSnapshots.isEmpty()) {
                            // Suggest an suggestAlternativeName
                            suggestAlternativeName(name, summary, dialog);
                        } else {
                            // Save the saveCommunityToFirestore
                            saveCommunityToFirestore(name, summary, dialog);
                        }
                    });
        });

        // Set up the btnCancel
        btnCancel.setOnClickListener(v -> dialog.dismiss());
        // Show the dialog
        dialog.show();
    }

    // Suggest an alternative name
    private void suggestAlternativeName(String originalName, String summary, AlertDialog
dialog) {
        // Set up the suggestedName
        findAvailableName(originalName, 1, summary, dialog);
    }

    // Find an available name
    private void findAvailableName(String baseName, int suffix, String summary,
AlertDialog dialog) {
        // Set up the suggestedName
        String suggestedName = baseName + " " + suffix;
        // firebaseDB
        tempFirebaseDB.collection("communities")
                .whereEqualTo("name", suggestedName)
                .get()
                .addOnSuccessListener(queryDocumentSnapshots -> {
                    if (queryDocumentSnapshots.isEmpty()) {
                        new MaterialAlertDialogBuilder(this)
                                .setTitle("Name already exists")
                                .setMessage("A community with that name already exists.
How about '" + suggestedName + "' instead?")
                                .setPositiveButton("Use Suggested", (d, w) -> {
                                    saveCommunityToFirestore(suggestedName, summary,
dialog);
                                })
                                .setNegativeButton("Change Manually", null)
                                .show();
```

```
                    } else {
                        // Suggest an alternative name
                        findAvailableName(baseName, suffix + 1, summary, dialog);
                    }
                });
    }

    // Save the community to Firestore
    private void saveCommunityToFirestore(String name, String summary, AlertDialog
dialog) {
        Map<String, Object> commMap = new HashMap<>();
        commMap.put("name", name);
        commMap.put("summary", summary);
        commMap.put("timestamp", System.currentTimeMillis());
        commMap.put("icon", "ic_launcher_foreground");
        // firebaseDB
        tempFirebaseDB.collection("communities")
                .add(commMap)
                .addOnSuccessListener(documentReference -> {
                    Toast.makeText(this, "Community created!",
Toast.LENGTH_SHORT).show();
                    dialog.dismiss();
                })
                // Handle error
                .addOnFailureListener(e -> {
                    Toast.makeText(this, "Failed to create community",
Toast.LENGTH_SHORT).show();
                });
    }

    // Save the post to Firestore
    private void savePostToFirestore(String title, String content, String communityId,
AlertDialog dialog) {
        // Get the current user
        com.google.firebase.auth.FirebaseUser currentUser =
FirebaseAuth.getInstance().getCurrentUser();
        // Check if the user is logged in
        if (currentUser == null) return;
        // Get the user's ID
        String uid = currentUser.getUid();
        // firebaseDB
        Map<String, Object> postMap = new HashMap<>();
        postMap.put("title", title);
        postMap.put("content", content);
        postMap.put("communityId", communityId);
        postMap.put("authorId", uid);
        postMap.put("authorName", tempCurrentUserName);
        postMap.put("authorEmail", currentUser.getEmail());
        postMap.put("timestamp", System.currentTimeMillis());
        postMap.put("upvotes", 0);
        postMap.put("downvotes", 0);
        postMap.put("commentCount", 0);
        postMap.put("bookmarkCount", 0);

        // firebaseDB
        tempFirebaseDB.collection("communities")
                .document(communityId)
                .collection("posts")
                .add(postMap)
                .addOnSuccessListener(documentReference -> {
                    Toast.makeText(this, "Post created!", Toast.LENGTH_SHORT).show();
                    dialog.dismiss();
                    triggerWidgetUpdate(); // Trigger widget refresh on the device
```

```
                    loadUserFeed();
                })
                // Handle error
                .addOnFailureListener(e -> {
                    Toast.makeText(this, "Failed to create post",
Toast.LENGTH_SHORT).show();
                });
    }

    // Load the user's feed
    private void loadUserFeed() {
        FirebaseAuth auth = FirebaseAuth.getInstance();
        // Check if the user is logged in
        if (auth.getCurrentUser() == null) return;
        // Get the user's ID
        String uid = auth.getCurrentUser().getUid();

        // firebaseDB
        tempFirebaseDB.collection("users")
                .document(uid)
                .get()
                .addOnSuccessListener(documentSnapshot -> {
                    // Set up the tempCurrentUserName
                    tempCurrentUserName = documentSnapshot.getString("name");
                    // Set up the tempCurrentUserEmail
                    tempCurrentUserEmail = documentSnapshot.getString("email");
                    // Set up the adapter
                    tempAdapter.setCurrentUser(tempCurrentUserName,
tempCurrentUserEmail);

                    // Check if the user has any communities
                    @SuppressWarnings("unchecked")
                    List<String> selected = (List<String>)
documentSnapshot.get("selectedCommunities");

                    // Check if the user has any communities
                    if (selected == null || selected.isEmpty()) {
                        return;
                    }

                    // Set up the tempUserCommunityIds
                    tempUserCommunityIds = selected;
                    // Load the user's feed
                    fetchPosts(selected);
                });
    }

    // Fetch the user's feed
    private void fetchPosts(List<String> communities) {
        // Clear the tempPostList
        tempPostList.clear();
        // Set up the pendingRequests
        tempPendingRequests = communities.size();

        // Check if there are any posts
        if (tempPendingRequests == 0) {
            // Set up the adapter
            tempAdapter.notifyDataSetChanged();
            return;
        }

        // firebaseDB
        String uid = FirebaseAuth.getInstance().getUid();
```

```java
        // Traverse the communities
        for (String communityId : communities) {
            // firebaseDB
            tempFirebaseDB.collection("communities").document(communityId).get()
                    .addOnSuccessListener(communityDoc -> {
                        // Get the community name
                        String communityName = communityDoc.getString("name");

                        tempFirebaseDB.collection("communities")
                                .document(communityId)
                                .collection("posts")
                                .orderBy("timestamp", Query.Direction.DESCENDING)
                                .limit(5)
                                .addSnapshotListener((queryDocumentSnapshots, e) -> {
                                    if (e != null || queryDocumentSnapshots == null) {
                                        // Handle error
                                        onRequestFinished();
                                        return;
                                    }

                                    // Traverse the dc
                                    for (DocumentChange dc :
queryDocumentSnapshots.getDocumentChanges()) {
                                        // Check if the doc is not null
                                        DocumentSnapshot doc = dc.getDocument();
                                        // Check if the doc is not null
                                        Post post = doc.toObject(Post.class);
                                        // Check if the post is not null
                                        if (post == null) continue;

                                        // Set up the post
                                        post.setPostId(doc.getId());
                                        // Set up the community ID
                                        post.setCommunityId(communityId);
                                        // Set up the community name
                                        post.setAuthorName(communityName);

                                        switch (dc.getType()) {
                                            // ADDED
                                            case ADDED:
                                                updateOrAddPost(post, uid);
                                                break;
                                            // MODIFIED
                                            case MODIFIED:
                                                updateOrAddPost(post, uid);
                                                break;
                                            // REMOVED
                                            case REMOVED:
                                                removePost(post.getPostId());
                                                break;
                                        }
                                    }
                                    // Check if all posts are finished
                                    onRequestFinished();
                                });
                    });
        }
    }

    // Update or add the post
    private void updateOrAddPost(Post post, String uid) {
        // Fetch extra info (bookmarks/comments/votes) then update UI
```

```
        post.getReference(tempFirebaseDB).collection("bookmarks").document(uid).get()
            .addOnSuccessListener(bookmarkDoc -> {
                // Check if the post is bookmarked
                post.setBookmarked(bookmarkDoc.exists());

                // Fetch user's vote for this post

tempFirebaseDB.collection("users").document(uid).collection("votes").document(post.getPos
tId()).get()
                    .addOnSuccessListener(voteDoc -> {
                        // Check if the voteDoc exists
                        if (voteDoc.exists()) {
                            // Set up the post
                            String type = voteDoc.getString("type");
                            // Set up the post
                            post.setUpvoted("up".equals(type));
                            // Set down the post
                            post.setDownvoted("down".equals(type));
                        }


post.getReference(tempFirebaseDB).collection("comments").count().get(AggregateSource.SERV
ER)
                            .addOnSuccessListener(aggregateQuerySnapshot -> {
                                post.setCommentCount((int)
aggregateQuerySnapshot.getCount());

                                synchronized (tempPostList) {
                                    // Set up the tempPostList
                                    int index = -1;
                                    // Traverse the tempPostList
                                    for (int i = 0; i < tempPostList.size(); i++) {
                                        // Check if the post is already in the
tempPostList
                                        if
(tempPostList.get(i).getPostId().equals(post.getPostId())) {
                                            // Set up the index
                                            index = i;
                                            // Break out of the loop
                                            break;
                                        }
                                    }

                                    // Check if the post is already in the tempPostList
                                    if (tempShowingBookmarks && !post.isBookmarked()) {
                                        // Remove the post from the tempPostList
                                        if (index != -1) {
                                            // tempPostList
                                            tempPostList.remove(index);
                                        }
                                        // Update the UI
                                    } else {
                                        // Update the post
                                        if (index != -1) {
                                            // Set up the post
                                            tempPostList.set(index, post);
                                        } else {
                                            // Add the post to the tempPostList
                                            if (!tempShowingBookmarks ||
post.isBookmarked()) {
                                                // Add the post to the tempPostList
                                                tempPostList.add(post);
                                            }
```

```
                                        }
                                    }
                                    tempPostList.sort((p1, p2) ->
Long.compare(p2.getTimestamp(), p1.getTimestamp()));
                                }
                                runOnUiThread(() -> tempAdapter.updateList(new
ArrayList<>(tempPostList)));
                            });
                    });
            });
    }

    // removePost
    private void removePost(String postId) {
        // Remove the post from the tempPostList
        synchronized (tempPostList) {
            // Remove the post from the tempPostList
            tempPostList.removeIf(p -> p.getPostId().equals(postId));
        }
        // Update the UI
        runOnUiThread(() -> tempAdapter.updateList(new ArrayList<>(tempPostList)));
    }

    // On request finished
    private void onRequestFinished() {
        // Decrement the pendingRequests
        tempPendingRequests--;
        // Check if all posts are finished
        if (tempPendingRequests <= 0) {
            // Sort the tempPostList
            tempPostList.sort((p1, p2) -> Long.compare(p2.getTimestamp(),
p1.getTimestamp()));
            // Update the UI
            tempAdapter.updateList(tempPostList);
        }
    }

    // Trigger the widget update
    private void triggerWidgetUpdate() {
        // Create a new one-time work request
        androidx.work.OneTimeWorkRequest updateRequest =
                // Set up the work request
                new androidx.work.OneTimeWorkRequest.Builder(WidgetUpdateWorker.class)
                        .build();

        // Schedule the update request
        androidx.work.WorkManager.getInstance(this).enqueueUniqueWork(
                // Set up the work name
                "widget_update_manual",
                // Replace the existing work
                androidx.work.ExistingWorkPolicy.REPLACE,
                // Set up the updateRequest
                updateRequest
        );
    }

    // Schedule the widget update
    private void scheduleWidgetUpdates() {
        // Create a new one-time work request
        Intent intent = new Intent(this, PostWidget.class);
        // Set the action
        intent.setAction("com.example.nexus.UPDATE_WIDGET_MANUAL");
        // Send the broadcast
```

```
        sendBroadcast(intent);
        // Log the widget update
        Log.d("DashboardActivity", "Widget update kickstarter");
    }
}
```

1.8. FeedAdapter.java

```
// Name: Harsh Patel (A20369913)

package com.example.nexus;

import android.annotation.SuppressLint;
import android.content.Context;
import android.content.Intent;
import android.net.Uri;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.core.content.ContextCompat;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.button.MaterialButton;
import com.google.android.material.dialog.MaterialAlertDialogBuilder;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.FirebaseFirestore;

import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;

public class FeedAdapter extends RecyclerView.Adapter<FeedAdapter.ViewHolder> {

    Context tempContext;
    List<Post> tempList;
    List<Post> tempListFull;
    FirebaseFirestore firebaseDB;
    private String tempCurrentUserName, tempCurrentUserEmail;
    private boolean tempShowingBookmarksMode;

    // Constructor FeedAdapter
    public FeedAdapter(Context context, List<Post> list) {
        this.tempContext = context;
        this.tempList = list;
        this.tempListFull = new ArrayList<>(list);
        this.firebaseDB = FirebaseFirestore.getInstance();
    }

    // ViewHolder
    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        // Inflate the layout for each item in the RecyclerView
        View tempView = LayoutInflater.from(tempContext).inflate(R.layout.item_post,
parent, false);
        // Create a new ViewHolder with the inflated view
        return new ViewHolder(tempView);
    }
```

```
    // Bind data to the ViewHolder
    @SuppressLint("SetTextI18n")
    @Override
    public void onBindViewHolder(@NonNull ViewHolder holder, int position) {
        // Get the post at the current position
        Post post = tempList.get(position);
        // Set the text for each view in the ViewHolder
        holder.title.setText(post.getTitle());
        // Set the text for each view in the ViewHolder
        holder.content.setText(post.getContent());
        // Set the text for each view in the ViewHolder
        holder.communityTag.setText(post.getAuthorName() != null ? post.getAuthorName() :
"nexus");
        // Set the text for each view in the ViewHolder
        holder.upvoteBtn.setText(String.valueOf(post.getUpvotes()));
        // Set the text for each view in the ViewHolder
        holder.downvoteBtn.setText(String.valueOf(post.getDownvotes()));
        // Set the text for each view in the ViewHolder
        holder.commentCount.setText(String.valueOf(post.getCommentCount()));
        // Set the text for each view in the ViewHolder
        holder.bookmarkBtn.setText(String.valueOf(post.getBookmarkCount()));

        // Set the timestamp
        if (post.getTimestamp() > 0) {
            // Set the timestamp
            java.text.SimpleDateFormat sdf = new java.text.SimpleDateFormat("MMM dd,
yyyy", java.util.Locale.US);
            // Set the timestamp
            holder.timestamp.setText(sdf.format(new
java.util.Date(post.getTimestamp())));
        }

        // UI State for UP Votes
        if (post.isUpvoted()) {
            // Set the icon and text color for the upvote button
            holder.upvoteBtn.setIconTintResource(R.color.upvote_green);
            // Set the icon and text color for the upvote button
            holder.upvoteBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.upvote_green));
        } else {
            // Set the icon and text color for the upvote button
            holder.upvoteBtn.setIconTintResource(R.color.subtitleText);
            // Set the icon and text color for the upvote button
            holder.upvoteBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.subtitleText));
        }

        // UI State for down Votes
        if (post.isDownvoted()) {
            // Set the icon and text icon for the downvote button
            holder.downvoteBtn.setIconTintResource(R.color.downvote_red);
            // Set the icon and text color for the downvote button
            holder.downvoteBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.downvote_red));
        } else {
            // Set the icon and text icon for the downvote button
            holder.downvoteBtn.setIconTintResource(R.color.subtitleText);
            // Set the icon and text color for the downvote button
            holder.downvoteBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.subtitleText));
        }
```

```
        // UI State for Bookmarks
        if (post.isBookmarked()) {
            // Set the icon and text icon for the bookmark button
            holder.bookmarkBtn.setIconResource(R.drawable.ic_bookmark);
            // Set the icon and text icon tint for the bookmark button
            holder.bookmarkBtn.setIconTintResource(R.color.orange);
            // Set the icon and text color for the bookmark button
            holder.bookmarkBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.orange));
        } else {
            // Set the icon and text icon for the bookmark button
            holder.bookmarkBtn.setIconResource(R.drawable.ic_bookmark_border);
            // Set the icon and text icon tint for the bookmark button
            holder.bookmarkBtn.setIconTintResource(R.color.subtitleText);
            // Set the icon and text color for the bookmark button
            holder.bookmarkBtn.setTextColor(ContextCompat.getColor(tempContext,
R.color.subtitleText));
        }

        // UPVOTE
        holder.upvoteBtn.setOnClickListener(v -> handleVote(post, true));

        // DOWNVOTE
        holder.downvoteBtn.setOnClickListener(v -> handleVote(post, false));

        // BOOKMARK
        holder.bookmarkBtn.setOnClickListener(v -> handleBookmark(post));

        // SHARE
        holder.shareBtn.setOnClickListener(v -> {
            // showShareOptions
            showShareOptions(post);
        });

        // COMMENT / DETAIL
        View.OnClickListener openDetail = v -> {
            // Create an intent to open the PostDetailActivity
            Intent intent = new Intent(tempContext, PostDetailActivity.class);
            // Pass the post data to the PostDetailActivity
            intent.putExtra("POST_DATA", post);
            // Start the PostDetailActivity
            tempContext.startActivity(intent);
        };
        // Set the click listener for the comment count
        holder.commentCount.setOnClickListener(openDetail);
        // Set the click listener for the community tag
        holder.itemView.setOnClickListener(openDetail);
    }

    // Handle Upvote/Downvote
    private void handleVote(Post post, boolean isUpvote) {
        // Get the user ID
        String uid = FirebaseAuth.getInstance().getUid();
        // If no user, return
        if (uid == null) return;

        // Update the upVote count
        if (isUpvote) {
            // Update the upVote count
            if (post.isUpvoted()) {
                post.setUpvotes(Math.max(0, post.getUpvotes() - 1));
                post.setUpvoted(false);
            } else {
```

```
                // Update the upVote count
                post.setUpvotes(post.getUpvotes() + 1);
                post.setUpvoted(true);
                if (post.isDownvoted()) {
                    post.setDownvotes(Math.max(0, post.getDownvotes() - 1));
                    post.setDownvoted(false);
                }
            }
        } else {
            // Update the downVote count
            if (post.isDownvoted()) {
                post.setDownvotes(Math.max(0, post.getDownvotes() - 1));
                post.setDownvoted(false);
            } else {
                // Update the downVote count
                post.setDownvotes(post.getDownvotes() + 1);
                post.setDownvoted(true);
                if (post.isUpvoted()) {
                    post.setUpvotes(Math.max(0, post.getUpvotes() - 1));
                    post.setUpvoted(false);
                }
            }
        }

        // Update the UI
        notifyDataSetChanged();

        // Update Firestore
        firebaseDB.collection("communities").document(post.getCommunityId())
                // Update the upVote count
                .collection("posts").document(post.getPostId())
                // Update the upVote count
                .update(
                        "upvotes", post.getUpvotes(),
                        "downvotes", post.getDownvotes()
                );

        // Update Firestore
        Map<String, Object> voteData = new HashMap<>();
        // Update Firestore
        voteData.put("type", isUpvote ? (post.isUpvoted() ? "up" : "none") :
(post.isDownvoted() ? "down" : "none"));

        // Update Firestore
        firebaseDB.collection("users").document(uid)
                // Update Firestore
                .collection("votes").document(post.getPostId())
                .set(voteData);
    }

    // Handle Bookmark
    private void handleBookmark(Post tempPost) {
        // Get the user ID
        String tempUID = FirebaseAuth.getInstance().getUid();
        // If no user, return
        if (tempUID == null) return;

        // Update the bookmark count
        if (tempPost.isBookmarked()) {
            // Update the bookmark count
            tempPost.setBookmarkCount(Math.max(0, tempPost.getBookmarkCount() - 1));
            // Update the bookmark count
            tempPost.setBookmarked(false);
```

```
        // If we are in Bookmarks Mode, remove the item from the list immediately
        if (tempShowingBookmarksMode) {
            // Remove the item from the list
            int position = tempList.indexOf(tempPost);
            // Remove the item from the list
            if (position != -1) {
                // Remove the item from the list
                tempList.remove(position);
                // Notify the adapter that the item has been removed
                notifyItemRemoved(position);

                // Also remove from listFull to ensure it doesn't reappear
                tempListFull.remove(tempPost);
            }
        } else {
            // Update the UI
            notifyDataSetChanged();
        }

        // Update Firestore
        firebaseDB.collection("communities").document(tempPost.getCommunityId())
                // Update the post count
                .collection("posts").document(tempPost.getPostId())
                // Update the bookmark count
                .collection("bookmarks").document(tempUID).delete();
    } else {
        // Update the bookmark count
        tempPost.setBookmarkCount(tempPost.getBookmarkCount() + 1);
        // Update the bookmark count
        tempPost.setBookmarked(true);
        // Update Firestore
        Map<String, Object> bookmark = new HashMap<>();
        // Update Firestore
        bookmark.put("timestamp", System.currentTimeMillis());
        // Update Firestore
        firebaseDB.collection("communities").document(tempPost.getCommunityId())
                // Update the post count
                .collection("posts").document(tempPost.getPostId())
                // Update the bookmark count
                .collection("bookmarks").document(tempUID).set(bookmark);
        // Update the UI
        notifyDataSetChanged();
    }

    // Update Firestore
    firebaseDB.collection("communities").document(tempPost.getCommunityId())
            // Update the post count
            .collection("posts").document(tempPost.getPostId())
            // Update the bookmark count
            .update("bookmarkCount", tempPost.getBookmarkCount());
}

// Get the number of items in the list
@Override
public int getItemCount() {
    return tempList.size();
}

// show share options
private void showShareOptions(Post post) {
    // Create a share link
    String baseUrl = "http://localhost:8080/post/" + post.getPostId();
```

```
        // queryParams link
        String queryParams = "?title=" + Uri.encode(post.getTitle()) +
                "&content=" + Uri.encode(post.getContent()) +
                "&author=" + Uri.encode(post.getAuthorName() != null ?
post.getAuthorName() : "nexus") +
                "&upvotes=" + post.getUpvotes() +
                "&comments=" + post.getCommentCount();
        // postLink
        String postLink = baseUrl + queryParams;

        // shareText
        String shareText = post.getTitle() + "\n\n" + post.getContent() + "\n\nRead more
at: " + postLink;
        // available options to share.
        String[] options = {"WhatsApp", "Messages", "Copy Link", "Open in Browser",
"Other Apps"};

        // new MaterialAlertDialogBuilder
        new MaterialAlertDialogBuilder(tempContext)
                .setTitle("Share Post")
                .setItems(options, (dialog, which) -> {
                    switch (which) {
                        case 0: // WhatsApp
                            shareToWhatsApp(shareText);
                            break;
                        case 1: // Messages
                            shareToMessages(shareText);
                            break;
                        case 2: // Copy Link
                            copyToClipboard(postLink);
                            break;
                        case 3: // Open in Browser
                            openInBrowser(postLink);
                            break;
                        case 4: // Other
                            shareToOther(shareText);
                            break;
                    }
                })
                .show();
    }

    // shareToWhatsApp
    private void shareToWhatsApp(String text) {
        Intent intent = new Intent(Intent.ACTION_SEND);
        intent.setType("text/plain");
        intent.setPackage("com.whatsapp");
        intent.putExtra(Intent.EXTRA_TEXT, text);
        try {
            tempContext.startActivity(intent);
        } catch (android.content.ActivityNotFoundException ex) {
            // Handle the exception
            Toast.makeText(tempContext, "WhatsApp not installed",
Toast.LENGTH_SHORT).show();
            shareToOther(text);
        }
    }

    // shareToMessages
    private void shareToMessages(String text) {
        Intent intent = new Intent(Intent.ACTION_VIEW);
        intent.setData(Uri.parse("sms:"));
        intent.putExtra("sms_body", text);
```

```
        try {
            tempContext.startActivity(intent);
        } catch (Exception e) {
            // Handle the exception
            Toast.makeText(tempContext, "Messaging app not found",
Toast.LENGTH_SHORT).show();
            shareToOther(text);
        }
    }

    // copyToClipboard
    private void copyToClipboard(String text) {
        // Create a new clipboard manager
        android.content.ClipboardManager clipboard = (android.content.ClipboardManager)
tempContext.getSystemService(android.content.Context.CLIPBOARD_SERVICE);
        // Create a new clip
        android.content.ClipData clip = android.content.ClipData.newPlainText("Post
Content", text);
        // Set the clip
        clipboard.setPrimaryClip(clip);
        Toast.makeText(tempContext, "Link copied to clipboard",
Toast.LENGTH_SHORT).show();
    }

    // openInBrowser on chrome
    private void openInBrowser(String url) {
        // Create a new intent to open a URL
        Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
        // Start the intent
        tempContext.startActivity(intent);
    }

    // shareToOther
    private void shareToOther(String text) {
        // Create a new intent to share text
        Intent intent = new Intent(Intent.ACTION_SEND);
        // Set the type of the intent
        intent.setType("text/plain");
        // Set the text to share
        intent.putExtra(Intent.EXTRA_TEXT, text);
        // Start the intent
        tempContext.startActivity(Intent.createChooser(intent, "Share via"));
    }

    // Update the list of posts
    public void updateList(List<Post> newList) {
        // Update the list
        this.tempList = newList;
        // Update the listFull
        this.tempListFull = new ArrayList<>(newList);
        // Notify the adapter
        notifyDataSetChanged();
    }

    // Filter the list of posts
    public void filter(String text) {
        // Clear the list
        tempList.clear();
        // If no text, show all posts
        if (text.isEmpty()) {
            // Add all posts to the list
            tempList.addAll(tempListFull);
        } else {
```

```java
            // Filter the list
            String filterPattern = text.toLowerCase().trim();
            // Loop through the list
            for (Post post : tempListFull) {
                //  if filterPattern
                if (post.getTitle().toLowerCase().contains(filterPattern) ||
                        post.getContent().toLowerCase().contains(filterPattern)) {
                    // add into tempList
                    tempList.add(post);
                }
            }
        }
        // Notify the adapter
        notifyDataSetChanged();
    }

    // Set the current user
    public void setCurrentUser(String name, String email) {
        // Set the current userName
        this.tempCurrentUserName = name;
        // Set the current user email
        this.tempCurrentUserEmail = email;
    }

    // Get the current user
    public void setTempShowingBookmarksMode(boolean mode) {
        this.tempShowingBookmarksMode = mode;
    }

    // ViewHolder
    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView title, content, communityTag, timestamp;
        MaterialButton upvoteBtn, downvoteBtn, commentCount, shareBtn, bookmarkBtn;

        // Constructor ViewHolder
        public ViewHolder(@NonNull View itemView) {
            // Constructor ViewHolder
            super(itemView);
            // Initialize title
            title = itemView.findViewById(R.id.postTitle);
            // Initialize content
            content = itemView.findViewById(R.id.postContent);
            // Initialize communityTag
            communityTag = itemView.findViewById(R.id.postTag);
            // Initialize timestamp
            timestamp = itemView.findViewById(R.id.timeText);
            // Initialize upvoteBtn
            upvoteBtn = itemView.findViewById(R.id.upvoteBtn);
            // Initialize downvoteBtn
            downvoteBtn = itemView.findViewById(R.id.downvoteBtn);
            // Initialize commentCount
            commentCount = itemView.findViewById(R.id.commentBtn);
            // Initialize shareBtn
            shareBtn = itemView.findViewById(R.id.shareBtn);
            // Initialize bookmarkBtn
            bookmarkBtn = itemView.findViewById(R.id.bookmarkBtn);
        }
    }
}
```

1.9. LoginActivity.java

```java
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.content.Intent;
import android.os.Bundle;
import android.content.Context;
import android.net.ConnectivityManager;
import android.net.Network;
import android.net.NetworkCapabilities;
import android.text.TextUtils;
import android.util.Patterns;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.textfield.TextInputEditText;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.FirebaseFirestore;

public class LoginActivity extends AppCompatActivity {

    TextInputEditText tempEmail, tempPassword;
    Button tempLoginBtn;
    TextView tempSignupSwitch;

    FirebaseAuth tempAuth;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        tempEmail =findViewById(R.id.email);
        tempPassword =findViewById(R.id.password);
        tempLoginBtn =findViewById(R.id.loginBtn);
        tempSignupSwitch =findViewById(R.id.signupSwitch);

        tempAuth = FirebaseAuth.getInstance();

        tempLoginBtn.setOnClickListener(v -> {

            String tempUserEmail = tempEmail.getText().toString().trim();
            String tempUserPassword = tempPassword.getText().toString().trim();

            // Email fill up check.
            if (TextUtils.isEmpty(tempUserEmail) ||
!Patterns.EMAIL_ADDRESS.matcher(tempUserEmail).matches()) {
                tempEmail.setError("Valid email is required");
                return;
            }

            // Password fill up check.
            if (TextUtils.isEmpty(tempUserPassword)) {
                tempPassword.setError("Password is required");
                return;
            }

            // Check for internet connection.
            if (!isNetworkAvailable()) {
```

```
                Toast.makeText(this, "No internet connection",
Toast.LENGTH_SHORT).show();
                return;
            }

            tempLoginBtn.setEnabled(false);

            // Sign in with email and password.
            tempAuth.signInWithEmailAndPassword(tempUserEmail,tempUserPassword)
                    .addOnCompleteListener(tempTask -> {
                        tempLoginBtn.setEnabled(true);
                        // if login is successful, the go to CustomizeFeedActivity.class
                        if(tempTask.isSuccessful())
                        {
                            Toast.makeText(this,"Login
Successful",Toast.LENGTH_SHORT).show();

                            String tempUserId = tempAuth.getCurrentUser().getUid();
                            // get all the users' details.

FirebaseFirestore.getInstance().collection("users").document(tempUserId).get()
                                    .addOnSuccessListener(documentSnapshot -> {
                                        Intent intent = new Intent(LoginActivity.this,
CustomizeFeedActivity.class);
                                        intent.putExtra("userName",
documentSnapshot.getString("name"));
                                        intent.putExtra("userEmail",
documentSnapshot.getString("email"));
                                        // start the CustomizeFeedActivity.class
                                        startActivity(intent);
                                        finish();
                                    })
                                    .addOnFailureListener(e -> {
                                        // start the CustomizeFeedActivity.class
                                        startActivity(new Intent(LoginActivity.this,
CustomizeFeedActivity.class));
                                        finish();
                                    });
                        }
                        else
                        {
                            // if login is not successful, then show error message.
                            Toast.makeText(this,"Login
Failed",Toast.LENGTH_SHORT).show();
                        }

                    });

        });

        // Switch to signup activity.
        tempSignupSwitch.setOnClickListener(v -> {
            // start the SignupActivity.class
            startActivity(new Intent(LoginActivity.this, SignupActivity.class));
        });

    }

    // Check for internet connection.
    private boolean isNetworkAvailable() {
        ConnectivityManager tempConnectivityManager = (ConnectivityManager)
getSystemService(Context.CONNECTIVITY_SERVICE);
        if (tempConnectivityManager != null) {
```

```
            // Check for active network.
            Network tempNetwork = tempConnectivityManager.getActiveNetwork();
            if (tempNetwork != null) {
                // Check for network capabilities.
                NetworkCapabilities tempNetworkCapabilities =
tempConnectivityManager.getNetworkCapabilities(tempNetwork);
                // Check for network type.
                return tempNetworkCapabilities != null &&
(tempNetworkCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_WIFI) ||

tempNetworkCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_CELLULAR) ||

tempNetworkCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_ETHERNET));
            }
        }
        return false;
    }
}
```

1.10. ExampleInstrumentedTest.java

```
// Name: Harsh Patel (A20369913

package com.example.tempapp;

import android.content.Context;

import androidx.test.platform.app.InstrumentationRegistry;
import androidx.test.ext.junit.runners.AndroidJUnit4;

import org.junit.Test;
import org.junit.runner.RunWith;

import static org.junit.Assert.*;

/**
 * Instrumented test, which will execute on an Android device.
 *
 * @see <a href="http://d.android.com/tools/testing">Testing documentation</a>
 */
@RunWith(AndroidJUnit4.class)
public class ExampleInstrumentedTest {
    @Test
    public void useAppContext() {
        // Context of the app under test.
        Context appContext =
InstrumentationRegistry.getInstrumentation().getTargetContext();
        assertEquals("com.example.tempapp", appContext.getPackageName());
    }
}
```

1.11. MyFirebaseMessagingService.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.appwidget.AppWidgetManager;
import android.content.ComponentName;
import android.content.Intent;
import android.util.Log;
```

```java
import androidx.annotation.NonNull;

import com.google.firebase.messaging.FirebaseMessagingService;
import com.google.firebase.messaging.RemoteMessage;

// WidgetUpdateWorker
public class MyFirebaseMessagingService extends FirebaseMessagingService {

    // Tag for logging
    private static final String TAG = "NexusFCM";

    @Override
    public void onMessageReceived(@NonNull RemoteMessage remoteMessage) {
        // Handle incoming messages
        super.onMessageReceived(remoteMessage);
        // Log the message
        Log.d(TAG, "From: " + remoteMessage.getFrom());

        // Check if message contains data payload
        if (remoteMessage.getData().size() > 0) {
            Log.d(TAG, "Message data payload: " + remoteMessage.getData());

            // Trigger widget update
            updateWidget();
        }

        // Check if message contains notification payload
        if (remoteMessage.getNotification() != null) {
            // Handle notification payload
            String notificationBody = remoteMessage.getNotification().getBody();
            Log.d(TAG, "Message Notification Body: " + notificationBody);

            // Show toast for foreground presentation
            new android.os.Handler(android.os.Looper.getMainLooper()).post(() -> {
                android.widget.Toast.makeText(getApplicationContext(),
                        "FCM Notification: " + notificationBody,
                        android.widget.Toast.LENGTH_LONG).show();
            });
        }
    }

    private void updateWidget() {
        // Create an intent to update the widget
        androidx.work.OneTimeWorkRequest tempUpdateRequest =
                // Create a new one-time work request
                new androidx.work.OneTimeWorkRequest.Builder(WidgetUpdateWorker.class)
                        // Set the input data
                        .build();

        // Use enqueueUniqueWork to prevent multiple simultaneous updates
        androidx.work.WorkManager.getInstance(this).enqueueUniqueWork(
                // Unique name for the work
                "widget_update",
                // REPLACE means if there's an existing work with the same name, it will
be replaced
                androidx.work.ExistingWorkPolicy.REPLACE,
                // The work request
                tempUpdateRequest
        );
    }

    @Override
```

```
    public void onNewToken(@NonNull String token) {
        // Handle new token
        super.onNewToken(token);
        // Log the token
        Log.d(TAG, "Refreshed token: " + token);
        // You might want to send this token to your Firestore user document
        // if you want to target specific users later.
    }
}
```

1.12. Post.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import com.google.firebase.firestore.PropertyName;
import java.io.Serializable;

public class Post implements Serializable {

    private String postId;
    private String communityId;

    private String title;
    private String content;
    private String imageUrl;

    private int upvotes = 0;
    private int downvotes = 0;
    private int commentCount = 0;
    private int bookmarkCount = 0;

    private long timestamp = 0;
    private long lastUpdated = 0;

    private String authorId;
    private String authorName;
    private String authorEmail;

    private boolean upvoted = false;
    private boolean downvoted = false;
    private boolean isBookmarked = false;

    public Post() {}

    // Getters
    public String getPostId() { return postId; }
    public String getCommunityId() { return communityId; }
    public String getTitle() { return title; }
    public String getContent() { return content; }
    public String getImageUrl() { return imageUrl; }
    public int getUpvotes() { return upvotes; }
    public int getDownvotes() { return downvotes; }
    public int getCommentCount() { return commentCount; }
    public int getBookmarkCount() { return bookmarkCount; }
    public long getTimestamp() { return timestamp; }
    public long getLastUpdated() { return lastUpdated; }
    public String getAuthorId() { return authorId; }
    public String getAuthorName() { return authorName; }
    public String getAuthorEmail() { return authorEmail; }
    public boolean isUpvoted() { return upvoted; }
    public boolean isDownvoted() { return downvoted; }
```

```java
    public boolean isBookmarked() { return isBookmarked; }

    // Setters with String-to-Number safety (Handles Firestore string numbers)
    public void setPostId(String postId) { this.postId = postId; }
    public void setCommunityId(String communityId) { this.communityId = communityId; }
    public void setTitle(String title) { this.title = title; }
    public void setContent(String content) { this.content = content; }
    public void setImageUrl(String imageUrl) { this.imageUrl = imageUrl; }
    public void setUpvoted(boolean upvoted) { this.upvoted = upvoted; }
    public void setDownvoted(boolean downvoted) { this.downvoted = downvoted; }
    public void setBookmarked(boolean bookmarked) { isBookmarked = bookmarked; }

    @PropertyName("upvotes")
    public void setUpvotes(Object upvotes) {
        this.upvotes = convertToInt(upvotes);
    }

    @PropertyName("downvotes")
    public void setDownvotes(Object downvotes) {
        this.downvotes = convertToInt(downvotes);
    }

    @PropertyName("commentCount")
    public void setCommentCount(Object commentCount) {
        this.commentCount = convertToInt(commentCount);
    }

    @PropertyName("bookmarkCount")
    public void setBookmarkCount(Object bookmarkCount) {
        this.bookmarkCount = convertToInt(bookmarkCount);
    }

    @PropertyName("timestamp")
    public void setTimestamp(Object timestamp) {
        this.timestamp = convertToLong(timestamp);
    }

    @PropertyName("lastUpdated")
    public void setLastUpdated(Object lastUpdated) {
        this.lastUpdated = convertToLong(lastUpdated);
    }

    // Setters with String-to-String safety setAuthorId
    public void setAuthorId(String authorId) { this.authorId = authorId; }
    // setAuthorName
    public void setAuthorName(String authorName) { this.authorName = authorName; }
    // setAuthorEmail
    public void setAuthorEmail(String authorEmail) { this.authorEmail = authorEmail; }

    // Helper conversion methods
    public com.google.firebase.firestore.DocumentReference
getReference(com.google.firebase.firestore.FirebaseFirestore db) {
        // If postId is null, return null
        return
db.collection("communities").document(communityId).collection("posts").document(postId);
    }

    // Helper conversion methods
    private int convertToInt(Object val) {
        // If val is null, return 0
        if (val == null) return 0;
        // If val is a Number, return its intValue
        if (val instanceof Number) return ((Number) val).intValue();
```

```
        // If val is a String, try to parse it as an integer
        if (val instanceof String) {
            // If parsing fails, return 0
            try { return Integer.parseInt((String) val); } catch (Exception e) { return
0; }
        }
        // If none of the above conditions are met, return 0
        return 0;
    }

    // Helper conversion methods
    private long convertToLong(Object val) {
        // If val is null, return 0L
        if (val == null) return 0L;
        // If val is a Number, return its longValue
        if (val instanceof Number) return ((Number) val).longValue();
        // If val is a String, try to parse it as a long
        if (val instanceof String) {
            // If parsing fails, return 0L
            try { return Long.parseLong((String) val); } catch (Exception e) { return 0L;
}
        }
        // If none of the above conditions are met, return 0L
        return 0L;
    }
}
```

1.13. PostDetailActivity.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.ContextCompat;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.bumptech.glide.Glide;
import com.google.android.material.button.MaterialButton;
import com.google.android.material.card.MaterialCardView;
import com.google.android.material.dialog.MaterialAlertDialogBuilder;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.DocumentSnapshot;
import com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.Query;
import com.google.firebase.messaging.FirebaseMessaging;

import java.text.SimpleDateFormat;
import java.util.ArrayList;
```

```java
import java.util.Date;
import java.util.HashMap;
import java.util.List;
import java.util.Locale;
import java.util.Map;

// MainActivity.java
public class PostDetailActivity extends AppCompatActivity {

    private TextView tempTvCommunityName, tempTvPostTime, tempTvPostTitle,
tempTvPostContent, tempTvAuthorEmail;
    private MaterialCardView tempCardPostImage;
    private ImageButton tempBtnBookmark, tempBtnSendComment;
    private MaterialButton tempBtnUpvote, tempBtnDownvote, tempBtnShare;
    private EditText tempEtComment;

    private FirebaseFirestore firebaseDB;
    private String tempPostId, tempCommunityId;
    private Post tempCurrentPost;

    @SuppressWarnings("deprecation")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        // Set theme before super.onCreate
        super.onCreate(savedInstanceState);
        // Set theme before super.onCreate
        setContentView(R.layout.activity_post_detail);
        // Set theme before super.onCreate
        firebaseDB = FirebaseFirestore.getInstance();

        // Try to get the post object passed from the adapter
        if (android.os.Build.VERSION.SDK_INT >= android.os.Build.VERSION_CODES.TIRAMISU)
{
            // New API
            tempCurrentPost = getIntent().getSerializableExtra("POST_DATA", Post.class);
        } else {
            // (Post) getIntent().getSerializableExtra("POST_DATA");
            tempCurrentPost = (Post) getIntent().getSerializableExtra("POST_DATA");
        }

        // Initialize views
        tempInitViews();
        // Set up toolbar
        tempSetupToolbar();

        // If post data is available, display it
        if (tempCurrentPost != null) {
            // Use cached data if available
            tempPostId = tempCurrentPost.getPostId();
            // Use cached data if available
            tempCommunityId = tempCurrentPost.getCommunityId();
            // Use cached data if available
            displayPost();
            // Fetch comments
            tempLoadComments();
            // Still fetch latest details in background to sync upvotes/downvotes
            refreshPostDetails();
        } else {
            // Fallback for deep links or notifications
            tempPostId = getIntent().getStringExtra("POST_ID");
            // Fallback for deep links or notifications
            tempCommunityId = getIntent().getStringExtra("COMMUNITY_ID");
            // tempPostId != null && tempCommunityId != null
```

```
            if (tempPostId != null && tempCommunityId != null) {
                // Fetch post details
                tempLoadPostDetails();
                // Fetch comments
                tempLoadComments();
            }
        }

        // Set up click listeners tempBtnSendComment
        tempBtnSendComment.setOnClickListener(v -> postComment());
        // Set up click listeners tempBtnShare
        tempBtnShare.setOnClickListener(v -> sharePost());
        // Set up click listeners tempBtnUpvote
        tempBtnUpvote.setOnClickListener(v -> handleVote(true));
        // Set up click listeners tempBtnDownvote
        tempBtnDownvote.setOnClickListener(v -> handleVote(false));
        // Set up click listeners tempBtnBookmark
        tempBtnBookmark.setOnClickListener(v -> handleBookmark());
    }

    // Fetch latest post details in background
    private void refreshPostDetails() {
        // Listen for changes to the post document

firebaseDB.collection("communities").document(tempCommunityId).collection("posts").docume
nt(tempPostId)
            .addSnapshotListener((snapshot, e) -> {
                // Handle errors
                if (snapshot != null && snapshot.exists()) {
                    // Update local post object
                    Post updatedPost = snapshot.toObject(Post.class);
                    if (updatedPost != null) {
                        // Update local state wasUpvoted
                        boolean wasUpvoted = tempCurrentPost.isUpvoted();
                        // Update local state wasDownvoted
                        boolean wasDownvoted = tempCurrentPost.isDownvoted();
                        // Update local state wasBookmarked
                        boolean wasBookmarked = tempCurrentPost.isBookmarked();
                        // Update local post object
                        tempCurrentPost = updatedPost;
                        // Set post ID
                        tempCurrentPost.setPostId(snapshot.getId());
                        // Set community ID
                        tempCurrentPost.setCommunityId(tempCommunityId);

                        // Re-apply local session state wasUpvoted
                        tempCurrentPost.setUpvoted(wasUpvoted);
                        // Re-apply local session state wasDownvoted
                        tempCurrentPost.setDownvoted(wasDownvoted);
                        // Re-apply local session state wasBookmarked
                        tempCurrentPost.setBookmarked(wasBookmarked);
                        // Re-display post
                        displayPost();
                    }
                }
            });
    }

    // Initialize views
    private void tempInitViews() {
        tempTvCommunityName = findViewById(R.id.tvCommunityName);
        tempTvPostTime = findViewById(R.id.tvPostTime);
        tempTvPostTitle = findViewById(R.id.tvPostTitle);
```

```
        tempTvPostContent = findViewById(R.id.tvPostContent);
        tempCardPostImage = findViewById(R.id.cardPostImage);
        tempBtnBookmark = findViewById(R.id.btnBookmark);
        tempBtnUpvote = findViewById(R.id.btnUpvote);
        tempBtnDownvote = findViewById(R.id.btnDownvote);
        tempBtnShare = findViewById(R.id.btnShare);
        tempEtComment = findViewById(R.id.etComment);
        tempBtnSendComment = findViewById(R.id.btnSendComment);
        tempTvAuthorEmail = findViewById(R.id.tvAuthorEmail);
        findViewById(R.id.btnUserProfile).setOnClickListener(v ->
showUserProfileDialog());

        RecyclerView rvComments = findViewById(R.id.rvComments);
        rvComments.setLayoutManager(new LinearLayoutManager(this));
    }

    // Show user profile dialog
    private void showUserProfileDialog() {
        // Check if user is logged in
        com.google.firebase.auth.FirebaseUser user =
FirebaseAuth.getInstance().getCurrentUser();
        // If not, return
        if (user == null) return;

        // Initialize dialog
        View dialogView = getLayoutInflater().inflate(R.layout.dialog_user_profile,
null);
        // Create dialog
        AlertDialog dialog = new MaterialAlertDialogBuilder(this)
                .setView(dialogView)
                .create();

        // Set dialog background to transparent
        if (dialog.getWindow() != null) {
            // Set dialog background to transparent

dialog.getWindow().setBackgroundDrawableResource(android.R.color.transparent);
        }

        TextView txtName = dialogView.findViewById(R.id.profileName);
        TextView txtEmail = dialogView.findViewById(R.id.profileEmail);
        Button btnLogout = dialogView.findViewById(R.id.btnLogout);
        Button btnShowFCMToken = dialogView.findViewById(R.id.btnShowFCMToken);

        // Show FCM Token logic
        btnShowFCMToken.setOnClickListener(v -> {
            FirebaseMessaging.getInstance().getToken()
                .addOnCompleteListener(task -> {
                    if (!task.isSuccessful()) {
                        Toast.makeText(this, "Fetching FCM token failed",
Toast.LENGTH_SHORT).show();
                        return;
                    }

                    String token = task.getResult();
                    new MaterialAlertDialogBuilder(this)
                        .setTitle("FCM Registration Token")
                        .setMessage(token)
                        .setPositiveButton("Copy", (d, w) -> {
                            android.content.ClipboardManager clipboard =
(android.content.ClipboardManager)
getSystemService(android.content.Context.CLIPBOARD_SERVICE);
                            android.content.ClipData clip =
```

```
android.content.ClipData.newPlainText("FCM Token", token);
                            clipboard.setPrimaryClip(clip);
                            Toast.makeText(this, "Token copied!",
Toast.LENGTH_SHORT).show();
                        })
                        .setNegativeButton("Close", null)
                        .show();
                });
        });

        // ALWAYS FETCH LATEST FROM FIRESTORE FOR ACCURACY
        FirebaseFirestore.getInstance().collection("users").document(user.getUid()).get()
                .addOnSuccessListener(doc -> {
                    if (doc.exists()) {
                        // Get user data
                        String tempName = doc.getString("name");
                        String tempEmail = doc.getString("email");
                        if (tempName != null) {
                            // Set name
                            txtName.setText(tempName);
                        }
                        if (tempEmail != null) {
                            // Set email
                            txtEmail.setText(tempEmail);
                        }
                    }
                });

        // Set user data
        txtName.setText(user.getDisplayName() != null ? user.getDisplayName() : "Email
ID");
        // Set email
        txtEmail.setText(user.getEmail());

        // Set up logout button
        btnLogout.setOnClickListener(v -> {
            // Sign out user
            FirebaseAuth.getInstance().signOut();
            // Go back to log in screen
            Intent tempIntent = new Intent(this, LoginActivity.class);
            tempIntent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
            startActivity(tempIntent);
            finish();
        });

        dialog.show();
    }

    // Set up toolbar
    private void tempSetupToolbar() {
        // Set up toolbar
        androidx.appcompat.widget.Toolbar toolbar = findViewById(R.id.toolbar);
        // Set toolbar as action bar
        setSupportActionBar(toolbar);
        if (getSupportActionBar() != null) {
            // Enable back button
            getSupportActionBar().setDisplayHomeAsUpEnabled(true);
            toolbar.setNavigationOnClickListener(v -> finish());
        }
    }

    // Fetch post details
```

```
    private void tempLoadPostDetails() {
        // Fetch post details

firebaseDB.collection("communities").document(tempCommunityId).collection("posts").docume
nt(tempPostId).get().addOnSuccessListener(documentSnapshot -> {
            // Handle errors
            tempCurrentPost = documentSnapshot.toObject(Post.class);
            // Set post ID
            if (tempCurrentPost != null) {
                // Set post ID
                tempCurrentPost.setPostId(documentSnapshot.getId());
                // Set community ID
                tempCurrentPost.setCommunityId(tempCommunityId);
                // Display post
                displayPost();
            }
            // Fetch comments
        }).addOnFailureListener(e -> {
            // Handle errors
            Toast.makeText(this, "Error loading post", Toast.LENGTH_SHORT).show();
            finish();
        });
    }

    // Display post
    private void displayPost() {
        // Set post data
        tempTvPostTitle.setText(tempCurrentPost.getTitle());
        tempTvPostContent.setText(tempCurrentPost.getContent());
        tempTvCommunityName.setText("r/" + (tempCurrentPost.getCommunityId() != null ?
tempCurrentPost.getCommunityId() : "nexus"));

        // Use author email as requested
        tempTvAuthorEmail.setText(tempCurrentPost.getAuthorEmail() != null ?
tempCurrentPost.getAuthorEmail() :
                            (tempCurrentPost.getAuthorName() != null ?
tempCurrentPost.getAuthorName() : "Anonymous"));

        if (tempCurrentPost.getTimestamp() > 0) {
            // Set post time
            SimpleDateFormat sdf = new SimpleDateFormat("MMM dd, yyyy • HH:mm",
Locale.getDefault());
            tempTvPostTime.setText(sdf.format(new Date(tempCurrentPost.getTimestamp())));
        }

        if (tempCurrentPost.getImageUrl() != null &&
!tempCurrentPost.getImageUrl().isEmpty()) {
            tempCardPostImage.setVisibility(View.VISIBLE);
            Glide.with(this)
                    .load(tempCurrentPost.getImageUrl())
                    .centerCrop()
                    .into((ImageView) findViewById(R.id.ivPostImage));
        } else {
            tempCardPostImage.setVisibility(View.GONE);
        }

        // Set bookmark count text
        tempBtnUpvote.setText(String.valueOf(tempCurrentPost.getUpvotes()));
        tempBtnDownvote.setText(String.valueOf(tempCurrentPost.getDownvotes()));
        // Set bookmark count text
        updateInteractionButtons();
    }
```

```
    // Set up interaction buttons
    private void updateInteractionButtons() {
        // Set up interaction buttons
        if (tempCurrentPost.isBookmarked()) {
            // Set bookmark icon
            tempBtnBookmark.setImageResource(R.drawable.ic_bookmark);
            // Set bookmark color
            tempBtnBookmark.setColorFilter(ContextCompat.getColor(this, R.color.orange));
        } else {
            tempBtnBookmark.setImageResource(R.drawable.ic_bookmark_border);
            tempBtnBookmark.clearColorFilter();
        }

        // Set up interaction buttons
        if (tempCurrentPost.isUpvoted()) {
            // setIconTintResource
            tempBtnUpvote.setIconTintResource(R.color.upvote_green);
            tempBtnUpvote.setTextColor(ContextCompat.getColor(this,
R.color.upvote_green));
        } else {
            tempBtnUpvote.setIconTintResource(R.color.subtitleText);
            tempBtnUpvote.setTextColor(ContextCompat.getColor(this,
R.color.subtitleText));
        }

        if (tempCurrentPost.isDownvoted()) {
            tempBtnDownvote.setIconTintResource(R.color.downvote_red);
            tempBtnDownvote.setTextColor(ContextCompat.getColor(this,
R.color.downvote_red));
        } else {
            tempBtnDownvote.setIconTintResource(R.color.subtitleText);
            tempBtnDownvote.setTextColor(ContextCompat.getColor(this,
R.color.subtitleText));
        }
    }

    // Comment adapter
    private Comment replyingTo = null;

    // tempLoadComments
    private void tempLoadComments() {
        // Set up comments
        RecyclerView rvComments = findViewById(R.id.rvComments);

firebaseDB.collection("communities").document(tempCommunityId).collection("posts")
                .document(tempPostId).collection("comments")
                .orderBy("timestamp", Query.Direction.ASCENDING)
                .addSnapshotListener((value, error) -> {
                    if (value != null) {
                        // Convert to list
                        List<Comment> tempComments = new ArrayList<>();
                        // Convert to list
                        for (DocumentSnapshot tempDoc : value.getDocuments()) {
                            // Convert to object
                            Comment tempComment = tempDoc.toObject(Comment.class);
                            // Set comment ID
                            if (tempComment != null) {
                                // Set comment ID
                                tempComment.setCommentId(tempDoc.getId());
                                // Set community ID
                                tempComments.add(tempComment);
                            }
                        }
```

```
                        // Thread comments
                        List<Comment> threaded = threadComments(tempComments);

                        // Set up adapter
                        String tempUID = FirebaseAuth.getInstance().getUid();
                        if (tempUID != null) {
                            // Fetch comment votes and setup adapter
                            fetchCommentVotesAndSetupAdapter(threaded, rvComments,
tempUID);
                        } else {
                            // Setup adapter
                            setupCommentAdapter(threaded, rvComments);
                        }
                    }
                });
    }

    // Fetch comment votes and setup adapter
    private void fetchCommentVotesAndSetupAdapter(List<Comment> comments, RecyclerView
rvComments, String uid) {
        // Fetch comment votes
        firebaseDB.collection("users").document(uid).collection("commentVotes").get()
                .addOnSuccessListener(querySnapshot -> {
                    // Convert to map
                    Map<String, String> tempUserVotes = new HashMap<>();
                    // Convert to map
                    for (DocumentSnapshot docIdx : querySnapshot.getDocuments()) {
                        // Set user vote
                        tempUserVotes.put(docIdx.getId(), docIdx.getString("type"));
                    }

                    // Set up adapter
                    for (Comment tempCommentIdx : comments) {
                        // Set up adapter
                        String voteType =
tempUserVotes.get(tempCommentIdx.getCommentId());
                        // Set up adapter
                        if (voteType != null) {
                            // Set up adapter
                            tempCommentIdx.setUpvoted("up".equals(voteType));
                            // Set down adapter
                            tempCommentIdx.setDownvoted("down".equals(voteType));
                        }
                    }
                    // Set up adapter
                    setupCommentAdapter(comments, rvComments);
                })
                // Handle errors
                .addOnFailureListener(e -> setupCommentAdapter(comments, rvComments));
    }

    // Set up adapter
    private void setupCommentAdapter(List<Comment> comments, RecyclerView tempTvComments)
{
        // Set up adapter
        tempTvComments.setAdapter(new CommentAdapter(comments, new
CommentAdapter.OnCommentActionListener() {
            // Handle comment click
            @Override
            public void onReply(Comment parentComment) {
                // Set up adapter
                replyingTo = parentComment;
```

```
                // Set up adapter
                tempEtComment.setHint("Replying to " + parentComment.getUserName() +
"...");
                // Set up adapter
                tempEtComment.requestFocus();
            }

            // Handle comment vote
            @Override
            public void onVote(Comment comment, boolean isUpvote) {
                // Handle comment vote
                handleCommentVote(comment, isUpvote);
            }
        }));
    }

    // Thread comments
    private List<Comment> threadComments(List<Comment> flat) {
        // Thread result
        List<Comment> tempResult = new ArrayList<>();
        // Thread childrenMap
        Map<String, List<Comment>> tempChildrenMap = new HashMap<>();
        // Thread roots
        List<Comment> tempRoots = new ArrayList<>();

        // Thread flat
        for (Comment commentIdx : flat) {
            // Thread roots
            if (commentIdx.getParentCommentId() == null) {
                // Thread tempRoots
                tempRoots.add(commentIdx);
            } else {
                // Thread tempChildrenMap
                tempChildrenMap.computeIfAbsent(commentIdx.getParentCommentId(), k -> new
ArrayList<>()).add(commentIdx);
            }
        }

        // Thread tempRoots
        for (Comment tempCommentIdx : tempRoots) {
            // Thread tempResult
            addCommentAndChildren(tempCommentIdx, tempChildrenMap, tempResult);
        }
        // Thread tempResult
        return tempResult;
    }

    // Add comment and children
    private void addCommentAndChildren(Comment parent, Map<String, List<Comment>> map,
List<Comment> result) {
        // Thread result add parents
        result.add(parent);
        // tempChildren
        List<Comment> tempChildren = map.get(parent.getCommentId());
        // tempChildren checker
        if (tempChildren != null) {
            // Thread tempChild loop.
            for (Comment tempChild : tempChildren) {
                // Thread addCommentAndChildren
                addCommentAndChildren(tempChild, map, result);
            }
        }
    }
```

```
    // Handle comment vote
    private void handleCommentVote(Comment tempCommentIdx, boolean isUpvote) {
        // tempUID
        String tempUID = FirebaseAuth.getInstance().getUid();
        // tempUID checker
        if (tempUID == null) return;

        // Handle comment vote
        if (isUpvote) {
            if (tempCommentIdx.isUpvoted()) {
                // Set up adapter
                tempCommentIdx.setUpvotes(Math.max(0, tempCommentIdx.getUpvotes() - 1));
                // Set up adapter false
                tempCommentIdx.setUpvoted(false);
            } else {
                // Set up adapter
                tempCommentIdx.setUpvotes(tempCommentIdx.getUpvotes() + 1);
                // Set up adapter upVoted
                tempCommentIdx.setUpvoted(true);
                // Set up adapter isDownvoted
                if (tempCommentIdx.isDownvoted()) {
                    // Set up adapter
                    tempCommentIdx.setDownvotes(Math.max(0, tempCommentIdx.getDownvotes()
- 1));
                    // Set up adapter isDownvoted
                    tempCommentIdx.setDownvoted(false);
                }
            }
        } else {
            if (tempCommentIdx.isDownvoted()) {
                tempCommentIdx.setDownvotes(Math.max(0, tempCommentIdx.getDownvotes() -
1));
                tempCommentIdx.setDownvoted(false);
            } else {
                tempCommentIdx.setDownvotes(tempCommentIdx.getDownvotes() + 1);
                tempCommentIdx.setDownvoted(true);
                if (tempCommentIdx.isUpvoted()) {
                    tempCommentIdx.setUpvotes(Math.max(0, tempCommentIdx.getUpvotes() -
1));
                    tempCommentIdx.setUpvoted(false);
                }
            }
        }

        // Update Firestore

firebaseDB.collection("communities").document(tempCommunityId).collection("posts")

.document(tempPostId).collection("comments").document(tempCommentIdx.getCommentId())
                .update("upvotes", tempCommentIdx.getUpvotes(),
                        "downvotes", tempCommentIdx.getDownvotes());

        // Sync local vote state to Firestore
        Map<String, Object> hashmapVoteData = new HashMap<>();
        // Set up adapter
        hashmapVoteData.put("type", isUpvote ? (tempCommentIdx.isUpvoted() ? "up" :
"none") : (tempCommentIdx.isDownvoted() ? "down" : "none"));

        // Sync local vote state to Firestore
        firebaseDB.collection("users").document(tempUID)
                .collection("commentVotes").document(tempCommentIdx.getCommentId())
                .set(hashmapVoteData);
```

```
    }

    // Post comment
    private void postComment() {
        // Get comment text
        String tempCommentText = tempEtComment.getText().toString().trim();
        // tempCommentText checker
        if (TextUtils.isEmpty(tempCommentText)) return;

        // tempUserId
        String tempUserId = FirebaseAuth.getInstance().getUid();
        // tempUserId checker
        String tempUserName = FirebaseAuth.getInstance().getCurrentUser() != null ?
                          (FirebaseAuth.getInstance().getCurrentUser().getDisplayName() !=
null ?
                           FirebaseAuth.getInstance().getCurrentUser().getDisplayName() :
                           FirebaseAuth.getInstance().getCurrentUser().getEmail()) :
"Anonymous";

        // Create tempComment
        Map<String, Object> tempComment = new HashMap<>();
        tempComment.put("text", tempCommentText);
        tempComment.put("userName", tempUserName);
        tempComment.put("timestamp", System.currentTimeMillis());
        tempComment.put("userId", tempUserId);
        tempComment.put("upvotes", 0);
        tempComment.put("downvotes", 0);

        // tempUID checker
        if (replyingTo != null) {
            tempComment.put("parentCommentId", replyingTo.getCommentId());
        }

        // Post comment

firebaseDB.collection("communities").document(tempCommunityId).collection("posts")
                .document(tempPostId).collection("comments")
                .add(tempComment)
                .addOnSuccessListener(documentReference -> {
                    tempEtComment.setText("");
                    tempEtComment.setHint("Add a comment...");
                    replyingTo = null;
                    Toast.makeText(this, "Comment posted!", Toast.LENGTH_SHORT).show();
                });
    }

    // Share post
    private void sharePost() {
        // tempCurrentPost is null checker
        if (tempCurrentPost == null) return;
        // Set up baseUrl
        String baseUrl = "http://localhost:8080/post/" + tempCurrentPost.getPostId();
        // Set up queryParams
        String queryParams = "?title=" + Uri.encode(tempCurrentPost.getTitle()) +
                "&content=" + Uri.encode(tempCurrentPost.getContent()) +
                "&author=" + Uri.encode(tempCurrentPost.getAuthorEmail() != null ?
tempCurrentPost.getAuthorEmail() : "nexus") +
                "&upvotes=" + tempCurrentPost.getUpvotes() +
                "&comments=" + (tempCurrentPost.getCommentCount() > 0 ?
tempCurrentPost.getCommentCount() : 0);
        // Set up postLink
        String postLink = baseUrl + queryParams;
```

```
        // Set up shareText
        String shareText = tempCurrentPost.getTitle() + "\n\n" +
tempCurrentPost.getContent() + "\n\nRead more at: " + postLink;
        // Set up options
        String[] options = {"WhatsApp", "Messages", "Copy Link", "Open in Browser",
"Other Apps"};

        // Show share dialog
        new MaterialAlertDialogBuilder(this)
                .setTitle("Share Post")
                .setItems(options, (dialog, which) -> {
                    switch (which) {
                        case 0: // WhatsApp
                            shareToWhatsApp(shareText);
                            break;
                        case 1: // Messages
                            shareToMessages(shareText);
                            break;
                        case 2: // Copy Link
                            copyToClipboard(postLink);
                            break;
                        case 3: // Open in Browser
                            openInBrowser(postLink);
                            break;
                        case 4: // Other
                            shareToOther(shareText);
                            break;
                    }
                })
                .show();
    }

    // shareToWhatsApp() method.
    private void shareToWhatsApp(String text) {
        // intent
        Intent tempIntent = new Intent(Intent.ACTION_SEND);
        // setType
        tempIntent.setType("text/plain");
        // setPackage
        tempIntent.setPackage("com.whatsapp");
        // putExtra
        tempIntent.putExtra(Intent.EXTRA_TEXT, text);
        try {
            // startActivity
            startActivity(tempIntent);
        } catch (android.content.ActivityNotFoundException ex) {
            // shareToOther
            Toast.makeText(this, "WhatsApp not installed", Toast.LENGTH_SHORT).show();
            shareToOther(text);
        }
    }

    // shareToMessages() method.
    private void shareToMessages(String text) {
        Intent tempIntent = new Intent(Intent.ACTION_VIEW);
        tempIntent.setData(Uri.parse("sms:"));
        tempIntent.putExtra("sms_body", text);
        try {
            startActivity(tempIntent);
        } catch (Exception e) {
            Toast.makeText(this, "Messaging app not found", Toast.LENGTH_SHORT).show();
            shareToOther(text);
        }
```

```
    }

    // copyToClipboard() method.
    private void copyToClipboard(String text) {
        // tempClipboard
        android.content.ClipboardManager tempClipboard =
(android.content.ClipboardManager)
getSystemService(android.content.Context.CLIPBOARD_SERVICE);
        // tempClip
        android.content.ClipData tempClip = android.content.ClipData.newPlainText("Post
Content", text);
        tempClipboard.setPrimaryClip(tempClip);
        // Toast
        Toast.makeText(this, "Link copied to clipboard", Toast.LENGTH_SHORT).show();
    }

    private void openInBrowser(String url) {
        // tempIntent
        Intent tempIntent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
        // startActivity
        startActivity(tempIntent);
    }

    // shareToOther() method.
    private void shareToOther(String text) {
        // Intent
        Intent tempIntent = new Intent(Intent.ACTION_SEND);
        // tempIntent
        tempIntent.setType("text/plain");
        // putExtra
        tempIntent.putExtra(Intent.EXTRA_TEXT, text);
        // startActivity
        startActivity(Intent.createChooser(tempIntent, "Share via"));
    }

    // Handle bookmark
    private void handleBookmark() {
        // tempCurrentPost checker
        if (tempCurrentPost == null) return;
        // tempUID checker
        String tempUID = FirebaseAuth.getInstance().getUid();
        // tempUID checker
        if (tempUID == null) return;

        // tempCurrentPost isBookmarked checker
        if (tempCurrentPost.isBookmarked()) {
            tempCurrentPost.setBookmarkCount(Math.max(0,
tempCurrentPost.getBookmarkCount() - 1));
            tempCurrentPost.setBookmarked(false);
            firebaseDB.collection("communities").document(tempCommunityId)
                    .collection("posts").document(tempPostId)
                    .collection("bookmarks").document(tempUID).delete();
        } else {
            tempCurrentPost.setBookmarkCount(tempCurrentPost.getBookmarkCount() + 1);
            tempCurrentPost.setBookmarked(true);
            Map<String, Object> bookmark = new HashMap<>();
            bookmark.put("timestamp", System.currentTimeMillis());
            firebaseDB.collection("communities").document(tempCommunityId)
                    .collection("posts").document(tempPostId)
                    .collection("bookmarks").document(tempUID).set(bookmark);
        }

        // Update Firestore
```

```
        updateInteractionButtons();
        // Update Firestore
        firebaseDB.collection("communities").document(tempCommunityId)
                .collection("posts").document(tempPostId)
                .update("bookmarkCount", tempCurrentPost.getBookmarkCount());
    }

    // handleVote() method.
    private void handleVote(boolean isUpvote) {
        // tempCurrentPost checker
        if (tempCurrentPost == null) return;

        // tempUID checker
        String tempUID = FirebaseAuth.getInstance().getUid();
        if (tempUID == null) return;

        // tempCurrentPost isUpvoted checker
        if (isUpvote) {
            if (tempCurrentPost.isUpvoted()) {
                tempCurrentPost.setUpvotes(Math.max(0, tempCurrentPost.getUpvotes() -
1));
                tempCurrentPost.setUpvoted(false);
            } else {
                // Set up adapter
                tempCurrentPost.setUpvotes(tempCurrentPost.getUpvotes() + 1);
                tempCurrentPost.setUpvoted(true);
                if (tempCurrentPost.isDownvoted()) {
                    tempCurrentPost.setDownvotes(Math.max(0,
tempCurrentPost.getDownvotes() - 1));
                    tempCurrentPost.setDownvoted(false);
                }
            }
        } else {
            // tempCurrentPost isDownvoted checker
            if (tempCurrentPost.isDownvoted()) {
                tempCurrentPost.setDownvotes(Math.max(0, tempCurrentPost.getDownvotes() -
1));
                tempCurrentPost.setDownvoted(false);
            } else {
                // Set up adapter
                tempCurrentPost.setDownvotes(tempCurrentPost.getDownvotes() + 1);
                tempCurrentPost.setDownvoted(true);
                if (tempCurrentPost.isUpvoted()) {
                    tempCurrentPost.setUpvotes(Math.max(0, tempCurrentPost.getUpvotes() -
1));
                    tempCurrentPost.setUpvoted(false);
                }
            }
        }

        // displayPost
        displayPost();
        // updateVotesInFirestore
        updateVotesInFirestore();

        // Sync local vote state to Firestore
        Map<String, Object> tempVoteData = new HashMap<>();
        // Set up adapter
        tempVoteData.put("type", isUpvote ? (tempCurrentPost.isUpvoted() ? "up" : "none")
: (tempCurrentPost.isDownvoted() ? "down" : "none"));

        // Sync local vote state to Firestore
        firebaseDB.collection("users").document(tempUID)
```

```
                .collection("votes").document(tempCurrentPost.getPostId())
                .set(tempVoteData);
    }

    // updateVotesInFirestore() method.
    private void updateVotesInFirestore() {
        // communities
        firebaseDB.collection("communities")
                .document(tempCommunityId)
                .collection("posts")
                .document(tempPostId)
                .update(
                        "upvotes", tempCurrentPost.getUpvotes(),
                        "downvotes", tempCurrentPost.getDownvotes()
                );
    }
}
```

1.14. PostWidget.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.app.AlarmManager;
import android.app.PendingIntent;
import android.appwidget.AppWidgetManager;
import android.appwidget.AppWidgetProvider;
import android.content.ComponentName;
import android.content.Context;
import android.content.Intent;
import android.util.Log;
import android.graphics.Color;
import android.widget.RemoteViews;

import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.Query;
import com.google.firebase.firestore.DocumentSnapshot;

import java.util.List;

// PostWidget class
public class PostWidget extends AppWidgetProvider {

    private static final String TAG = "PostWidget";
    private static final String ACTION_UPDATE_MANUAL =
"com.example.nexus.UPDATE_WIDGET_MANUAL";
    private static final long REFRESH_INTERVAL_MS = 60000; // 1 minute to ensure system
stability

    // Called when the widget is first created
    @Override
    public void onUpdate(Context context, AppWidgetManager appWidgetManager, int[]
appWidgetIds) {
        // Start the update cycle
        startUpdateCycle(context);
    }

    // Called when the widget is updated
    @Override
    public void onReceive(Context context, Intent intent) {
        // Handle manual updates
```

```java
        String action = intent.getAction();
        // Log the action for debugging
        if (ACTION_UPDATE_MANUAL.equals(action)) {
            // Log the action for debugging
            startUpdateCycle(context);
        } else {
            // Handle other actions if needed
            super.onReceive(context, intent);
        }
    }

    // Called when the widget is enabled
    private void startUpdateCycle(Context context) {
        // tempAppWidgetManager
        AppWidgetManager tempAppWidgetManager = AppWidgetManager.getInstance(context);
        // tempComponentName
        ComponentName tempComponentName = new ComponentName(context, PostWidget.class);
        // tempAppWidgetIds
        int[] tempAppWidgetIds = tempAppWidgetManager.getAppWidgetIds(tempComponentName);

        // Log the action for debugging
        if (tempAppWidgetIds.length > 0) {
            // fetchDataAndUpdate
            fetchDataAndUpdate(context, tempAppWidgetManager);
            // scheduleNextUpdate
            scheduleNextUpdate(context);
        }
    }

    // Helper method to create the base RemoteViews
    private static RemoteViews createBaseRemoteViews(Context context) {
        // Create the base RemoteViews
        RemoteViews tempViews = new RemoteViews(context.getPackageName(),
R.layout.post_widget);

        // Set colors for icons
        int tempSecondaryColor = Color.parseColor("#A0AABF");
        int tempAccentColor = Color.parseColor("#6C7BFF");
        // Set the tempViews and set setColorFilter
        tempViews.setInt(R.id.widgetUpvoteIcon, "setColorFilter", tempSecondaryColor);
        tempViews.setInt(R.id.widgetDownvoteIcon, "setColorFilter", tempSecondaryColor);
        tempViews.setInt(R.id.widgetBookmarkIcon, "setColorFilter", tempSecondaryColor);
        tempViews.setInt(R.id.widgetCommentIcon, "setColorFilter", tempAccentColor);

        // Set the click action
        Intent tempIntent = new Intent(context, DashboardActivity.class);
        // Set the pending intent
        PendingIntent tempPendingIntent = PendingIntent.getActivity(context, 0,
tempIntent,
                PendingIntent.FLAG_UPDATE_CURRENT | PendingIntent.FLAG_IMMUTABLE);
        // Set the click action
        tempViews.setOnClickPendingIntent(R.id.widgetRoot, tempPendingIntent);
        // Return the tempViews
        return tempViews;
    }

    // fetchDataAndUpdate
    private void fetchDataAndUpdate(Context context, AppWidgetManager appWidgetManager) {
        // Check if user is logged in
        FirebaseAuth fbAuth = FirebaseAuth.getInstance();
        // If not logged in, update UI
        if (fbAuth.getCurrentUser() == null) {
            // updateUI
```

```
            updateUI(context, appWidgetManager, "Please log in", "");
            return;
        }

        // Fetch user data
        String tempUID = fbAuth.getCurrentUser().getUid();
        // FirebaseFirestore
        FirebaseFirestore firebaseDB = FirebaseFirestore.getInstance();

        // Fetch user profile

firebaseDB.collection("users").document(tempUID).get().addOnSuccessListener(userDoc -> {
            if (!userDoc.exists()) {
                // updateUI
                updateUI(context, appWidgetManager, "User profile not found", "");
                return;
            }
            // Check if communities are selected
            @SuppressWarnings("unchecked")
            List<String> tempSelected = (List<String>)
userDoc.get("selectedCommunities");
            // If not selected, update UI
            if (tempSelected == null || tempSelected.isEmpty()) {
                // updateUI
                updateUI(context, appWidgetManager, "No communities selected", "");
                return;
            }

            // Fetch latest post
            String tempCommunityId = tempSelected.get(0);
            // Fetch community name

firebaseDB.collection("communities").document(tempCommunityId).get().addOnSuccessListener
(commDoc -> {
                // If community doesn't exist, update UI
                String tempCommunityName = commDoc.getString("name");

                // Fetch latest post

firebaseDB.collection("communities").document(tempCommunityId).collection("posts")
                        .orderBy("timestamp", Query.Direction.DESCENDING)
                        .limit(1)
                        .get()
                        .addOnSuccessListener(querySnapshot -> {
                            // If no posts, update UI
                            RemoteViews tempViews = createBaseRemoteViews(context);
                            // Set community name
                            tempViews.setTextViewText(R.id.widgetCommunityName,
tempCommunityName != null ? tempCommunityName.toUpperCase() : "COMMUNITY");

                            // If posts exist, update UI
                            if (!querySnapshot.isEmpty()) {
                                // Set post details
                                DocumentSnapshot tempDoc =
querySnapshot.getDocuments().get(0);
                                // Title
                                tempViews.setTextViewText(R.id.widgetPostTitle,
tempDoc.getString("title"));
                                // Content
                                tempViews.setTextViewText(R.id.widgetPostContent,
tempDoc.getString("content"));
                                // Upvotes
                                tempViews.setTextViewText(R.id.widgetUpvoteCount,
```

```
String.valueOf(tempDoc.getLong("upvotes") != null ? tempDoc.getLong("upvotes") : 0));
                                // Downvotes
                                tempViews.setTextViewText(R.id.widgetDownvoteCount,
String.valueOf(tempDoc.getLong("downvotes") != null ? tempDoc.getLong("downvotes") : 0));
                                // Bookmark count
                                tempViews.setTextViewText(R.id.widgetBookmarkCount,
String.valueOf(tempDoc.getLong("bookmarkCount") != null ?
tempDoc.getLong("bookmarkCount") : 0));

                                // Fetch comment count separately as it might not be
synced in the post document

tempDoc.getReference().collection("comments").get().addOnSuccessListener(commentSnapshot
-> {
                                    // Set comment count
                                    tempViews.setTextViewText(R.id.widgetCommentCount,
String.valueOf(commentSnapshot.size()));
                                    // Update widget
                                    appWidgetManager.updateAppWidget(new
ComponentName(context, PostWidget.class), tempViews);
                                });
                            } else {
                                // Set default UI
                                tempViews.setTextViewText(R.id.widgetPostTitle, "No posts
yet");
                                // Set default UI
                                tempViews.setTextViewText(R.id.widgetPostContent, "Be the
first to post!");
                            }
                            // Update widget
                            appWidgetManager.updateAppWidget(new ComponentName(context,
PostWidget.class), tempViews);
                        });
            });
        }).addOnFailureListener(e -> updateUI(context, appWidgetManager, "Network Error",
""));
    }

    // updateUI
    private void updateUI(Context context, AppWidgetManager manager, String title, String
content) {
        // tempViews
        RemoteViews tempViews = createBaseRemoteViews(context);
        // setTextViewText title
        tempViews.setTextViewText(R.id.widgetPostTitle, title);
        // setTextViewText content
        tempViews.setTextViewText(R.id.widgetPostContent, content);
        // updateAppWidget
        manager.updateAppWidget(new ComponentName(context, PostWidget.class), tempViews);
    }

    // scheduleNextUpdate
    private void scheduleNextUpdate(Context context) {
        // intent
        Intent intent = new Intent(context, PostWidget.class);
        // intent
        intent.setAction(ACTION_UPDATE_MANUAL);
        // pendingIntent
        PendingIntent tempPendingIntent = PendingIntent.getBroadcast(context, 0, intent,
                PendingIntent.FLAG_UPDATE_CURRENT | PendingIntent.FLAG_IMMUTABLE);

        // alarmManager
        AlarmManager tempAlarmManager = (AlarmManager)
```

```
context.getSystemService(Context.ALARM_SERVICE);
        if (tempAlarmManager != null) {
            // nextUpdate
            long nextUpdate = System.currentTimeMillis() + REFRESH_INTERVAL_MS;
            // setAndAllowWhileIdle is safer than setExact for avoiding ANRs
            tempAlarmManager.setAndAllowWhileIdle(AlarmManager.RTC_WAKEUP, nextUpdate,
tempPendingIntent);
        }
    }

    // onDisabled() method.
    @Override
    public void onDisabled(Context context) {
        // Cancel the alarm
        super.onDisabled(context);
        // Cancel the alarm
        Intent intent = new Intent(context, PostWidget.class);
        // ACTION_UPDATE_MANUAL
        intent.setAction(ACTION_UPDATE_MANUAL);
        // pendingIntent
        PendingIntent pendingIntent = PendingIntent.getBroadcast(context, 0, intent,
                PendingIntent.FLAG_NO_CREATE | PendingIntent.FLAG_IMMUTABLE);
        if (pendingIntent != null) {
            // alarmManager
            AlarmManager alarmManager = (AlarmManager)
context.getSystemService(Context.ALARM_SERVICE);
            // cancel
            alarmManager.cancel(pendingIntent);
            // pendingIntent.cancel();
            pendingIntent.cancel();
        }
    }
}
```

1.15. SignUpActivity.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.content.Intent;
import android.os.Bundle;
import android.content.Context;
import android.net.ConnectivityManager;
import android.net.Network;
import android.net.NetworkCapabilities;
import android.text.TextUtils;
import android.util.Patterns;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.textfield.TextInputEditText;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.FirebaseFirestore;

import java.util.HashMap;

public class SignupActivity extends AppCompatActivity {

    TextInputEditText tempFullName, tempEmail, tempPassword;
```

```
    Button tempJoinBtn;
    TextView tempLoginSwitch;

    FirebaseAuth tempAuth;
    FirebaseFirestore tempFirebaseDB;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        // Layout file: activity_signup.xml
        setContentView(R.layout.activity_signup);

        tempFullName =findViewById(R.id.fullName);
        tempEmail =findViewById(R.id.email);
        tempPassword =findViewById(R.id.password);
        tempJoinBtn =findViewById(R.id.joinBtn);
        tempLoginSwitch =findViewById(R.id.loginSwitch);

        // Firebase Auth check
        tempAuth = FirebaseAuth.getInstance();
        // Firebase Firestore check
        tempFirebaseDB = FirebaseFirestore.getInstance();

        // Signup button click listener.
        tempJoinBtn.setOnClickListener(v -> {
            // Get all the user details.
            String tempName= tempFullName.getText().toString().trim();
            // Get all the user details.
            String tempUserEmail= tempEmail.getText().toString().trim();
            // Get all the user details.
            String tempUserPassword= tempPassword.getText().toString().trim();

            if (TextUtils.isEmpty(tempName)) {
                // If name is empty, then show error message.
                tempFullName.setError("Name is required");
                return;
            }

            if (TextUtils.isEmpty(tempUserEmail) ||
!Patterns.EMAIL_ADDRESS.matcher(tempUserEmail).matches()) {
                // If email is empty or invalid, then show error message.
                tempEmail.setError("Valid email is required");
                return;
            }

            if (TextUtils.isEmpty(tempUserPassword) || tempUserPassword.length() < 6) {
                // If password is empty or less than 6 characters, then show error
message.
                tempPassword.setError("Password must be at least 6 characters");
                return;
            }

            if (!isNetworkAvailable()) {
                // If no internet connection, then show error message.
                Toast.makeText(this, "No internet connection. Please check your
network.", Toast.LENGTH_LONG).show();
                return;
            }

            tempJoinBtn.setEnabled(false); // Prevent multiple clicks during network call

            tempAuth.createUserWithEmailAndPassword(tempUserEmail,tempUserPassword)
```

```
                    .addOnCompleteListener(task -> {
                        // If signup is successful, then go to
CustomizeFeedActivity.class
                        tempJoinBtn.setEnabled(true);
                        // If signup is successful, then go to
CustomizeFeedActivity.class
                        if(task.isSuccessful())
                        {
                            // Get the user id.
                            String uid= tempAuth.getCurrentUser().getUid();
                            // Create a hashmap.
                            HashMap<String,Object> tempUser=new HashMap<>();
                            // Put all the userName in the hashmap.
                            tempUser.put("name",tempName);
                            // Put all the user email in the hashmap.
                            tempUser.put("email",tempUserEmail);

                            // Put all the user id in the hashmap.

tempFirebaseDB.collection("users").document(uid).set(tempUser)
                                    .addOnSuccessListener(aVoid -> {
                                        // If signup is successful, then go to
CustomizeFeedActivity.class
                                        Toast.makeText(SignupActivity.this, "Account
Created", Toast.LENGTH_SHORT).show();
                                        // Get the user id.
                                        Intent intent = new Intent(SignupActivity.this,
CustomizeFeedActivity.class);
                                        // Get the useName.
                                        intent.putExtra("userName", tempName);
                                        // Get the user email.
                                        intent.putExtra("userEmail", tempUserEmail);
                                        // start the CustomizeFeedActivity.class
                                        startActivity(intent);
                                        // finish the SignupActivity.class
                                        finish();
                                    })
                                    .addOnFailureListener(e -> {
                                        // If signup is not successful, then show error
message.
                                        Toast.makeText(SignupActivity.this, "Failed to
save user info", Toast.LENGTH_SHORT).show();
                                    });
                        }
                        else
                        {
                            // If signup is not successful, then show error message.
                            String error = task.getException() != null ?
task.getException().getMessage() : "Unknown error";
                            // Log the error
                            android.util.Log.e("NexusDebug", "Signup Error: " + error);
                            // Show the error message
                            Toast.makeText(this,"Signup Failed: " +
error,Toast.LENGTH_LONG).show();
                        }
                    });
        });

        tempLoginSwitch.setOnClickListener(v -> {
            // start the LoginActivity.class
            startActivity(new Intent(SignupActivity.this,LoginActivity.class));
        });
```

```
    }

    private boolean isNetworkAvailable() {
        // Check for internet connection.
        ConnectivityManager tempConnectivityManager = (ConnectivityManager)
getSystemService(Context.CONNECTIVITY_SERVICE);
        // If connectivity manager is not null, then check for network.
        if (tempConnectivityManager != null) {
            // Check for active network.
            Network tempNetwork = tempConnectivityManager.getActiveNetwork();
            // If network is not null, then check for network capabilities.
            if (tempNetwork != null) {
                // Check for network capabilities.
                NetworkCapabilities tempCapabilities =
tempConnectivityManager.getNetworkCapabilities(tempNetwork);
                // Check for network type.
                return tempCapabilities != null &&
(tempCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_WIFI) ||

tempCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_CELLULAR) ||

tempCapabilities.hasTransport(NetworkCapabilities.TRANSPORT_ETHERNET));
            }
        }
        // If no network is available, then return false.
        return false;
    }
}
```

1.16. WidgetUpdateWorker.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.appwidget.AppWidgetManager;
import android.content.ComponentName;
import android.content.Context;
import android.content.Intent;
import android.util.Log;

import androidx.annotation.NonNull;
import androidx.work.Worker;
import androidx.work.WorkerParameters;

// WidgetUpdateWorker.java
public class WidgetUpdateWorker extends Worker {

    // Constructor
    public WidgetUpdateWorker(@NonNull Context context, @NonNull WorkerParameters
workerParams) {
        // Call super
        super(context, workerParams);
    }

    // Override doWork
    @NonNull
    @Override
    public Result doWork() {
        // Log the action for debugging
        Log.d("WidgetUpdateWorker", "Performing periodic widget update");

        // Create an intent to update the widget
```

```
        Context tempContext = getApplicationContext();
        // ACTION_UPDATE_MANUAL
        Intent tempIntent = new Intent(tempContext, PostWidget.class);
        // ACTION_UPDATE_MANUAL
        tempIntent.setAction("com.example.nexus.UPDATE_WIDGET_MANUAL");

        // Update the widget
        AppWidgetManager tempAppWidgetManager =
AppWidgetManager.getInstance(tempContext);
        // Get the widget IDs
        int[] tempIDs = tempAppWidgetManager.getAppWidgetIds(new
ComponentName(tempContext, PostWidget.class));
        // Set the widget IDs
        tempIntent.putExtra(AppWidgetManager.EXTRA_APPWIDGET_IDS, tempIDs);
        // Send the broadcast
        tempContext.sendBroadcast(tempIntent);
        // Return success
        return Result.success();
    }
}
```

1.17. ExampleUnitTest.java

```
// Name: Harsh Patel (A20369913

package com.example.tempapp;

import org.junit.Test;

import static org.junit.Assert.*;

/**
 * Example local unit test, which will execute on the development machine (host).
 *
 * @see <a href="http://d.android.com/tools/testing">Testing documentation</a>
 */
public class ExampleUnitTest {
    @Test
    public void addition_isCorrect() {
        assertEquals(4, 2 + 2);
    }
}
```

1.18. ExampleInstrumentedTest.java

```
// Name: Harsh Patel (A20369913

package com.example.nexus;

import android.content.Context;

import androidx.test.platform.app.InstrumentationRegistry;
import androidx.test.ext.junit.runners.AndroidJUnit4;

import org.junit.Test;
import org.junit.runner.RunWith;

import static org.junit.Assert.*;

/**
 * Instrumented test, which will execute on an Android device.
 *
 * @see <a href="http://d.android.com/tools/testing">Testing documentation</a>
```

```
 */
@RunWith(AndroidJUnit4.class)
public class ExampleInstrumentedTest {
    @Test
    public void useAppContext() {
        // Context of the app under test.
        Context appContext =
InstrumentationRegistry.getInstrumentation().getTargetContext();
        assertEquals("com.example.nexus", appContext.getPackageName());
    }
}
```

1.19. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFFFFF"
    android:fitsSystemWindows="true">

    <!-- App Title -->
    <TextView
        android:id="@+id/appTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Nexus"
        android:textSize="36sp"
        android:textStyle="bold"
        android:textColor="#000000"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="120dp"
        tools:ignore="HardcodedText" />

    <!-- Logo -->
    <ImageView
        android:id="@+id/logo"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:layout_below="@id/appTitle"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:src="@drawable/nexus_logo"
        tools:ignore="ContentDescription" />

    <!-- Loading Spinner -->
    <ProgressBar
        android:id="@+id/loadingSpinner"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/logo"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"/>

    <!-- Loading Text -->
    <TextView
        android:id="@+id/loadingText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Loading Feed..."
        android:textSize="16sp"
        android:textColor="#666666"
        android:layout_below="@id/loadingSpinner"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="10dp"
        tools:ignore="HardcodedText" />

</RelativeLayout>
```

1.20. activity_customize_feed.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#F6F8FC"
    android:padding="20dp"
    android:fitsSystemWindows="true">

    <!-- Header with Profile -->
    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp">

        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:layout_alignParentStart="true"
            android:layout_toStartOf="@+id/btnUserProfile">

            <!-- Title -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/customize_your_feed"
                android:textSize="26sp"
                android:textStyle="bold"
                android:textColor="#222222" />

            <!-- Subtitle -->
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="@string/customize_subtitle"
                android:textSize="14sp"
                android:textColor="#777777"
                android:layout_marginTop="4dp" />
        </LinearLayout>

        <!-- Profile Button -->
        <com.google.android.material.imageview.ShapeableImageView
            android:id="@+id/btnUserProfile"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:layout_alignParentEnd="true"
```

```
        android:layout_centerVertical="true"
        android:padding="2dp"
        android:src="@drawable/ic_launcher_foreground"
        android:background="#E0E7FF"
        app:shapeAppearanceOverlay="@style/CircleAppearance"
        app:strokeColor="#6C7BFF"
        app:strokeWidth="2dp"
        android:contentDescription="@string/user_profile"
        android:clickable="true"
        android:focusable="true" />
</RelativeLayout>

<!-- SEARCH BAR -->
<LinearLayout
    android:id="@+id/searchBarLayout"
    android:layout_width="match_parent"
    android:layout_height="46dp"
    android:layout_marginBottom="16dp"
    android:background="@drawable/search_bar_bg"
    android:gravity="center_vertical"
    android:paddingHorizontal="14dp">

    <EditText
        android:id="@+id/searchCommunities"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:background="@null"
        android:hint="@string/search_communities_hint"
        android:autofillHints="none"
        android:textColor="#111827"
        android:textColorHint="#9CA3AF"
        android:textSize="14sp"
        android:maxLines="1"
        android:inputType="text"/>
</LinearLayout>

<!-- Selected count (Center Position) -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
    android:layout_marginBottom="16dp">

    <TextView
        android:id="@+id/selectedCountText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="@drawable/bg_selected_count"
        android:paddingHorizontal="16dp"
        android:paddingVertical="8dp"
        android:text="@string/selected_count_initial"
        android:textColor="#3F6DF6"
        android:textSize="20sp"
        android:textStyle="bold"
        android:gravity="center" />
</LinearLayout>

<!-- Dynamic Community Grid -->
<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/communityRecycler"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
```

```
        android:clipToPadding="false"
        android:paddingBottom="10dp"/>

    <!-- Continue Button (renamed for clarity) -->
    <com.google.android.material.button.MaterialButton
        android:id="@+id/startFeedBtn"
        android:layout_width="match_parent"
        android:layout_height="55dp"
        android:text="@string/start_my_feed"
        android:textAllCaps="false"
        android:textSize="16sp"
        android:layout_marginTop="16dp"
        android:backgroundTint="#3F6DF6"
        android:textColor="#FFFFFF"
        app:cornerRadius="14dp" />

</LinearLayout>
```

1.21. activity_dashboard.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/appBackground">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:fitsSystemWindows="true">

        <!-- HEADER -->
        <RelativeLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:paddingStart="20dp"
            android:paddingEnd="20dp"
            android:paddingTop="24dp"
            android:paddingBottom="16dp"
            android:background="@color/headerBackground">

            <!-- Title & Subtitle Group -->
            <LinearLayout
                android:id="@+id/headerTextLayout"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:layout_alignParentStart="true"
                android:layout_toStartOf="@+id/actionButtonsContainer"
                android:layout_marginEnd="10dp">

                <TextView
                    android:id="@+id/titleNexus"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="@string/nexus_ai"
                    android:textStyle="bold"
                    android:textSize="26sp"
                    android:textColor="@color/titleText"
                    android:letterSpacing="0.05"/>
```

```
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/nexus_subtitle"
            android:textSize="14sp"
            android:textColor="@color/subtitleText"
            android:layout_marginTop="4dp"/>

    </LinearLayout>

    <!-- Bookmark & Profile Group -->
    <LinearLayout
        android:id="@+id/actionButtonsContainer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentEnd="true"
        android:layout_alignTop="@+id/headerTextLayout"
        android:orientation="horizontal"
        android:gravity="center_vertical">

        <!-- Bookmark Button -->
        <ImageButton
            android:id="@+id/btnShowBookmarks"
            android:layout_width="44dp"
            android:layout_height="44dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@drawable/ic_bookmark"
            app:tint="@color/titleText"
            android:padding="10dp"
            android:scaleType="centerInside"
            android:contentDescription="@string/show_bookmarks"/>

        <!-- Profile Button -->
        <com.google.android.material.imageview.ShapeableImageView
            android:id="@+id/btnUserProfile"
            android:layout_width="36dp"
            android:layout_height="36dp"
            android:layout_marginStart="8dp"
            android:padding="2dp"
            android:src="@drawable/ic_launcher_foreground"
            android:background="#E0E7FF"
            app:shapeAppearanceOverlay="@style/CircleAppearance"
            app:strokeColor="#6C7BFF"
            app:strokeWidth="2dp"
            android:contentDescription="@string/user_profile"
            android:clickable="true"
            android:focusable="true" />

    </LinearLayout>

    <!-- SEARCH & FILTER ROW -->
    <LinearLayout
        android:id="@+id/searchFilterRow"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@+id/headerTextLayout"
        android:layout_marginTop="16dp"
        android:orientation="horizontal"
        android:gravity="center_vertical">

        <!-- SEARCH BAR -->
        <LinearLayout
```

```
                android:id="@+id/searchBarLayout"
                android:layout_width="0dp"
                android:layout_height="46dp"
                android:layout_weight="1"
                android:background="@drawable/search_bar_bg"
                android:gravity="center_vertical"
                android:paddingHorizontal="14dp">

                <EditText
                    android:id="@+id/searchPosts"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:background="@null"
                    android:hint="@string/search_hint"
                    android:autofillHints="none"
                    android:textColor="@color/inputText"
                    android:textColorHint="@color/inputHint"
                    android:textSize="14sp"
                    android:maxLines="1"
                    android:inputType="text"/>
            </LinearLayout>

            <!-- Filter Button (Communities) -->
            <ImageButton
                android:id="@+id/btnFilterCommunities"
                android:layout_width="48dp"
                android:layout_height="48dp"
                android:layout_marginStart="8dp"
                android:background="?attr/selectableItemBackgroundBorderless"
                android:src="@drawable/ic_filter"
                app:tint="@color/titleText"
                android:padding="12dp"
                android:scaleType="centerInside"
                android:contentDescription="@string/select_communities"/>

            <!-- Filter Button (Date) -->
            <ImageButton
                android:id="@+id/btnFilterDate"
                android:layout_width="48dp"
                android:layout_height="48dp"
                android:layout_marginStart="4dp"
                android:background="?attr/selectableItemBackgroundBorderless"
                android:src="@android:drawable/ic_menu_today"
                app:tint="@color/titleText"
                android:padding="12dp"
                android:scaleType="centerInside"
                android:contentDescription="Filter by date"/>
        </LinearLayout>

    </RelativeLayout>

    <!-- FEED -->
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/feedRecycler"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:paddingStart="14dp"
        android:paddingEnd="14dp"
        android:paddingTop="12dp"
        android:clipToPadding="false"/>

</LinearLayout>
```

```
    <!-- ADD POST BUTTON -->
    <com.google.android.material.floatingactionbutton.FloatingActionButton
        android:id="@+id/addPostFab"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="bottom|end"
        android:layout_margin="24dp"
        android:src="@drawable/ic_add"
        app:backgroundTint="#6C7BFF"
        app:tint="#FFFFFF"
        app:elevation="8dp"
        android:contentDescription="@string/add_new_post"/>

</FrameLayout>
```

1.22. activity_login.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F6F8FC"
    android:fitsSystemWindows="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="28dp">

        <!-- Logo -->
        <ImageView
            android:layout_width="80dp"
            android:layout_height="80dp"
            android:layout_gravity="center"
            android:src="@drawable/nexus_logo"
            android:layout_marginTop="30dp"/>

        <!-- Title -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Welcome Back"
            android:textSize="26sp"
            android:textStyle="bold"
            android:textColor="#222222"
            android:layout_gravity="center"
            android:layout_marginTop="18dp"
            tools:ignore="HardcodedText"/>

        <!-- Subtitle -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Login to continue your personalized feed..."
            android:textSize="14sp"
            android:textColor="#777777"
            android:layout_gravity="center"
```

```
        android:layout_marginBottom="28dp"
        tools:ignore="HardcodedText"/>

    <!-- Card Container -->
    <com.google.android.material.card.MaterialCardView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:cardCornerRadius="18dp"
        app:cardElevation="6dp"
        android:padding="18dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="298dp"
            android:orientation="vertical">

            <!-- Email -->
            <com.google.android.material.textfield.TextInputLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Email"
                app:boxBackgroundMode="outline">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/email"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:inputType="textEmailAddress" />

            </com.google.android.material.textfield.TextInputLayout>

            <!-- Password -->
            <com.google.android.material.textfield.TextInputLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="14dp"
                android:hint="Password"
                app:boxBackgroundMode="outline"
                app:endIconMode="password_toggle">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/password"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:inputType="textPassword" />

            </com.google.android.material.textfield.TextInputLayout>

            <!-- Remember + Forgot -->
            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="10dp"
                android:gravity="center_vertical"
                android:orientation="horizontal">

                <CheckBox
                    android:id="@+id/rememberMe"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Remember me"
                    tools:ignore="HardcodedText" />
```

```
                </LinearLayout>

                <!-- Login Button -->
                <com.google.android.material.button.MaterialButton
                    android:id="@+id/loginBtn"
                    android:layout_width="match_parent"
                    android:layout_height="50dp"
                    android:layout_marginTop="18dp"
                    android:layout_marginBottom="10dp"
                    android:backgroundTint="#3F6DF6"
                    android:text="Login"
                    android:textAllCaps="false"
                    android:textColor="#FFFFFF"
                    android:textSize="16sp"
                    app:cornerRadius="12dp" />

                <TextView
                    android:id="@+id/forgotPassword"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Forgot password?"
                    android:textColor="#3F6DF6"
                    tools:ignore="HardcodedText" />

            </LinearLayout>

        </com.google.android.material.card.MaterialCardView>

        <!-- Signup Link -->
        <TextView
            android:id="@+id/signupSwitch"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="New here? Create an account"
            android:textStyle="bold"
            android:textColor="#3F6DF6"
            android:layout_marginTop="24dp"
            android:layout_gravity="center"
            tools:ignore="HardcodedText"/>

    </LinearLayout>

</ScrollView>
```

1.23. activity_post_detail.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/appBackground">

    <com.google.android.material.appbar.AppBarLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/headerBackground"
        android:fitsSystemWindows="true"
        app:elevation="0dp">

        <com.google.android.material.appbar.MaterialToolbar
```

```
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        app:navigationIcon="@drawable/ic_arrow_back"
        app:title="Post Detail"
        app:titleCentered="true"
        app:titleTextAppearance="@style/TextAppearance.Material3.TitleLarge">

        <com.google.android.material.imageview.ShapeableImageView
            android:id="@+id/btnUserProfile"
            android:layout_width="32dp"
            android:layout_height="32dp"
            android:layout_gravity="end"
            android:layout_marginEnd="16dp"
            android:background="#E0E7FF"
            android:clickable="true"
            android:contentDescription="@string/user_profile"
            android:focusable="true"
            android:padding="2dp"
            android:src="@drawable/ic_launcher_foreground"
            app:shapeAppearanceOverlay="@style/CircleAppearance"
            app:strokeColor="#6C7BFF"
            app:strokeWidth="2dp" />

    </com.google.android.material.appbar.MaterialToolbar>

</com.google.android.material.appbar.AppBarLayout>

<androidx.core.widget.NestedScrollView
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:layout_behavior="@string/appbar_scrolling_view_behavior">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp"
        android:paddingBottom="88dp"
        android:clipToPadding="false">

        <!-- Community and Time Info -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_vertical"
            android:orientation="horizontal"
            android:layout_marginBottom="12dp">

            <com.google.android.material.imageview.ShapeableImageView
                android:id="@+id/communityIcon"
                android:layout_width="32dp"
                android:layout_height="32dp"
                android:src="@drawable/nexus_logo"
                app:shapeAppearanceOverlay="@style/CircleAppearance" />

            <LinearLayout
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_marginStart="8dp"
                android:layout_weight="1"
                android:orientation="vertical">
```

```
            <TextView
                android:id="@+id/tvCommunityName"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="r/community"
                android:textStyle="bold"
                android:textColor="@color/titleText"
                android:textSize="14sp" />

            <LinearLayout
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:orientation="horizontal">

                <TextView
                    android:id="@+id/tvAuthorEmail"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="user@example.com"
                    android:textColor="@color/subtitleText"
                    android:textSize="12sp" />

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text=" • "
                    android:textColor="@color/subtitleText"
                    android:textSize="12sp" />

                <TextView
                    android:id="@+id/tvPostTime"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="2 hours ago"
                    android:textColor="@color/subtitleText"
                    android:textSize="12sp" />
            </LinearLayout>
        </LinearLayout>

        <ImageButton
            android:id="@+id/btnBookmark"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@drawable/ic_bookmark"
            app:tint="@color/subtitleText" />
    </LinearLayout>

    <!-- Post Title -->
    <TextView
        android:id="@+id/tvPostTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="BREAKING NEWS: MAJOR TECH MERGER"
        android:textColor="@color/titleText"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="12dp" />

    <!-- Post Image (if any) -->
    <com.google.android.material.card.MaterialCardView
        android:id="@+id/cardPostImage"
        android:layout_width="match_parent"
```

```
                android:layout_height="200dp"
                android:layout_marginBottom="16dp"
                app:cardCornerRadius="12dp"
                app:cardElevation="0dp"
                app:strokeWidth="1dp"
                app:strokeColor="@color/divider">

                <ImageView
                    android:id="@+id/ivPostImage"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:scaleType="centerCrop"
                    android:src="@drawable/ic_gaming" />
            </com.google.android.material.card.MaterialCardView>

            <!-- Post Content -->
            <TextView
                android:id="@+id/tvPostContent"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="AI generated concise summary. All major stakeholders are in
agreement. This merger will reshape the tech landscape for years to come."
                android:textColor="@color/inputText"
                android:textSize="16sp"
                android:lineSpacingExtra="4dp"
                android:layout_marginBottom="24dp" />

            <!-- Interaction Buttons (Reddit Style like image) -->
            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="horizontal"
                android:gravity="center"
                android:layout_marginBottom="24dp">

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnUpvote"
                    style="@style/Widget.Material3.Button.TextButton"
                    android:layout_width="0dp"
                    android:layout_height="wrap_content"
                    android:layout_weight="1"
                    android:text="Upvote"
                    android:textColor="@color/subtitleText"
                    app:icon="@android:drawable/arrow_up_float"
                    app:iconTint="@color/subtitleText"
                    app:iconSize="18dp"
                    app:iconGravity="top"
                    android:layout_marginEnd="8dp"
                    android:paddingVertical="8dp"
                    app:cornerRadius="12dp"
                    android:textSize="12sp" />

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnShare"
                    style="@style/Widget.Material3.Button.TextButton"
                    android:layout_width="0dp"
                    android:layout_height="wrap_content"
                    android:layout_weight="1"
                    android:text="Share"
                    android:textColor="@color/subtitleText"
                    app:icon="@android:drawable/ic_menu_share"
                    app:iconTint="@color/subtitleText"
                    app:iconSize="18dp"
```

```
                app:iconGravity="top"
                android:layout_marginHorizontal="4dp"
                android:paddingVertical="8dp"
                app:cornerRadius="12dp"
                android:textSize="12sp" />

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnDownvote"
                style="@style/Widget.Material3.Button.TextButton"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:text="Downvote"
                android:textColor="@color/subtitleText"
                app:icon="@android:drawable/arrow_down_float"
                app:iconTint="@color/subtitleText"
                app:iconSize="18dp"
                app:iconGravity="top"
                android:layout_marginStart="8dp"
                android:paddingVertical="8dp"
                app:cornerRadius="12dp"
                android:textSize="12sp" />
        </LinearLayout>

        <View
            android:layout_width="match_parent"
            android:layout_height="1dp"
            android:background="@color/divider"
            android:layout_marginBottom="16dp" />

        <!-- Comments Section -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Comments"
            android:textStyle="bold"
            android:textColor="@color/titleText"
            android:textSize="18sp"
            android:layout_marginBottom="16dp" />

        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/rvComments"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:nestedScrollingEnabled="false"
            android:paddingBottom="100dp"
            android:clipToPadding="false"
            tools:listitem="@layout/item_comment" />

    </LinearLayout>
</androidx.core.widget.NestedScrollView>

<!-- Floating Comment Input (Modern style) -->
<com.google.android.material.card.MaterialCardView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom"
    android:layout_margin="16dp"
    app:cardCornerRadius="24dp"
    app:cardElevation="8dp">

    <LinearLayout
        android:layout_width="match_parent"
```

```
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:gravity="center_vertical"
            android:paddingHorizontal="16dp"
            android:paddingVertical="8dp">

            <EditText
                android:id="@+id/etComment"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:hint="Add a comment..."
                android:background="@null"
                android:maxLines="4"
                android:textSize="14sp" />

            <ImageButton
                android:id="@+id/btnSendComment"
                android:layout_width="40dp"
                android:layout_height="40dp"
                android:background="?attr/selectableItemBackgroundBorderless"
                android:src="@android:drawable/ic_menu_send"
                app:tint="@color/orange" />
        </LinearLayout>
    </com.google.android.material.card.MaterialCardView>

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

1.24. activity_signup.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F6F8FC"
    android:fitsSystemWindows="true">

    <LinearLayout
        android:orientation="vertical"
        android:padding="28dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <!-- Logo -->
        <ImageView
            android:layout_width="80dp"
            android:layout_height="80dp"
            android:layout_gravity="center"
            android:src="@drawable/nexus_logo"
            android:layout_marginTop="30dp"/>

        <!-- Title -->
        <TextView
            android:text="Create Nexus Account"
            android:textSize="26sp"
            android:textStyle="bold"
            android:textColor="#222222"
            android:layout_marginTop="20dp"
            android:layout_gravity="center"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        tools:ignore="HardcodedText"/>

    <!-- Subtitle -->
    <TextView
        android:text="Join communities and personalize your feed..."
        android:textSize="14sp"
        android:textColor="#777777"
        android:layout_gravity="center"
        android:layout_marginBottom="30dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        tools:ignore="HardcodedText"/>

    <!-- Card container -->
    <com.google.android.material.card.MaterialCardView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:cardCornerRadius="18dp"
        app:cardElevation="6dp"
        android:padding="18dp">

        <LinearLayout
            android:orientation="vertical"
            android:layout_width="match_parent"
            android:layout_height="wrap_content">

            <!-- Full Name -->
            <com.google.android.material.textfield.TextInputLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Full Name"
                app:boxBackgroundMode="outline">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/fullName"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"/>

            </com.google.android.material.textfield.TextInputLayout>

            <!-- Email -->
            <com.google.android.material.textfield.TextInputLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="14dp"
                android:hint="Email"
                app:boxBackgroundMode="outline">

                <com.google.android.material.textfield.TextInputEditText
                    android:id="@+id/email"
                    android:inputType="textEmailAddress"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"/>

            </com.google.android.material.textfield.TextInputLayout>

            <!-- Password -->
            <com.google.android.material.textfield.TextInputLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="14dp"
```

```
                    android:hint="Password"
                    app:endIconMode="password_toggle"
                    app:boxBackgroundMode="outline">

                    <com.google.android.material.textfield.TextInputEditText
                        android:id="@+id/password"
                        android:inputType="textPassword"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"/>

                </com.google.android.material.textfield.TextInputLayout>

                <!-- Sign Up Button -->
                <com.google.android.material.button.MaterialButton
                    android:id="@+id/joinBtn"
                    android:text="Create Account"
                    android:textAllCaps="false"
                    android:textSize="16sp"
                    android:layout_marginTop="22dp"
                    android:layout_width="match_parent"
                    android:layout_height="50dp"
                    android:backgroundTint="#3F6DF6"
                    android:textColor="#FFFFFF"
                    app:cornerRadius="12dp"/>

            </LinearLayout>

        </com.google.android.material.card.MaterialCardView>

        <!-- Login Link -->
        <TextView
            android:id="@+id/loginSwitch"
            android:text="Already have an account? Login"
            android:textColor="#3F6DF6"
            android:textStyle="bold"
            android:layout_marginTop="24dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center"
            tools:ignore="HardcodedText"/>

    </LinearLayout>

</ScrollView>
```

1.25. community_item.xml

```
<?xml version="1.0" encoding="utf-8"?>

<com.google.android.material.card.MaterialCardView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/cardView"
    android:layout_width="match_parent"
    android:layout_height="140dp"
    android:layout_margin="10dp"
    app:cardCornerRadius="22dp"
    app:cardElevation="6dp"
    app:strokeWidth="2dp"
    app:strokeColor="#E6E8F0"
    android:foreground="?attr/selectableItemBackground">
```

```
    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:orientation="vertical"
            android:gravity="center"
            android:padding="14dp">

            <!-- Icon background circle -->
            <FrameLayout
                android:layout_width="64dp"
                android:layout_height="64dp"
                android:background="@drawable/community_icon_bg"
                android:gravity="center">

                <ImageView
                    android:id="@+id/communityIcon"
                    android:layout_width="34dp"
                    android:layout_height="34dp"
                    android:tint="#3F6DF6"
                    android:scaleType="centerInside"
                    tools:ignore="ContentDescription,UseAppTint" />

            </FrameLayout>

            <!-- Community Name -->
            <TextView
                android:id="@+id/communityName"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Community"
                android:textStyle="bold"
                android:textColor="#2B2B2B"
                android:textSize="15sp"
                android:layout_marginTop="10dp"
                android:gravity="center"
                tools:ignore="HardcodedText" />

        </LinearLayout>

        <!-- Green Checkmark (Hidden by default) -->
        <ImageView
            android:id="@+id/checkMark"
            android:layout_width="24dp"
            android:layout_height="24dp"
            android:layout_alignParentEnd="true"
            android:layout_margin="12dp"
            android:src="@drawable/ic_check_circle"
            android:visibility="gone"
            app:tint="#4CAF50"
            tools:ignore="ContentDescription" />

    </RelativeLayout>

</com.google.android.material.card.MaterialCardView>
```

1.26. dialog_create_community.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    app:cardCornerRadius="28dp"
    app:cardElevation="16dp"
    app:cardBackgroundColor="#1A1F2A"
    android:layout_margin="16dp">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:overScrollMode="never"
        android:scrollbars="none">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="28dp">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="New Community"
                android:textSize="24sp"
                android:textStyle="bold"
                android:textColor="#FFFFFF"
                android:letterSpacing="0.02"
                android:layout_marginBottom="24dp"/>

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="COMMUNITY NAME"
                android:textSize="11sp"
                android:textColor="#6C7BFF"
                android:textStyle="bold"
                android:letterSpacing="0.1"
                android:layout_marginBottom="8dp"/>

            <EditText
                android:id="@+id/editCommunityName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="e.g. Android Developers"
                android:textColor="#FFFFFF"
                android:textColorHint="#565E71"
                android:background="@drawable/bg_post_input"
                android:textSize="16sp"
                android:padding="16dp"
                android:layout_marginBottom="20dp"
                android:inputType="textCapWords"
                android:importantForAutofill="no"/>

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="SUMMARY"
                android:textSize="11sp"
                android:textColor="#6C7BFF"
                android:textStyle="bold"
                android:letterSpacing="0.1"
```

```
            android:layout_marginBottom="8dp"/>

        <EditText
            android:id="@+id/editCommunitySummary"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="What is this community about?"
            android:textColor="#B0B8C5"
            android:textColorHint="#565E71"
            android:background="@drawable/bg_post_input"
            android:textSize="15sp"
            android:padding="16dp"
            android:minLines="4"
            android:gravity="top"
            android:layout_marginBottom="20dp"
            android:inputType="textMultiLine|textCapSentences"
            android:importantForAutofill="no"
            android:lineSpacingExtra="4dp"/>

        <!-- IMAGE SELECTION AREA -->
        <LinearLayout
            android:id="@+id/layoutSelectCommImage"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/bg_image_select"
            android:padding="20dp"
            android:gravity="center"
            android:layout_marginBottom="32dp"
            android:clickable="true"
            android:focusable="true">

            <ImageView
                android:id="@+id/imgCommPreview"
                android:layout_width="match_parent"
                android:layout_height="120dp"
                android:scaleType="centerCrop"
                android:visibility="gone"
                android:layout_marginBottom="12dp"
                android:background="@drawable/bg_post_input"/>

            <TextView
                android:id="@+id/textAddCommImage"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="+ Add Icon"
                android:textColor="#6C7BFF"
                android:textStyle="bold"
                android:textSize="14sp"
                android:letterSpacing="0.05"/>
        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:gravity="end">

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnCancelComm"
                style="@style/Widget.MaterialComponents.Button.TextButton"
                android:layout_width="wrap_content"
                android:layout_height="54dp"
```

```
                    android:text="Discard"
                    android:textColor="#9CA3AF"
                    android:textAllCaps="false"
                    android:textSize="15sp"/>

                <com.google.android.material.button.MaterialButton
                    android:id="@+id/btnCreateComm"
                    android:layout_width="140dp"
                    android:layout_height="54dp"
                    android:text="Create"
                    android:backgroundTint="#6C7BFF"
                    android:textColor="#FFFFFF"
                    android:layout_marginStart="16dp"
                    app:cornerRadius="16dp"
                    android:textAllCaps="false"
                    android:textSize="16sp"
                    android:textStyle="bold"
                    app:elevation="4dp"/>
            </LinearLayout>

        </LinearLayout>
    </ScrollView>

</androidx.cardview.widget.CardView>
```

1.27. dialog_create_post.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    app:cardCornerRadius="28dp"
    app:cardElevation="16dp"
    app:cardBackgroundColor="#1A1F2A"
    android:layout_margin="16dp">

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:overScrollMode="never"
        android:scrollbars="none">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="28dp">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="New Post"
                android:textSize="24sp"
                android:textStyle="bold"
                android:textColor="#FFFFFF"
                android:letterSpacing="0.02"
                android:layout_marginBottom="24dp"/>

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
```

```
                android:text="COMMUNITY"
                android:textSize="11sp"
                android:textColor="#6C7BFF"
                android:textStyle="bold"
                android:letterSpacing="0.1"
                android:layout_marginBottom="8dp"/>

            <com.google.android.material.textfield.TextInputLayout
                android:id="@+id/communityInputLayout"

style="@style/Widget.Material3.TextInputLayout.FilledBox.ExposedDropdownMenu"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginBottom="20dp"
                app:boxBackgroundColor="#2A3350"
                app:boxCornerRadiusTopStart="12dp"
                app:boxCornerRadiusTopEnd="12dp"
                app:boxCornerRadiusBottomStart="12dp"
                app:boxCornerRadiusBottomEnd="12dp"
                app:boxStrokeWidth="0dp"
                app:boxStrokeWidthFocused="0dp"
                app:hintEnabled="false"
                app:endIconTint="#6C7BFF">

                <AutoCompleteTextView
                    android:id="@+id/communityAutoComplete"
                    android:layout_width="match_parent"
                    android:layout_height="wrap_content"
                    android:inputType="none"
                    android:padding="16dp"
                    android:text="Select Community"
                    android:textColor="#FFFFFF"
                    android:textSize="15sp" />

            </com.google.android.material.textfield.TextInputLayout>

            <EditText
                android:id="@+id/editPostTitle"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Give it a catchy title..."
                android:textColor="#FFFFFF"
                android:textColorHint="#565E71"
                android:background="@drawable/bg_post_input"
                android:textSize="16sp"
                android:padding="16dp"
                android:layout_marginBottom="12dp"
                android:inputType="textCapSentences"
                android:importantForAutofill="no"/>

            <EditText
                android:id="@+id/editPostContent"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="What's happening?"
                android:textColor="#B0B8C5"
                android:textColorHint="#565E71"
                android:background="@drawable/bg_post_input"
                android:textSize="15sp"
                android:padding="16dp"
                android:minLines="5"
                android:gravity="top"
                android:layout_marginBottom="20dp"
```

```
            android:inputType="textMultiLine|textCapSentences"
            android:importantForAutofill="no"
            android:lineSpacingExtra="4dp"/>

        <!-- IMAGE SELECTION AREA -->
        <LinearLayout
            android:id="@+id/layoutSelectImage"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/bg_image_select"
            android:padding="20dp"
            android:gravity="center"
            android:layout_marginBottom="32dp"
            android:clickable="true"
            android:focusable="true">

            <ImageView
                android:id="@+id/imgPostPreview"
                android:layout_width="match_parent"
                android:layout_height="180dp"
                android:scaleType="centerCrop"
                android:visibility="gone"
                android:layout_marginBottom="12dp"
                android:clipToOutline="true"
                android:background="@drawable/bg_post_input"/>

            <TextView
                android:id="@+id/textAddImage"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="+ Add Media"
                android:textColor="#6C7BFF"
                android:textStyle="bold"
                android:textSize="14sp"
                android:letterSpacing="0.05"/>
        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:gravity="end">

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnCancel"
                style="@style/Widget.MaterialComponents.Button.TextButton"
                android:layout_width="wrap_content"
                android:layout_height="54dp"
                android:text="Discard"
                android:textColor="#9CA3AF"
                android:textAllCaps="false"
                android:textSize="15sp"
                app:rippleColor="#33FFFFFF"/>

            <com.google.android.material.button.MaterialButton
                android:id="@+id/btnPost"
                android:layout_width="120dp"
                android:layout_height="54dp"
                android:text="Post"
                android:backgroundTint="#6C7BFF"
                android:textColor="#FFFFFF"
                android:layout_marginStart="16dp"
```

```
                    app:cornerRadius="16dp"
                    android:textAllCaps="false"
                    android:textSize="16sp"
                    android:textStyle="bold"
                    app:elevation="4dp"/>
            </LinearLayout>

        </LinearLayout>
    </ScrollView>

</androidx.cardview.widget.CardView>
```

1.28. dialog_select_option.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="@drawable/bg_dialog_options"
    android:elevation="8dp"
    android:minWidth="180dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Option"
        android:textSize="17sp"
        android:textStyle="bold"
        android:textColor="#1A1C1E"
        android:layout_marginBottom="12dp"
        android:layout_marginStart="4dp"/>

    <LinearLayout
        android:id="@+id/optionCreatePost"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical"
        android:paddingVertical="10dp"
        android:paddingHorizontal="8dp"
        android:background="?attr/selectableItemBackground"
        android:clickable="true"
        android:focusable="true">

        <ImageView
            android:layout_width="20dp"
            android:layout_height="20dp"
            android:src="@drawable/ic_add"
            app:tint="#44474E"
            android:layout_marginEnd="12dp"
            android:contentDescription="Create Post Icon"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Create Post"
            android:textSize="15sp"
            android:textColor="#44474E"
            android:layout_marginEnd="8dp"/>
    </LinearLayout>
```

```
    <View
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:background="#1A000000"
        android:layout_marginVertical="2dp"/>

    <LinearLayout
        android:id="@+id/optionCreateCommunity"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical"
        android:paddingVertical="10dp"
        android:paddingHorizontal="8dp"
        android:background="?attr/selectableItemBackground"
        android:clickable="true"
        android:focusable="true">

        <ImageView
            android:layout_width="20dp"
            android:layout_height="20dp"
            android:src="@drawable/ic_add"
            app:tint="#44474E"
            android:layout_marginEnd="12dp"
            android:contentDescription="Create Community Icon"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Create Community"
            android:textSize="15sp"
            android:textColor="#44474E"
            android:layout_marginEnd="8dp"/>
    </LinearLayout>

</LinearLayout>
```

1.29. dialog_user_profile.xml

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="16dp"
    app:cardBackgroundColor="@android:color/white"
    app:cardCornerRadius="28dp"
    app:cardElevation="0dp"
    app:strokeWidth="0dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_horizontal"
        android:orientation="vertical"
        android:padding="24dp">

        <!-- Handle/Decoration -->
        <com.google.android.material.card.MaterialCardView
            android:layout_width="40dp"
            android:layout_height="4dp"
```

```
        android:layout_marginBottom="24dp"
        app:cardBackgroundColor="#E5E7EB"
        app:cardCornerRadius="2dp"
        app:cardElevation="0dp"
        app:strokeWidth="0dp" />

    <!-- Profile Image Container -->
    <FrameLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <com.google.android.material.imageview.ShapeableImageView
            android:id="@+id/profileImage"
            android:layout_width="96dp"
            android:layout_height="96dp"
            android:background="#F3F4FF"
            android:padding="12dp"
            android:src="@drawable/ic_launcher_foreground"
            app:shapeAppearanceOverlay="@style/CircleAppearance"
            app:strokeColor="#6C7BFF"
            app:strokeWidth="3dp" />

        <!-- Online Status Indicator (Circle) -->
        <com.google.android.material.card.MaterialCardView
            android:layout_width="18dp"
            android:layout_height="18dp"
            android:layout_gravity="bottom|end"
            android:layout_margin="6dp"
            app:cardBackgroundColor="#10B981"
            app:cardCornerRadius="9dp"
            app:cardElevation="2dp"
            app:strokeColor="@android:color/white"
            app:strokeWidth="2dp" />
    </FrameLayout>

    <TextView
        android:id="@+id/profileName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:fontFamily="sans-serif-medium"
        android:text="Email ID"
        android:textColor="#111827"
        android:textSize="22sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/profileEmail"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="4dp"
        android:fontFamily="sans-serif"
        android:text="email@example.com"
        android:textColor="#6B7280"
        android:textSize="14sp" />

    <!-- Stats Section -->
    <com.google.android.material.card.MaterialCardView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        app:cardBackgroundColor="#F9FAFB"
        app:cardCornerRadius="16dp"
```

```
        app:cardElevation="0dp"
        app:strokeWidth="0dp">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:padding="16dp"
            android:weightSum="2">

            <LinearLayout
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:gravity="center"
                android:orientation="vertical">
                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="MEMBER"
                    android:textColor="#9CA3AF"
                    android:textSize="10sp"
                    android:textStyle="bold"
                    android:letterSpacing="0.1"/>
                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginTop="2dp"
                    android:text="Free Tier"
                    android:textColor="#374151"
                    android:textSize="13sp"
                    android:textStyle="bold"/>
            </LinearLayout>

            <View
                android:layout_width="1dp"
                android:layout_height="match_parent"
                android:layout_marginVertical="4dp"
                android:background="#E5E7EB"/>

            <LinearLayout
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:gravity="center"
                android:orientation="vertical">
                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="JOINED"
                    android:textColor="#9CA3AF"
                    android:textSize="10sp"
                    android:textStyle="bold"
                    android:letterSpacing="0.1"/>
                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginTop="2dp"
                    android:text="Jan 2024"
                    android:textColor="#374151"
                    android:textSize="13sp"
                    android:textStyle="bold"/>
            </LinearLayout>
```

```
            </LinearLayout>
        </com.google.android.material.card.MaterialCardView>

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnShowFCMToken"
            style="@style/Widget.Material3.Button.TonalButton"
            android:layout_width="match_parent"
            android:layout_height="56dp"
            android:layout_marginTop="16dp"
            android:text="Show FCM Token"
            android:textColor="#10B981"
            android:textSize="16sp"
            android:textStyle="bold"
            app:backgroundTint="#ECFDF5"
            app:cornerRadius="16dp"
            app:icon="@android:drawable/ic_dialog_info"
            app:iconGravity="textStart"
            app:iconPadding="12dp"
            app:iconSize="20dp"
            app:iconTint="#10B981"
            app:rippleColor="#D1FAE5" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnRefreshWidget"
            style="@style/Widget.Material3.Button.TonalButton"
            android:layout_width="match_parent"
            android:layout_height="56dp"
            android:layout_marginTop="16dp"
            android:text="Refresh Widget"
            android:textColor="#6C7BFF"
            android:textSize="16sp"
            android:textStyle="bold"
            app:backgroundTint="#F3F4FF"
            app:cornerRadius="16dp"
            app:icon="@android:drawable/ic_menu_rotate"
            app:iconGravity="textStart"
            app:iconPadding="12dp"
            app:iconSize="20dp"
            app:iconTint="#6C7BFF"
            app:rippleColor="#E5E7EB" />

        <com.google.android.material.button.MaterialButton
            android:id="@+id/btnLogout"
            style="@style/Widget.Material3.Button.TonalButton"
            android:layout_width="match_parent"
            android:layout_height="56dp"
            android:layout_marginTop="24dp"
            android:text="Log Out"
            android:textColor="#EF4444"
            android:textSize="16sp"
            android:textStyle="bold"
            app:backgroundTint="#FEF2F2"
            app:cornerRadius="16dp"
            app:icon="@android:drawable/ic_lock_power_off"
            app:iconGravity="textStart"
            app:iconPadding="12dp"
            app:iconSize="20dp"
            app:iconTint="#EF4444"
            app:rippleColor="#FEE2E2" />

    </LinearLayout>
</com.google.android.material.card.MaterialCardView>
```

1.30. item_comment.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:paddingVertical="8dp">

    <!-- Thread Line -->
    <View
        android:id="@+id/threadLine"
        android:layout_width="2dp"
        android:layout_height="match_parent"
        android:layout_marginEnd="12dp"
        android:layout_marginStart="4dp"
        android:background="@color/comment_line" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_vertical"
            android:orientation="horizontal"
            android:layout_marginBottom="4dp">

            <com.google.android.material.imageview.ShapeableImageView
                android:id="@+id/ivCommentUser"
                android:layout_width="24dp"
                android:layout_height="24dp"
                android:src="@drawable/ic_launcher_foreground"
                android:background="#E0E7FF"
                app:shapeAppearanceOverlay="@style/CircleAppearance" />

            <TextView
                android:id="@+id/tvCommentUser"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="8dp"
                android:text="u/UserHandle"
                android:textColor="@color/titleText"
                android:textSize="12sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/tvCommentTime"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginStart="8dp"
                android:text="1h"
                android:textColor="@color/subtitleText"
                android:textSize="11sp" />
        </LinearLayout>

        <TextView
            android:id="@+id/tvCommentText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:text="Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do
eiusmod tempor incididunt ut labore et dolore magna aliqua."
            android:textColor="@color/inputText"
            android:textSize="14sp"
            android:lineSpacingExtra="2dp" />

        <!-- Comment Actions -->
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:layout_marginTop="8dp"
            android:gravity="center_vertical">

            <ImageView
                android:id="@+id/ivCommentUpvote"
                android:layout_width="16dp"
                android:layout_height="16dp"
                android:src="@android:drawable/arrow_up_float"
                app:tint="@color/subtitleText" />

            <TextView
                android:id="@+id/tvCommentUpvotes"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="0"
                android:textSize="12sp"
                android:textColor="@color/subtitleText"
                android:layout_marginStart="4dp"
                android:layout_marginEnd="12dp" />

            <ImageView
                android:id="@+id/ivCommentDownvote"
                android:layout_width="16dp"
                android:layout_height="16dp"
                android:src="@android:drawable/arrow_down_float"
                app:tint="@color/subtitleText" />

            <TextView
                android:id="@+id/tvCommentDownvotes"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="0"
                android:textSize="12sp"
                android:textColor="@color/subtitleText"
                android:layout_marginStart="4dp"
                android:layout_marginEnd="12dp" />

            <TextView
                android:id="@+id/tvCommentReply"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Reply"
                android:textSize="12sp"
                android:textColor="@color/subtitleText"
                android:layout_marginStart="16dp"
                android:clickable="true"
                android:focusable="true" />

        </LinearLayout>
    </LinearLayout>
</LinearLayout>
```

1.31. item_post.xml

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="12dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="2dp"
    app:strokeWidth="1dp"
    app:strokeColor="#2A3350"
    app:cardBackgroundColor="#1A1F2A">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:id="@+id/postTag"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="COMMUNITY"
            android:textSize="10sp"
            android:textStyle="bold"
            android:textColor="#6C7BFF"
            android:background="#121620"
            android:paddingHorizontal="8dp"
            android:paddingVertical="2dp"
            android:layout_marginBottom="8dp"/>

        <TextView
            android:id="@+id/postTitle"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Post Title"
            android:textSize="18sp"
            android:textStyle="bold"
            android:textColor="#FFFFFF"/>

        <TextView
            android:id="@+id/postContent"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Post content preview..."
            android:textSize="14sp"
            android:textColor="#B0B8C5"
            android:layout_marginTop="6dp"
            android:maxLines="3"
            android:ellipsize="end"
            android:lineSpacingExtra="2dp"/>

        <TextView
            android:id="@+id/timeText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Just now"
            android:textSize="11sp"
            android:textColor="#5C667E"
            android:layout_marginTop="8dp"/>
```

```
    <View
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:background="#2A3350"
        android:layout_marginTop="12dp"
        android:layout_marginBottom="8dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical">

        <com.google.android.material.button.MaterialButton
            android:id="@+id/upvoteBtn"
            style="@style/Widget.Material3.Button.TextButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="0"
            android:textColor="#B0B8C5"
            app:icon="@android:drawable/arrow_up_float"
            app:iconTint="#B0B8C5"
            app:iconSize="18dp"
            android:padding="0dp"
            android:minWidth="0dp"
            android:insetTop="0dp"
            android:insetBottom="0dp"/>

        <com.google.android.material.button.MaterialButton
            android:id="@+id/downvoteBtn"
            style="@style/Widget.Material3.Button.TextButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="0"
            android:textColor="#B0B8C5"
            app:icon="@android:drawable/arrow_down_float"
            app:iconTint="#B0B8C5"
            app:iconSize="18dp"
            android:layout_marginStart="4dp"
            android:padding="0dp"
            android:minWidth="0dp"
            android:insetTop="0dp"
            android:insetBottom="0dp"/>

        <View
            android:layout_width="0dp"
            android:layout_height="0dp"
            android:layout_weight="1"/>

        <com.google.android.material.button.MaterialButton
            android:id="@+id/commentBtn"
            style="@style/Widget.Material3.Button.TextButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="0"
            android:textColor="#B0B8C5"
            app:icon="@android:drawable/stat_notify_chat"
            app:iconTint="#6C7BFF"
            app:iconSize="18dp"
            android:padding="0dp"
            android:minWidth="0dp"
            android:insetTop="0dp"
```

```
                android:insetBottom="0dp"/>

            <com.google.android.material.button.MaterialButton
                android:id="@+id/shareBtn"
                style="@style/Widget.Material3.Button.IconButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                app:icon="@android:drawable/ic_menu_share"
                app:iconTint="#B0B8C5"
                app:iconSize="18dp"
                android:layout_marginStart="8dp"
                android:padding="0dp"
                android:minWidth="0dp"
                android:insetTop="0dp"
                android:insetBottom="0dp"/>

            <com.google.android.material.button.MaterialButton
                android:id="@+id/bookmarkBtn"
                style="@style/Widget.Material3.Button.IconButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                app:icon="@drawable/ic_bookmark"
                app:iconTint="#B0B8C5"
                app:iconSize="18dp"
                android:layout_marginStart="8dp"
                android:padding="0dp"
                android:minWidth="0dp"
                android:insetTop="0dp"
                android:insetBottom="0dp"/>

        </LinearLayout>
    </LinearLayout>
</com.google.android.material.card.MaterialCardView>
```

1.32. post_widget.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/widgetRoot"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/bg_widget"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Community Chip -->
    <TextView
        android:id="@+id/widgetCommunityName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="@drawable/bg_community_tag"
        android:paddingLeft="10dp"
        android:paddingRight="10dp"
        android:paddingTop="4dp"
        android:paddingBottom="4dp"
        android:text="COMMUNITY"
        android:textColor="#6C7BFF"
        android:textSize="10sp"
        android:textStyle="bold" />

    <!-- Post Title -->
    <TextView
        android:id="@+id/widgetPostTitle"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:ellipsize="end"
        android:maxLines="2"
        android:text="Post Title"
        android:textColor="#FFFFFF"
        android:textSize="18sp"
        android:textStyle="bold" />

    <!-- Post Content Summary -->
    <TextView
        android:id="@+id/widgetPostContent"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_marginTop="6dp"
        android:layout_weight="1"
        android:ellipsize="end"
        android:lineSpacingExtra="3dp"
        android:maxLines="3"
        android:text="Post content preview..."
        android:textColor="#A0AABF"
        android:textSize="13sp" />

    <!-- Horizontal Divider -->
    <FrameLayout
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:layout_marginTop="16dp"
        android:layout_marginBottom="12dp"
        android:background="#2A3350" />

    <!-- Engagement Stats Footer -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_vertical"
        android:orientation="horizontal">

        <!-- Upvotes -->
        <ImageView
            android:id="@+id/widgetUpvoteIcon"
            android:layout_width="16dp"
            android:layout_height="16dp"
            android:src="@android:drawable/arrow_up_float"
            android:contentDescription="Upvotes" />

        <TextView
            android:id="@+id/widgetUpvoteCount"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="4dp"
            android:text="0"
            android:textColor="#A0AABF"
            android:textSize="12sp"
            android:textStyle="bold" />

        <!-- Downvotes -->
        <ImageView
            android:id="@+id/widgetDownvoteIcon"
            android:layout_width="16dp"
            android:layout_height="16dp"
            android:layout_marginStart="16dp"
```

```
            android:src="@android:drawable/arrow_down_float"
            android:contentDescription="Downvotes" />

        <TextView
            android:id="@+id/widgetDownvoteCount"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="4dp"
            android:text="0"
            android:textColor="#A0AABF"
            android:textSize="12sp"
            android:textStyle="bold" />

        <FrameLayout
            android:layout_width="0dp"
            android:layout_height="1dp"
            android:layout_weight="1" />

        <!-- Comments -->
        <ImageView
            android:id="@+id/widgetCommentIcon"
            android:layout_width="16dp"
            android:layout_height="16dp"
            android:src="@android:drawable/stat_notify_chat"
            android:contentDescription="Comments" />

        <TextView
            android:id="@+id/widgetCommentCount"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="4dp"
            android:text="0"
            android:textColor="#A0AABF"
            android:textSize="12sp"
            android:textStyle="bold" />

        <!-- Bookmarks -->
        <ImageView
            android:id="@+id/widgetBookmarkIcon"
            android:layout_width="16dp"
            android:layout_height="16dp"
            android:layout_marginStart="16dp"
            android:src="@drawable/ic_bookmark"
            android:contentDescription="Bookmarks" />

        <TextView
            android:id="@+id/widgetBookmarkCount"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="4dp"
            android:text="0"
            android:textColor="#A0AABF"
            android:textSize="12sp"
            android:textStyle="bold" />

    </LinearLayout>
</LinearLayout>
```

1.33. bg_community_tag.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#121620"/>
```

```
    <corners android:radius="6dp"/>
</shape>
```

1.34. bg_dialog_options.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#E9E7F7" />
    <corners
        android:topLeftRadius="32dp"
        android:topRightRadius="32dp"
        android:bottomLeftRadius="32dp"
        android:bottomRightRadius="4dp" />
</shape>
```

1.35. bg_image_select.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#2A3350"/>
    <corners android:radius="16dp"/>
    <stroke android:width="1dp" android:color="#3F4B75" android:dashWidth="6dp"
android:dashGap="6dp"/>
</shape>
```

1.36. bg_post_input.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#2A3350"/>
    <corners android:radius="12dp"/>
</shape>
```

1.37. bg_selected_count.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#EEF2FF" />
    <corners android:radius="20dp" />
</shape>
```

1.38. bg_widget.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#1A1F2A"/>
    <corners android:radius="24dp"/>
    <stroke android:width="1dp" android:color="#2A3350"/>
</shape>
```

1.39. community_icon_bg.xml

```
<?xml version="1.0" encoding="utf-8"?>

<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">

    <gradient
        android:startColor="#EEF2FF"
        android:endColor="#E4EBFF"
        android:angle="90"/>
```

```
</shape>
```

1.40. ic_add.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#FFFFFF"
        android:pathData="M19,13h-6v6h-2v-6H5v-2h6V5h2v6h6v2z"/>
</vector>
```

1.41. ic_arrow_back.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24"
    android:tint="?attr/colorControlNormal">
    <path
        android:fillColor="@android:color/white"
        android:pathData="M20,11H7.83l5.59,-5.59L12,4l-8,8 8,8 1.41,-1.41L7.83,13H20v-
2z"/>
</vector>
```

1.42. ic_bookmark.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#FFFFFF"
        android:pathData="M17,3H7C5.9,3 5,3.9 5,5V21L12,18L19,21V5C19,3.9 18.1,3 17,3Z"/>
</vector>
```

1.43. ic_bookmark_border.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24"
    android:tint="?attr/colorControlNormal">
  <path
      android:fillColor="@android:color/white"
      android:pathData="M17,3H7c-1.1,0 -2,0.9 -2,2v16l7,-3 7,3V5c0,-1.1 -0.9,-2 -2,-
2zM17,18l-5,- 8L7,18V5h10v13z"/>
</vector>
```

1.44. ic_check_circle.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24"
```

```
    android:tint="#4CAF50">
    <path
        android:fillColor="#4CAF50"
        android:pathData="M12,2C6.48,2 2,6.48 2,12s4.48,10 10,10 10,-4.48 10,-10S17.52,2
12,2zM10,17.41l-4.71,-4.71l1.41,-1.41L10,14.59l7.3,-7.29 1.41,1.41L10,17.41z"/>
</vector>
```

1.45. ic_filter.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#FFFFFF"
        android:pathData="M10,18H14V16H10V18ZM3,6V8H21V6H3ZM6,13H18V11H6V13Z"/>
</vector>
```

1.46. ic_launcher_background.xml

```
<?xml version="1.0" encoding="utf-8"?>
<vector
    android:height="108dp"
    android:width="108dp"
    android:viewportHeight="108"
    android:viewportWidth="108"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <path android:fillColor="#3DDC84"
          android:pathData="M0,0h108v108h-108z"/>
    <path android:fillColor="#00000000" android:pathData="M9,0L9,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,0L19,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M29,0L29,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M39,0L39,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M49,0L49,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M59,0L59,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M69,0L69,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M79,0L79,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M89,0L89,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M99,0L99,108"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,9L108,9"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,19L108,19"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,29L108,29"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,39L108,39"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,49L108,49"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,59L108,59"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,69L108,69"
```

```
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,79L108,79"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,89L108,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M0,99L108,99"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,29L89,29"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,39L89,39"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,49L89,49"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,59L89,59"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,69L89,69"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M19,79L89,79"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M29,19L29,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M39,19L39,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M49,19L49,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M59,19L59,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M69,19L69,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
    <path android:fillColor="#00000000" android:pathData="M79,19L79,89"
          android:strokeColor="#33FFFFFF" android:strokeWidth="0.8"/>
</vector>
```

1.47. ic_launcher_foreground.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:aapt="http://schemas.android.com/aapt"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path android:pathData="M31,63.928c0,0 6.4,-11 12.1,-13.1c7.2,-2.6 26,-1.4 26,-
1.4138.1,38.1L107,108.9281-32,-1L31,63.928z">
        <aapt:attr name="android:fillColor">
            <gradient
                android:endX="85.84757"
                android:endY="92.4963"
                android:startX="42.9492"
                android:startY="49.59793"
                android:type="linear">
                <item
                    android:color="#44000000"
                    android:offset="0.0" />
                <item
                    android:color="#00000000"
                    android:offset="1.0" />
            </gradient>
        </aapt:attr>
    </path>
    <path
        android:fillColor="#FFFFFF"
        android:fillType="nonZero"
        android:pathData="M65.3,45.828l3.8,-6.6c0.2,-0.4 0.1,-0.9 -0.3,-1.1c-0.4,-0.2 -
```

```
0.9,-0.1 -1.1,0.3l-3.9,6.7c-6.3,-2.8 -13.4,-2.8 -19.7,0l-3.9,-6.7c-0.2,-0.4 -0.7,-0.5 -
1.1,-0.3C38.8,38.328 38.7,38.828 38.9,39.228l3.8,6.6C36.2,49.428 31.7,56.028
31,63.928h46C76.3,56.028 71.8,49.428 65.3,45.828zM43.4,57.328c-0.8,0 -1.5,-0.5 -1.8,-
1.2c-0.3,-0.7 -0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1
1.2,1.8C45.3,56.528 44.5,57.328 43.4,57.328L43.4,57.328zM64.6,57.328c-0.8,0 -1.5,-0.5 -
1.8,-1.2s-0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1 1.2,1.8C66.5,56.528
65.6,57.328 64.6,57.328L64.6,57.328z"
        android:strokeWidth="1"
        android:strokeColor="#00000000" />
</vector>
```

1.48. search_bar_bg.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="@color/searchBackground"/>
    <corners android:radius="14dp"/>
    <stroke
        android:width="1dp"
        android:color="@color/searchBorder"/>
</shape>
```

1.49. post_widget_info.xml

```
<?xml version="1.0" encoding="utf-8"?>
<appwidget-provider xmlns:android="http://schemas.android.com/apk/res/android"
    android:initialLayout="@layout/post_widget"
    android:minWidth="250dp"
    android:minHeight="110dp"
    android:updatePeriodMillis="30000"
    android:widgetCategory="home_screen" />
```

1.50. values.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="firebase_database_url" translatable="false">https://camscannerai-
default-rtdb.firebaseio.com</string>
    <string name="gcm_defaultSenderId" translatable="false">333825133998</string>
    <string name="google_api_key"
translatable="false">AIzaSyCH6qgOnnctFA0sejk5_kbSvXzqQZRYjCk</string>
    <string name="google_app_id"
translatable="false">1:333825133998:android:2e0126415d5062989d0c3f</string>
    <string name="google_crash_reporting_api_key"
translatable="false">AIzaSyCH6qgOnnctFA0sejk5_kbSvXzqQZRYjCk</string>
    <string name="google_storage_bucket"
translatable="false">camscannerai.firebasestorage.app</string>
    <string name="project_id" translatable="false">camscannerai</string>
</resources>
```

1.51. google-services.json

```
        {
  "project_info": {
    "project_number": "333825133998",
    "firebase_url": "https://camscannerai-default-rtdb.firebaseio.com",
    "project_id": "camscannerai",
    "storage_bucket": "camscannerai.firebasestorage.app"
  },
  "client": [
    {
      "client_info": {
        "mobilesdk_app_id": "1:333825133998:android:2e0126415d5062989d0c3f",
```

```
        "android_client_info": {
          "package_name": "com.example.nexus"
        }
      },
      "oauth_client": [],
      "api_key": [
        {
          "current_key": "AIzaSyCH6qgOnnctFA0sejk5_kbSvXzqQZRYjCk"
        }
      ],
      "services": {
        "appinvite_service": {
          "other_platform_oauth_client": []
        }
      }
    }
  ],
  "configuration_version": "1"
}
```


1.52. colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="appBackground">#FFFFFF</color>
    <color name="headerBackground">#F5F7FB</color>
    <color name="titleText">#111827</color>
    <color name="subtitleText">#6B7280</color>
    <color name="inputText">#111827</color>
    <color name="inputHint">#9CA3AF</color>
    <color name="searchBackground">#FFFFFF</color>
    <color name="searchBorder">#E5E7EB</color>
    <color name="orange">#FF9800</color>
    <color name="upvote_green">#4CAF50</color>
    <color name="downvote_red">#F44336</color>
    <color name="share_gray">#374151</color>
    <color name="divider">#E5E7EB</color>
    <color name="comment_line">#D1D5DB</color>
</resources>
```

1.53. colors.xml (night)

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="appBackground">#0B0F19</color>
    <color name="headerBackground">#101726</color>
    <color name="titleText">#FFFFFF</color>
    <color name="subtitleText">#9AA4B2</color>
    <color name="inputText">#FFFFFF</color>
    <color name="inputHint">#7C8698</color>
    <color name="searchBackground">#1A2133</color>
    <color name="searchBorder">#2A3448</color>
</resources>
```

1.54. themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Nexus" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>
```

```
    <style name="Theme.Nexus" parent="Base.Theme.Nexus" />

    <style name="CircleAppearance">
        <item name="cornerSize">50%</item>
    </style>
</resources>
```

1.55. themes.xml (night)

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Nexus" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your dark theme here. -->
        <!-- <item name="colorPrimary">@color/my_dark_primary</item> -->
    </style>
</resources>
```

1.56. strings.xml

```
<resources>
    <string name="app_name">Nexus</string>
    <string name="selected_count">%d selected</string>
    <string name="nexus_ai">NEXUS AI</string>
    <string name="nexus_subtitle">Your personalized intelligence feed</string>
    <string name="show_bookmarks">Show Bookmarks</string>
    <string name="user_profile">User Profile</string>
    <string name="search_hint">Search posts, communities&#8230;</string>
    <string name="select_communities">Select Communities</string>
    <string name="add_new_post">Add new post</string>
    <string name="customize_your_feed">Customize Your Feed</string>
    <string name="customize_subtitle">Select communities you want to follow</string>
    <string name="search_communities_hint">Search communities&#8230;</string>
    <string name="start_my_feed">START MY FEED</string>
    <string name="user_name">Email ID</string>
    <string name="user_email_placeholder">user@example.com</string>
    <string name="user_id_prefix">ID: %1$s</string>
    <string name="logout">Logout</string>
    <string name="selected_count_initial">0 selected</string>
    <string name="community_format">r/%1$s</string>
</resources>
```

1.57. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <!-- Internet permission (required for Firebase) -->
    <uses-permission android:name="android.permission.INTERNET"/>
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
    <uses-permission android:name="android.permission.SCHEDULE_EXACT_ALARM" />
    <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />


    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
```

```
        android:supportsRtl="true"
        android:theme="@style/Theme.Nexus"
        tools:targetApi="31">

        <!-- Launcher (Splash / Main) -->
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <!-- Login Screen -->
        <activity
            android:name=".LoginActivity"
            android:exported="false" />

        <!-- Signup Screen -->
        <activity
            android:name=".SignupActivity"
            android:exported="false" />

        <!-- Customize Feed Screen -->
        <activity
            android:name=".CustomizeFeedActivity"
            android:exported="false" />

        <activity
            android:name=".DashboardActivity"
            android:exported="false" />

        <activity
            android:name=".PostDetailActivity"
            android:exported="false"
            android:theme="@style/Theme.Nexus" />

        <receiver
            android:name=".PostWidget"
            android:exported="true">
            <intent-filter>
                <action android:name="android.appwidget.action.APPWIDGET_UPDATE" />
                <action android:name="com.example.nexus.UPDATE_WIDGET_MANUAL" />
            </intent-filter>
            <meta-data
                android:name="android.appwidget.provider"
                android:resource="@xml/post_widget_info" />
        </receiver>

        <service
            android:name=".MyFirebaseMessagingService"
            android:exported="false">
            <intent-filter>
                <action android:name="com.google.firebase.MESSAGING_EVENT" />
            </intent-filter>
        </service>

    </application>

</manifest>
```

1.58. build.gradle.kts (Nexus)

```
// Top-level build file where you can add configuration options common to all sub-
projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
    id("com.google.gms.google-services") version "4.4.4" apply false
}
```

1.59. build.gradle.kts (:app)

```
plugins {
    alias(libs.plugins.android.application)
    id("com.google.gms.google-services")
}

android {
    namespace = "com.example.nexus"
    compileSdk {
        version = release(36) {
            minorApiLevel = 1
        }
    }

    defaultConfig {
        applicationId = "com.example.nexus"
        minSdk = 26
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

tasks.withType<JavaCompile> {
    options.compilerArgs.add("-Xlint:deprecation")
}


dependencies {
    implementation(libs.appcompat)
    implementation(libs.material)
    implementation(libs.activity)
    implementation(libs.constraintlayout)

    // Firebase BOM first to manage versions
    implementation(platform("com.google.firebase:firebase-bom:34.12.0"))
    implementation(libs.firebase.auth)
```

```
    implementation(libs.firebase.firestore)
    implementation(libs.firebase.messaging)
    implementation(libs.work.runtime)
    implementation("com.google.firebase:firebase-analytics")
    implementation(libs.glide)

    testImplementation(libs.junit)
    androidTestImplementation(libs.ext.junit)
    androidTestImplementation(libs.espresso.core)
}
```

1.60. libs.versions.toml

```
[versions]
agp = "9.1.1"
junit = "4.13.2"
junitVersion = "1.1.5"
espressoCore = "3.5.1"
appcompat = "1.6.1"
material = "1.10.0"
activity = "1.8.0"
constraintlayout = "2.1.4"
firebaseAuth = "24.0.1"
firebaseFirestore = "26.1.2"
firebaseMessaging = "24.1.0"
glide = "4.16.0"
workManager = "2.9.0"

[libraries]
junit = { group = "junit", name = "junit", version.ref = "junit" }
ext-junit = { group = "androidx.test.ext", name = "junit", version.ref = "junitVersion" }
espresso-core = { group = "androidx.test.espresso", name = "espresso-core", version.ref =
"espressoCore" }
appcompat = { group = "androidx.appcompat", name = "appcompat", version.ref = "appcompat"
}
material = { group = "com.google.android.material", name = "material", version.ref =
"material" }
activity = { group = "androidx.activity", name = "activity", version.ref = "activity" }
constraintlayout = { group = "androidx.constraintlayout", name = "constraintlayout",
version.ref = "constraintlayout" }
firebase-auth = { group = "com.google.firebase", name = "firebase-auth", version.ref =
"firebaseAuth" }
firebase-firestore = { group = "com.google.firebase", name = "firebase-firestore",
version.ref = "firebaseFirestore" }
firebase-messaging = { group = "com.google.firebase", name = "firebase-messaging",
version.ref = "firebaseMessaging" }
glide = { group = "com.github.bumptech.glide", name = "glide", version.ref = "glide" }
work-runtime = { group = "androidx.work", name = "work-runtime", version.ref =
"workManager" }

[plugins]
android-application = { id = "com.android.application", version.ref = "agp" }
```

1.61. settings.gradle.kts

```
pluginManagement {
    repositories {
        google {
            content {
                includeGroupByRegex("com\\.android.*")
                includeGroupByRegex("com\\.google.*")
                includeGroupByRegex("androidx.*")
```

```
            }
        }
        mavenCentral()
        gradlePluginPortal()
    }
}
plugins {
    id("org.gradle.toolchains.foojay-resolver-convention") version "1.0.0"
}
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
        google()
        mavenCentral()
    }
}

rootProject.name = "Nexus"
include(":app")
```

1.62. server.py

```
from http.server import HTTPServer, BaseHTTPRequestHandler
import urllib.parse

class MockServer(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header("Content-type", "text/html; charset=utf-8")
        self.end_headers()

        # Parse URL and query parameters
        parsed_path = urllib.parse.urlparse(self.path)
        params = urllib.parse.parse_qs(parsed_path.query)

        post_id = parsed_path.path.split('/')[-1]
        title = params.get('title', ['Post Title'])[0]
        content = params.get('content', ['Post content goes here...'])[0]
        author = params.get('author', ['Anonymous'])[0]
        upvotes = params.get('upvotes', ['0'])[0]
        comments = params.get('comments', ['0'])[0]

        html = f"""
        <!DOCTYPE html>
        <html>
        <head>
            <meta charset="UTF-8">
            <meta name="viewport" content="width=device-width, initial-scale=1">
            <title>Nexus - {title}</title>
            <style>
                body {{
                    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
Helvetica, Arial, sans-serif;
                    background-color: #0F111A;
                    color: #E0E7FF;
                    margin: 0;
                    padding: 20px;
                    display: flex;
                    justify-content: center;
                }}
                .card {{
```

```
                background-color: #1A1D29;
                border-radius: 16px;
                padding: 24px;
                box-shadow: 0 10px 25px rgba(0,0,0,0.5);
                max-width: 500px;
                width: 100%;
                border: 1px solid #2D3748;
            }}
            .header {{
                display: flex;
                align-items: center;
                margin-bottom: 20px;
            }}
            .avatar {{
                width: 40px;
                height: 40px;
                border-radius: 50%;
                background: linear-gradient(135deg, #6C7BFF, #3B82F6);
                margin-right: 12px;
            }}
            .title {{
                font-size: 22px;
                font-weight: 800;
                margin-bottom: 16px;
                color: #FFFFFF;
                line-height: 1.3;
            }}
            .content {{
                font-size: 16px;
                line-height: 1.6;
                color: #CBD5E0;
                white-space: pre-wrap;
            }}
            .divider {{
                height: 1px;
                background-color: #2D3748;
                margin: 24px 0;
            }}
            .actions {{
                display: flex;
                justify-content: space-around;
                color: #718096;
                font-weight: 600;
                font-size: 14px;
            }}
            .action-item {{
                display: flex;
                align-items: center;
                gap: 6px;
            }}
            .badge {{
                background-color: #2D3748;
                color: #6C7BFF;
                padding: 4px 12px;
                border-radius: 20px;
                font-size: 12px;
                margin-bottom: 8px;
                display: inline-block;
            }}
        </style>
    </head>
    <body>
        <div class="card">
```

```
                <div class="header">
                    <div class="avatar"></div>
                    <div>
                        <div style="font-weight: 700; color: #FFFFFF;">{author}</div>
                        <div style="font-size: 12px; color: #718096;">Posted to Nexus •
Just now</div>
                    </div>
                </div>
                <div class="badge">Post ID: {post_id}</div>
                <div class="title">{title}</div>
                <div class="content">{content}</div>
                <div class="divider"></div>
                <div class="actions">
                    <div class="action-item"><span>▲</span> {upvotes}</div>
                    <div class="action-item"><span>▼</span></div>
                    <div class="action-item"><span>💬</span> {comments}</div>
                    <div class="action-item"><span>🔗</span> Share</div>
                </div>
            </div>
        </body>
        </html>
        """

        self.wfile.write(bytes(html, "utf8"))

print("Server starting on http://localhost:8080...")
print("Make sure to run 'adb reverse tcp:8080 tcp:8080' to connect your device.")
HTTPServer(('0.0.0.0', 8080), MockServer).serve_forever()
```